

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\128-C-AK-With 4 nozzle.pvdb

Date of Analysis : Sep 9, 2026 9:28am

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User
 FileName : 128-C-AK-With 4 nozzle Page 2 of 222
 Vessel Design Summary: Step: 36 9:28am Sep 9,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1755.000 mm ID
 Vessel Design Length, Tangent to Tangent 8411.35 mm
 Specified Datum Line Distance 0.00 mm
 Shell Side Design Temperature 150 °C
 Channel Side Design Temperature 82 °C
 Shell Side Design Pressure 1550.000 KPa.
 Channel Side Design Pressure 1000.000 KPa.
 Shop Shell Side Test Pressure 2548.008 KPa.
 Field Shell Side Test Pressure 2015.000 KPa.
 Shop Channel Side Test Pressure 1584.308 KPa.
 Field Channel Side Test Pressure 1300.000 KPa.
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-266 2	...	...	K03506	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-266 2	...	...	K03506	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Blind Flang	1019.330	103.42	1102.000	3.0000	N/A	No
LHS Ch Flg	1017.203	103.42	1437.000	3.0000	N/A	No
LHS Ch Shell	1017.232	103.42	1437.000	3.0000	N/A	No
Main Shell	1567.232	0.00	2922.000	3.0000	N/A	No
Main Shell	1567.232	0.00	1694.000	3.0000	N/A	No
Main Shell	1567.232	0.00	2922.000	3.0000	N/A	No
RHS Ch Shell	1017.232	103.42	1437.000	3.0000	N/A	No
RHS Ch Flg	1017.203	103.42	1437.000	3.0000	N/A	No
RHS Blind Flang	1019.330	103.42	1063.000	3.0000	N/A	No

Liquid Level: 1972.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:



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Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	126.2	123.0	131.0	123.0	117.4	...	0.85	0.85
Body Flg	331.2	205.0	165.0	157.0	152.4	...	0.85	0.85
Cylinder	1131.2	800.0	14.0	14.0	10.7	...	0.85	0.85
Cylinder	3181.2	2000.0	22.0	22.0	13.1	...	1.00	1.00
Cylinder	5531.2	2350.0	14.0	14.0	13.1	...	1.00	1.00
Cylinder	7531.2	2000.0	22.0	22.0	13.1	...	1.00	1.00
Cylinder	8081.2	500.0	14.0	14.0	10.7	...	0.85	0.85
Body Flg	8289.3	205.0	165.0	157.0	152.4	...	0.85	0.85
Body Flg	8411.3	122.0	130.0	122.0	117.4	...	0.85	0.85

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	123	88.9	103.419	No Calc
20	30	157	152.425	103.419	No Calc
30	Tsht	14	7.09621	103.419	802.015
Tsht	40	14	No Calc	103.419	No Calc
40	50	22	No Calc	...	536.781
50	60	14	No Calc	...	141.089
60	Tsht	22	No Calc	...	536.781
Tsht	70	22	No Calc	...	No Calc
70	80	14	6.36151	103.419	899.915
80	90	157	152.425	103.419	No Calc
90	100	122	88.9	103.419	No Calc

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	799.969	8482.19	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	6350	No Calc	No Calc	No Calc
50	60	6350	No Calc	No Calc	No Calc
60	Tsht	6350	No Calc	No Calc	No Calc
Tsht	70	No Calc	No Calc	No Calc	No Calc
70	80	500.03	8480.55	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-Tube Inlet	WRC-297	0.935	Shell	Passed
N2-Tube Outlet	WRC-297	0.937	Shell	Passed
S3-Shell outle	WRC-297	0.983	Shell	Passed
S1-Shell Inlet	WRC-297	0.983	Shell	Passed
S2-Shell Inlet	WRC-297	0.983	Shell	Passed
S4-Shell outle	WRC-297	0.983	Shell	Passed



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Vessel Design Summary:

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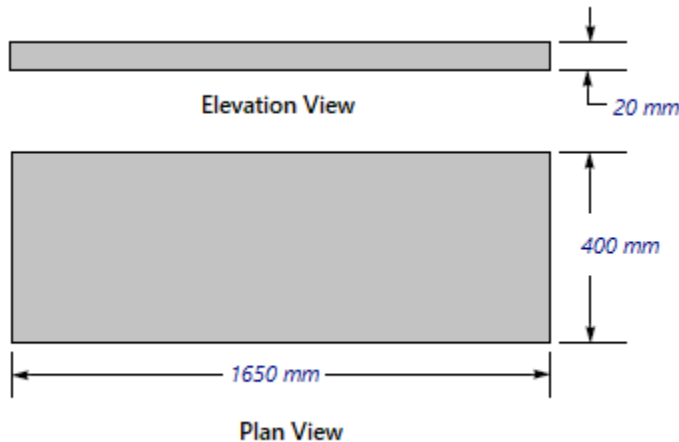
Foundation/Support Summary

Saddle Parameters:

Saddle Width	375.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	1500.000	mm
Wear Pad Width	610.000	mm
Wear Pad Thickness	14.000	mm
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	1300.000	mm

Baseplate Length	1650.000	mm
Baseplate Thickness	20.000	mm
Baseplate Width	400.000	mm
Number of Ribs (including outside ribs)	4	
Rib Thickness	10.000	mm
Web Thickness	10.000	mm
Height of Center Web	625.000	mm
Number of Bolts in Baseplate	6	

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	34627.95	Kgf
Maximum Transverse Saddle Shear Load	2111.92	Kgf
Maximum Longitudinal Saddle Shear Load	3197.49	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	29123.50	Kgf
Maximum Transverse Saddle Shear Load	197.12	Kgf
Maximum Longitudinal Saddle Shear Load	157.27	Kgf

Weights:

Fabricated - Bare w/o removable internals	35179.8	kg.
Shop Test - Fabricated + water (full)	54038.3	kg.
Shipping - Fab. + rem. intls.+ shipping app.	35179.8	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	35179.8	kg.
Empty - Fab. + intls. + details + wghts.	36510.6	kg.
Operating - Empty + operating liquid (no ca)	55369.2	kg.
Field Test - Empty weight + test medium (full)	55369.2	kg.



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Nozzle Summary:

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Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1-Tube Inlet	1437	OK	...	OK	9.41	OK	Passed
N2-Tube Outlet	1437	OK	...	OK	9.52	OK	Passed
S3-Shell outlet	2505	...	...	OK	11.33	OK	Passed
S1-Shell Inlet	2572	...	...	OK	11.33	OK	Passed
N5-Vent	1694	...	...	OK	6.42	OK	No Calc[*]
N6-Drain	1694	...	...	OK	6.42	OK	No Calc[*]
S2-Shell Inlet	2499	...	...	OK	11.33	OK	Passed
S4-Shell outlet	2505	...	...	OK	11.33	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1694 Nozzle : N5-Vent [Shellside]
 Minimum MAWP Nozzles : 1437 Nozzle : N1-Tube Inlet [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N1-Tube Inlet	731.175	0.000	399.350
30	N2-Tube Outlet	731.175	180.000	399.350
40	S3-Shell outlet	1706.175	180.000	907.280
40	S1-Shell Inlet	1706.175	0.000	907.280
50	N5-Vent	4441.130	0.000	88.352
50	N6-Drain	4524.534	180.000	88.352
60	S2-Shell Inlet	7106.175	0.000	907.280
60	S4-Shell outlet	7106.175	180.000	907.280

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 33 9:28am Sep 9,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N5-Vent	50 mm	XXS	WNF	60.3	11.074	...	...	164	150
N6-Drain	50 mm	XXS	WNF	60.3	11.074	...	...	164	150
N1-Tube Inlet	200 mm	80	WNF	219.1	12.700	379.08	16.00	320	150
N2-Tube Outlet	200 mm	80	WNF	219.1	12.700	379.08	16.00	320	150
S3-Shell outlet	477 mm Actual	WNF	WNF	477.8	15.090	...	...	455	150
S1-Shell Inlet	477 mm Actual	WNF	WNF	477.8	15.090	...	...	455	150
S2-Shell Inlet	477 mm Actual	WNF	WNF	477.8	15.090	...	...	455	150
S4-Shell outlet	477 mm Actual	WNF	WNF	477.8	15.090	...	...	455	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N5-Vent	SA-106 B	14.000	10.000	...	...	...
N6-Drain	SA-106 B	14.000	10.000	...	...	...
N1-Tube Inl	SA-106 B	14.000	10.000	12.000	16.000	...
N2-Tube Out	SA-106 B	14.000	10.000	12.000	16.000	...
S3-Shell ou	SA-266 2	14.000	10.000	...	...	...
S1-Shell In	SA-266 2	14.000	10.000	...	...	...
S2-Shell In	SA-266 2	14.000	10.000	...	...	...
S4-Shell ou	SA-266 2	14.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N5-Vent	4441.130	0.0	150.00	0.00	Main Shell
N6-Drain	4524.534	180.0	150.00	0.00	Main Shell
N1-Tube Inlet	731.175	0.0	300.00	0.00	LHS Ch Shell
N2-Tube Outlet	731.175	180.0	300.00	0.00	LHS Ch Shell
S3-Shell outlet	1706.175	180.0	400.00	0.00	Main Shell
S1-Shell Inlet	1706.175	0.0	400.00	0.00	Main Shell
S2-Shell Inlet	7106.175	0.0	400.00	0.00	Main Shell
S4-Shell outlet	7106.175	180.0	400.00	0.00	Main Shell

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Bill of Material:

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 131.0mm THK X 1955.0mm OD	SA-266 2
2	WELD NECK FLANGE: 165.0mm THK X 1955.0mm OD	SA-266 2
1	CYLINDER: 14.0mm THK X 1755.0mm ID X 800.0mm	SA-516 70
1	BLIND FLANGE: 130.0mm THK X 1955.0mm OD	SA-266 2
11	BAFFLES: 1742.1mm DIA X 9.5mm THK	SA-516 70
12	TIE RODS: 15mm DIA X 5304mm	SA-193 B7M
2	TUBESHEET: 1783.0mm X 50.0mm THK	SA-266 2
2408	TUBES: 6450.0mm X 19.0mm DIA X 2.1mm THK	SA-179
2	SADDLE: 375.0mm X 120 DEG	SA-516 70
1	INSULATION: 2000mm X 100mm THK	
1	INSULATION: 2350mm X 100mm THK	Hot Insulation
1	INSULATION: 2000mm X 100mm THK	Hot Insulation
2	GASKET: 1825.0mm OD X 1787.0mm ID	...
176	BODY FLANGE BOLTS: 28.6mm DIA	SA-193 B7
352	NUTS FOR BODY FLANGE BOLTS: 28.6mm DIA	...
1	CYLINDRICAL SEGMENT 22.0mm THK X 3047.7mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 2465.7mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 3047.7mm X 2350.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 2465.7mm X 2350.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 3063.0mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 2450.4mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 3056.7mm X 500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 2456.7mm X 500.0mm	SA-516 70
1	NAMEPLATE	...



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MDMT Summary:

Step: 35 9:28am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Main Shell [8]	B	-5	-33	-29	0.534	22.000	1.00	Yes
Main Shell [8]	B	-18	-22	-29	0.923	14.000	1.00	Yes
Main Shell [8]	B	-5	-33	-29	0.534	22.000	1.00	Yes
Left Saddle [24]	B	-35	-48		0.534	14.000	1.00	Yes
RH fixed sadd[24]	B	-35	-48		0.534	14.000	1.00	Yes
S3-Shell outle[1]	B	-5	-33	-29	0.530	22.000	1.00	Yes
Nozzle Flg [5]	A	-18	-29					
S1-Shell Inlet[1]	B	-5	-34	-29	0.524	22.000	1.00	Yes
Nozzle Flg [5]	A	-18	-29					
N5-Vent [1]	B	-8	-104	-29	0.044	9.690	1.00	Yes
Nozzle Flg [5]	A	-18	-29					
N6-Drain [1]	B	-8	-104	-29	0.044	9.690	1.00	Yes
Nozzle Flg [5]	A	-18	-29					
S2-Shell Inlet[1]	B	-5	-34	-29	0.524	22.000	1.00	Yes
Nozzle Flg [5]	A	-18	-29					
S4-Shell outle[1]	B	-5	-33	-29	0.530	22.000	1.00	Yes
Nozzle Flg [5]	A	-18	-29					
Tubesheet: SS[13]	B	-21	-45	-29	0.580	22.000	1.00	Yes
Warmest MDMT:		-5	-22					
LHS Blind Fla[11]	B	5	-5		0.833	30.750	1.00	No
LHS Ch Flg [11]	B	-18	-43	-29	0.573	14.000	1.00	No
RHS Ch Flg [11]	B	-18	-43	-29	0.573	14.000	1.00	No
RHS Blind Fla[11]	B	4	-3		0.861	30.500	1.00	No
LHS Ch Shell [8]	B	-18	-39	-29	0.620	14.000	0.85	No
RHS Ch Shell [8]	B	-18	-39	-29	0.620	14.000	0.85	No
N1-Tube Inlet [1]	B	-18	-42	-29	0.583	14.000	1.00	No
Nozzle Flg [5]	A	-18	-48					
N2-Tube Outlet[1]	B	-18	-41	-29	0.593	14.000	1.00	No
Nozzle Flg [5]	A	-18	-48					
Tubesheet: CS[14]	B	-18	-48	-29	0.411	14.000	1.00	No
Bolting [21]		-48						
Bolting [21]		-48						
Warmest MDMT:		5	-3					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-29.0	-12.0	Pass
Channel/Tube	-3.0	-12.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material



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MDMT Summary:

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- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 9:28am Sep 9,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1 9:28am Sep 9,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	1550	KPa.
Channel Side Design Pressure	999.99	KPa.
Shell Side Design Temperature	150.0	°C
Channel Side Design Temperature	82.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Lethal	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



PV Elite 28 Licensee: SL User

FileName : 128-C-AK-With 4 nozzle

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Input Echo:

Step: 1 9:28am Sep 9,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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FileName : 128-C-AK-With 4 nozzle

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Input Echo:

Step: 1 9:28am Sep 9,2026

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes

Do not use Nozzle MDMT Interpretation VIII-1 01-37 No

Use Table G instead of exact equation for "A" Yes

Shell Head Joints are Tapered Yes

Compute "K" in corroded condition Yes

Use Code Case 2286 No

Use the MAWP to compute the MDMT Yes

For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes

For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/9
Element From Node	10
Element To Node	20
Element Type	Flange
Description	LHS Blind Flang
Distance "FROM" to "TO"	123 mm
Flange Outside Diameter	1955 mm
Element Thickness	123 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	131 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230.08 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1972 mm
Liquid Density	0.0009984 kg./cm ³

Element#	2/9
Element From Node	20



PV Elite 28 Licensee: SL User

FileName : 128-C-AK-With 4 nozzle

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Input Echo:

Step: 1 9:28am Sep 9,2026

Element To Node	30
Element Type	Flange
Description	LHS Ch Flg
Distance "FROM" to "TO"	205 mm
Flange Inside Diameter	1755 mm
Element Thickness	157 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	165 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1755 mm
Liquid Density	0.0009984 kg./cm ³

Element#	3/9
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	LHS Ch Shell
Distance "FROM" to "TO"	800 mm
Inside Diameter	1755 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	242.6 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1755 mm
Liquid Density	0.0009984 kg./cm ³



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FileName : 128-C-AK-With 4 nozzle

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Input Echo:

Step: 1 9:28am Sep 9,2026

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N1-Tube Inlet	
Dist. from "FROM" Node / Offset dist	400	mm
Nozzle Diameter	200	mm
Nozzle Schedule	80	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	48.561	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N2-Tube Outlet	
Dist. from "FROM" Node / Offset dist	400	mm
Nozzle Diameter	200	mm
Nozzle Schedule	80	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	48.561	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	30	
Detail Type	Weight	
Detail ID	Pass partition	
Dist. from "FROM" Node / Offset dist	400	mm
Miscellaneous Weight	200	Kgf
Offset from Element Centerline	0	mm

Element#	4/9	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Main Shell	
Distance "FROM" to "TO"	2000	mm
Inside Diameter	1755	mm
Element Thickness	22	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	22	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1550	KPa.
Design Temperature Internal Pressure	150	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	1300	mm
Width of Saddle	375	mm
Height of Saddle at Bottom	1500	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	610	mm



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FileName : 128-C-AK-With 4 nozzle

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Input Echo:

Step: 1 9:28am Sep 9,2026

Thickness of Wear Plate	14	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.3000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Cumferential Eff. over Saddle	1.0	
Cumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1755	mm
Liquid Density	0.0009984	kg./cm ³
Element From Node	40	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2000	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	S3-Shell outlet	
Dist. from "FROM" Node / Offset dist	525	mm
Nozzle Diameter	477.8	mm
Nozzle Schedule	40	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	224.98	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	S1-Shell Inlet	
Dist. from "FROM" Node / Offset dist	525	mm
Nozzle Diameter	477.8	mm
Nozzle Schedule	40	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	233.48	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	
Element From Node	40	
Detail Type	Weight	
Detail ID	Wght:[1 of 1]	
Dist. from "FROM" Node / Offset dist	1270	mm
Miscellaneous Weight	500	Kgf
Offset from Element Centerline	1270	mm

Element#	5/9
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	2350 mm



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FileName : 128-C-AK-With 4 nozzle

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Input Echo:

Step: 1 9:28am Sep 9,2026

Inside Diameter	1755	mm
Element Thickness	14	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	14	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1550	KPa.
Design Temperature Internal Pressure	150	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1755	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2350	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N5-Vent	
Dist. from "FROM" Node / Offset dist	1260	mm
Nozzle Diameter	50	mm
Nozzle Schedule	XXS	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	4.0589	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N6-Drain	
Dist. from "FROM" Node / Offset dist	1343.4	mm
Nozzle Diameter	50	mm
Nozzle Schedule	XXS	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	4.0589	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	6/9	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	Main Shell	
Distance "FROM" to "TO"	2000	mm
Inside Diameter	1755	mm



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FileName : 128-C-AK-With 4 nozzle

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Input Echo:

Step: 1 9:28am Sep 9,2026

Element Thickness	22	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	22	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1550	KPa.
Design Temperature Internal Pressure	150	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516	70
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Saddle	
Detail ID	RH fixed saddle	
Dist. from "FROM" Node / Offset dist	700	mm
Width of Saddle	375	mm
Height of Saddle at Bottom	1500	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	610	mm
Thickness of Wear Plate	14	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1755	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	60	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2000	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	60	
Detail Type	Nozzle	
Detail ID	S2-Shell Inlet	
Dist. from "FROM" Node / Offset dist	1575	mm
Nozzle Diameter	477.8	mm
Nozzle Schedule	40	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	223.02	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266	2

Element From Node	60	
Detail Type	Nozzle	
Detail ID	S4-Shell outlet	
Dist. from "FROM" Node / Offset dist	1575	mm
Nozzle Diameter	477.8	mm



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Input Echo:

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Nozzle Schedule	40	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	224.98	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element From Node	60	
Detail Type	Weight	
Detail ID	Wght:[1 of 1]	
Dist. from "FROM" Node / Offset dist	730	mm
Miscellaneous Weight	500	Kgf
Offset from Element Centerline	1500	mm

Element#	7/9	
Element From Node	70	
Element To Node	80	
Element Type	Cylinder	
Description	RHS Ch Shell	
Distance "FROM" to "TO"	500	mm
Inside Diameter	1755	mm
Element Thickness	14	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	14	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1755	mm
Liquid Density	0.0009984	kg./cm³

Element#	8/9	
Element From Node	80	
Element To Node	90	
Element Type	Flange	
Description	RHS Ch Flg	
Distance "FROM" to "TO"	205	mm
Flange Inside Diameter	1755	mm
Element Thickness	157	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	165	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm²



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Input Echo:

Step: 1 9:28am Sep 9, 2026

Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	230.08	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1755	mm
Liquid Density	0.0009984	kg./cm ³

Element#	9/9	
Element From Node	90	
Element To Node	100	
Element Type	Flange	
Description	RHS Blind Flang	
Distance "FROM" to "TO"	122	mm
Flange Outside Diameter	1955	mm
Element Thickness	122	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	130	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	90	
Detail Type	Liquid	
Detail ID	Liquid: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1972	mm
Liquid Density	0.0009984	kg./cm ³



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XY Coordinate Calculations:

Step: 2 9:28am

Sep 9,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Blind Flang		126.175	...	123	...
LHS Ch Flg		331.175	...	205	...
LHS Ch Shell		1131.18	...	800	...
Main Shell		3181.18	...	2000	...
Main Shell		5531.17	...	2350	...
Main Shell		7531.17	...	2000	...
RHS Ch Shell		8081.17	...	500	...
RHS Ch Flg		8289.35	...	205	...
RHS Blind Flang		8411.35	...	122	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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FileName : 128-C-AK-With 4 nozzle

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Flg Calc [Int P]: RHS CH Cover

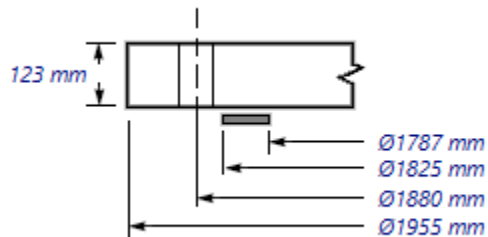
Flng: 9 9:28am Sep 9,2026

Flange Input Data Values

Description: RHS CH Cover :

LHS Blind Flang

Flange Type	TEMA Channel Cover		
Design Pressure	P	1019.33	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1955.000	mm
Flange Thickness	t	123.0000	mm
Flange Material	SA-266 2		
Flange Material UNS number	K03506		
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	5906.194	mm
Youngs Modulus for Blind Flange Matl.	Sy	197846.66	N./mm ²
Allowed Channel Cover Deflection	dfAllow	2.1937	mm
Diameter of Bolt Circle	C	1880.000	mm
Nominal Bolt Diameter	a	28.5750	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	88		
Flange Face Outside Diameter	Fod	1836.000	mm
Flange Face Inside Diameter	Fid	1755.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1825.000	mm
Gasket Inside Diameter	Gi	1787.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Go - Gi)/2$	19.000 mm
Basic Gasket Width,	$bo = N/2$	9.500 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.767 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1809.466 mm



PV Elite 28 Licensee: SL User
 FileName : 128-C-AK-With 4 nozzle
 Flg Calc [Int P]: RHS CH Cover Flng: 9 9:28am Page 23 of 222
 Sep 9, 2026

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1809.4663^2 \cdot 1019.33$$

$$= 267301.906 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.7669 \cdot \pi \cdot 1809.4663 \cdot 3.0 \cdot 1019.33$$

$$= 27536.676 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(267302 + 27537 + 0, 0)$$

$$= 294838.562 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 7.7669 \cdot \pi \cdot 1809.466 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 310423.531 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg / Sa)$$

$$= \max((294839 + 0 + 4 \cdot 0 / 1809.47) / 172, 310424 / 172)$$

$$= 176.996 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	176.996	413.315	
Radial Distance between Bolts and Edge:	28.575	37.500	
Circ. Spacing between the Bolts:	63.500	67.102	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 413.315 \cdot 172.0 / (68.95 \cdot \pi (1825.0 + 1787.0))$$

$$= 9.086 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 413.3153$$

$$= 724892.12 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 294838.56 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1880.0 - 1809.4663) / 2$$

$$= 35.2668 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 294839 \cdot 35.2668 \cdot 1.0 / (120.0^2 \cdot 1809.4663) + 0.3 \cdot 1.0 \cdot 1019.33 \cdot 1809.4663^2 / 120.0^2$$

$$= 76.97 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:



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$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 724892 \cdot 35.2668 \cdot 1.0 / (120.0^2 \cdot 1809.4663)$$

$$= 18.28 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 294839 / 413.3153$$

$$= 69.96 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (310424 / 413.3153)$$

$$= 73.65 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	76.97	138.00	18.28	138.00
Bolting	69.96	172.00	73.65	172.00

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 2.0539 mm

Allowed deflection at center of Cover 2.1937 mm

Reqd. Blind Flange Thickness at Center 100.207 mm

Reqd. Blind Flange Thickness at Gasket 50.374 mm

Min. Required Flange Thickness per TEMA 117.394 mm

Estimated M.A.W.P. (Operating) 1122 KPa.

Estimated Finished Weight of Flange at given Thk. 2861.7 kg.

Estimated Unfinished Weight of Forging at given Thk 2861.7 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.833, Temperature Reduction per Fig. UCS 66.1 = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 5 °C

Min Metal Temp. at Required thickness (UCS 66.1) -5 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	73.26	138.00	17.40	138.00
Bolting	69.96	172.00	73.65	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 116.713 mm

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Flange Input Data Values

Description: FLANGE :

LHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	1017.20	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1755.000	mm
Flange Outside Diameter	A	1955.000	mm
Flange Thickness	t	157.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	gl	21.0000	mm
Length of Hub	h	48.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1880.000	mm
Nominal Bolt Diameter	a	28.5750	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		88	
Flange Face Outside Diameter	Fod	1836.000	mm
Flange Face Inside Diameter	Fid	1755.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1825.000	mm
Gasket Inside Diameter	Gi	1787.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	1787.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



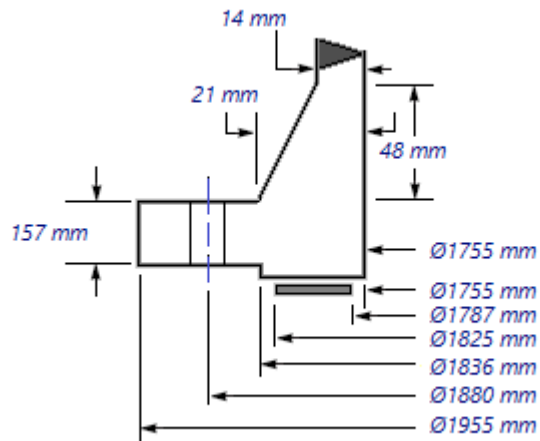
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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1017.2 \cdot (1755.0/2 + 3.0)) / (138.0 \cdot 0.85 - 0.6 \cdot 1017.2) + C_a$$

$$= 10.6760 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 11.0) / (880.5 + 0.6 \cdot 11.0)$$

$$= 1454.404 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 14.0) / (877.5 + 0.6 \cdot 14.0)$$

$$= 1853.567 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	154.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1761.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	18.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	11.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	41.500 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	19.000 mm
Basic Gasket Width, $b_o = N/2$	9.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	7.767 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1809.466 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1809.4663^2 \cdot 1017.203$$

$$= 266744.125 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 7.7669 \cdot \pi \cdot 1809.4663 \cdot 3.0 \cdot 1017.2$$

$$+ 2 \cdot 1787.0 \cdot 5.0 \cdot 3.0 \cdot 1017.2025$$

$$= 33040.172 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1761.0^2 \cdot 1017.2025$$

$$= 252646.094 \text{ Kg}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 266744 - 252646$$

$$= 14098.029 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(266744 + 33040 + 0, 0)$$

$$= 299784.281 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 7.7669 \cdot \pi \cdot 1809.466 + 68.95 \cdot 5.0 \cdot 1787.0$$

$$= 373243.844 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((299784 + 0 + 4 \cdot 0 / 1809.47) / 172, 373244 / 172)$$

$$= 212.814 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 28.575 + 6 \cdot 154.0 / (3.0 + 0.5)$$

$$= 321.150 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1880.0 \cdot \sin(\pi / 88)$$

$$= 67.102 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 67.102 / (2 \cdot 28.575 + 154.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	212.814	413.315	
Radial Distance between Hub and Bolts:	38.100	41.500	
Radial Distance between Bolts and Edge:	28.575	37.500	
Circ. Spacing between the Bolts:	63.500	67.102	321.150

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 413.315 \cdot 172.0 / (68.95 \cdot \pi (1825.0 + 1787.0))$$

$$= 9.086 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 413.3153$$

$$= 724892.12 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 299784 - 266744$$

$$= 33040.17 \text{ Kgf}$$



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Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1880.0 - 1809.4663) / 2$$

$$= 35.2668 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((1880.0 - 1761.0) / 2 + 35.2668)$$

$$= 47.3834 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1880.0 - 1761.0 - 18.0) / 2$$

$$= 50.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	252646.	50.5000	1.0000	12759.
Face Pressure, Mt	14098.	47.3834	1.0000	668.
Gasket Load, Mg	33040.	35.2668	1.0000	1165.
Gasket Seating, Matm	724892.	35.2668	1.0000	25565.

Total Moment for Operation, Mo 14592. Kg-m.

Total Moment for Gasket seating, Mg 25565. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 139.180 mm

Hub Ratio, h/h0 = HL / H0 0.345

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.636

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.868

Factor V per 4.16.5 0.330

Factor f per 4.16.5 1.239

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1955.0 / 1761.0$$

$$= 1.110$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.11^2 \cdot \log_{10}(1.11))) / (1.11^2 - 1)) / (1.11 - 1)$$

$$= 18.555$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.11^2 (1 + 8.55246 \cdot \log_{10}(1.11)) - 1) / ((1.0472 + 1.9448 \cdot 1.11^2) \cdot (1.11 - 1))$$

$$= 1.874$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.11^2 (1 + 8.55246 \cdot \log_{10}(1.11)) - 1) / (1.36136 (1.11^2 - 1) \cdot (1.11 - 1))$$

$$= 20.390$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.11^2 + 1) / (1.11^2 - 1)$$

$$= 9.603$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$



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$$= 20.39 \cdot 11.0^2 \cdot 139.18 / 0.33$$

$$= 1041516.375 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.868 / 139.18$$

$$= 0.006 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (154.0 \cdot 0.006 + 1) / 1.874 + 154.0^3 / 1041516$$

$$= 4.553$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g^{12})$$

$$= (1.239 \cdot 14592 / 1761.0) / (4.553 \cdot 18.0^2)$$

$$= 68.25 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g^{12})$$

$$= (1.239 \cdot 25565 / 1761.0) / (4.553 \cdot 18.0^2)$$

$$= 119.58 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.2804 \cdot 14592 / 1761.0) / (4.553 \cdot 154.0^2)$$

$$= 1.72 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.2804 \cdot 25565 / 1761.0) / (4.553 \cdot 154.0^2)$$

$$= 3.01 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (18.5551 \cdot 14592 / (154.0^2 \cdot 1761.0)) - 9.6034 \cdot 2$$

$$= 47.10 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (18.5551 \cdot 25565 / (154.0^2 \cdot 1761.0)) - 9.6034 \cdot 3$$

$$= 82.51 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (68 + \max(2, 47)) / 2$$

$$= 57.67 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (120 + \max(3, 83)) / 2$$

$$= 101.04 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 299784 / 413.3153$$

$$= 71.13 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (373244 / 413.3153)$$

$$= 88.56 \text{ N./mm}^2$$



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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	68.25	207.00	119.58	207.00
Radial Flange	1.72	138.00	3.01	138.00
Tangential Flange	47.10	138.00	82.51	138.00
Maximum Average	57.67	138.00	101.04	138.00
Bolting	71.13	172.00	88.56	172.00

Minimum Required Flange Thickness [Rigidity] 152.425 mm

Estimated M.A.W.P. (Operating) 1454 KPa.

Estimated Finished Weight of Flange at given Thk. 745.4 kg.

Estimated Unfinished Weight of Forging at given Thk 925.9 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	344.74	374.99	0.919	344.74	374.99	0.919
Tangential Flange	Sto	158.71	250.00	0.635	158.71	250.00	0.635
Combined	Sh/f + St	747.97	749.99	0.997	747.97	749.99	0.997
Combined	Sr + St	357.30	749.99	0.476	357.30	749.99	0.476
Combined	Sh + Sto	658.70	749.99	0.878	658.70	749.99	0.878
Radial Flange	Sr	12.56	499.99	0.025	12.56	499.99	0.025
Maximum Stress Ratio				1.000			1.000



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Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.6321/1.3162)

= 1.2400

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.4991/1.2681) if $h/h0 < 0.35$, else max (0.25; 0.3032/0.7587)

= 0.3936

In this case, the hub ratio $h/h0$, 0.3449 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	718.6 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	40.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Root Area of a Single Bolt	Abs	4.6968 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.639 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (1825.0^2 - 1787.0^2) + (0.5 \cdot 1787.0 \cdot 5.0)$
 $= 1122.7 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 207.0 \cdot 1122.683 / (88 \cdot 4.697)$
 $= 562.3 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(562.271; 507.5)$
 $= 507.5 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(507.5; 290.0)$
 $= 507.5 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(507.5; 718.624)$
 $= 507.5 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$
 $:: 507.5 \geq 40.0(1122.683 / (88 \cdot 4.697))$
 $:: 507.5 \geq 108.7 \text{ N./mm}^2$ Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 507.5 \geq (97.0 \cdot 1122.683 + \pi/4 \cdot 1017.203 \cdot 1787.0^2) / (0.7 \cdot 4.697 \cdot 88)$
 $:: 507.5 \geq 464.6 \text{ N./mm}^2$ Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$
 $:: 507.5 \leq 276.0(1122.683 / (88 \cdot 4.697))$
 $:: 507.5 \leq 749.7 \text{ N./mm}^2$ Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax(\text{thetagmax} / \text{thetafmax})$
 $:: 507.5 \leq 718.624(1.0/0.639)$
 $:: 507.5 \leq 1123.9 \text{ N./mm}^2$ Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 507.5 \cdot 0.2 \cdot 4.697 \cdot 28.575$
 $= 138.9 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



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Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 25565.2 / 1.0 \cdot 9806.637 \cdot 0.33 / (4.234 \cdot 201334 \cdot 11.0^2 \cdot 139.18 \cdot 0.3)$
 $= 1.000$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 14592.1 / 1.0 \cdot 9806.637 \cdot 0.33 / (4.234 \cdot 197847 \cdot 11.0^2 \cdot 139.18 \cdot 0.3)$
 $= 0.581$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 25565.2 / 1.0 \cdot 9806.637 \cdot 0.33 / (4.553 \cdot 201334 \cdot 11.0^2 \cdot 139.18 \cdot 0.3)$
 $= 0.930$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 14592.1 / 1.0 \cdot 9806.637 \cdot 0.33 / (4.553 \cdot 197847 \cdot 11.0^2 \cdot 139.18 \cdot 0.3)$
 $= 0.540$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.573, Temperature Reduction per Fig. UCS 66.1 = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	65.29	207.00	111.38	207.00
Radial Flange	2.20	138.00	3.76	138.00
Tangential Flange	40.57	138.00	69.20	138.00
Maximum Average	52.93	138.00	90.29	138.00
Bolting	71.13	172.00	88.56	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 139.421 mm

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Flg Calc [Int P]: RHSCH Flange

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Flange Input Data Values

Description: RHSCH Flange :

RHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	1017.20	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1755.000	mm
Flange Outside Diameter	A	1955.000	mm
Flange Thickness	t	157.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	gl	21.0000	mm
Length of Hub	h	48.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1880.000	mm
Nominal Bolt Diameter	a	28.5750	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		88	
Flange Face Outside Diameter	Fod	1836.000	mm
Flange Face Inside Diameter	Fid	1755.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1825.000	mm
Gasket Inside Diameter	Gi	1787.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	1787.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



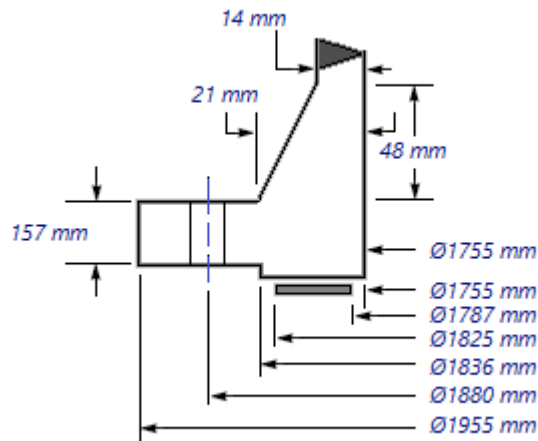
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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1017.2 \cdot (1755.0/2 + 3.0)) / (138.0 \cdot 0.85 - 0.6 \cdot 1017.2) + C_a$$

$$= 10.6760 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 11.0) / (880.5 + 0.6 \cdot 11.0)$$

$$= 1454.404 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 14.0) / (877.5 + 0.6 \cdot 14.0)$$

$$= 1853.567 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	154.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1761.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	18.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	11.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	41.500 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	19.000 mm
Basic Gasket Width, $b_o = N/2$	9.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	7.767 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1809.466 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1809.4663^2 \cdot 1017.203$$

$$= 266744.125 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 7.7669 \cdot \pi \cdot 1809.4663 \cdot 3.0 \cdot 1017.2$$

$$+ 2 \cdot 1787.0 \cdot 5.0 \cdot 3.0 \cdot 1017.2025$$

$$= 33040.172 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1761.0^2 \cdot 1017.2025$$

$$= 252646.094 \text{ Kg}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 266744 - 252646$$

$$= 14098.029 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(266744 + 33040 + 0, 0)$$

$$= 299784.281 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 7.7669 \cdot \pi \cdot 1809.466 + 68.95 \cdot 5.0 \cdot 1787.0$$

$$= 373243.844 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((299784 + 0 + 4 \cdot 0 / 1809.47) / 172, 373244 / 172)$$

$$= 212.814 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 28.575 + 6 \cdot 154.0 / (3.0 + 0.5)$$

$$= 321.150 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1880.0 \cdot \sin(\pi / 88)$$

$$= 67.102 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ (Bs / (2a + t)) }, 1)$$

$$= \max(\sqrt{ (67.102 / (2 \cdot 28.575 + 154.0)) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	212.814	413.315	
Radial Distance between Hub and Bolts:	38.100	41.500	
Radial Distance between Bolts and Edge:	28.575	37.500	
Circ. Spacing between the Bolts:	63.500	67.102	321.150

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 413.315 \cdot 172.0 / (68.95 \cdot \pi (1825.0 + 1787.0))$$

$$= 9.086 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 413.3153$$

$$= 724892.12 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 299784 - 266744$$

$$= 33040.17 \text{ Kgf}$$



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Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1880.0 - 1809.4663) / 2$$

$$= 35.2668 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((1880.0 - 1761.0) / 2 + 35.2668)$$

$$= 47.3834 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1880.0 - 1761.0 - 18.0) / 2$$

$$= 50.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	252646.	50.5000	1.0000	12759.
Face Pressure, Mt	14098.	47.3834	1.0000	668.
Gasket Load, Mg	33040.	35.2668	1.0000	1165.
Gasket Seating, Matm	724892.	35.2668	1.0000	25565.

Total Moment for Operation, Mo 14592. Kg-m.

Total Moment for Gasket seating, Mg 25565. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 139.180 mm

Hub Ratio, h/h0 = HL / H0 0.345

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.636

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.868

Factor V per 4.16.5 0.330

Factor f per 4.16.5 1.239

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1955.0 / 1761.0$$

$$= 1.110$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.11^2 \cdot \log_{10}(1.11))) / (1.11^2 - 1)) / (1.11 - 1)$$

$$= 18.555$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.11^2 (1 + 8.55246 \cdot \log_{10}(1.11)) - 1) / ((1.0472 + 1.9448 \cdot 1.11^2) \cdot (1.11 - 1))$$

$$= 1.874$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.11^2 (1 + 8.55246 \cdot \log_{10}(1.11)) - 1) / (1.36136 (1.11^2 - 1) \cdot (1.11 - 1))$$

$$= 20.390$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.11^2 + 1) / (1.11^2 - 1)$$

$$= 9.603$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$



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$$= 20.39 \cdot 11.0^2 \cdot 139.18 / 0.33$$

$$= 1041516.375 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.868 / 139.18$$

$$= 0.006 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (154.0 \cdot 0.006 + 1) / 1.874 + 154.0^3 / 1041516$$

$$= 4.553$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g^{12})$$

$$= (1.239 \cdot 14592 / 1761.0) / (4.553 \cdot 18.0^2)$$

$$= 68.25 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g^{12})$$

$$= (1.239 \cdot 25565 / 1761.0) / (4.553 \cdot 18.0^2)$$

$$= 119.58 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.2804 \cdot 14592 / 1761.0) / (4.553 \cdot 154.0^2)$$

$$= 1.72 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.2804 \cdot 25565 / 1761.0) / (4.553 \cdot 154.0^2)$$

$$= 3.01 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$$

$$= (18.5551 \cdot 14592 / (154.0^2 \cdot 1761.0)) - 9.6034 \cdot 2$$

$$= 47.10 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (18.5551 \cdot 25565 / (154.0^2 \cdot 1761.0)) - 9.6034 \cdot 3$$

$$= 82.51 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (68 + \max(2, 47)) / 2$$

$$= 57.67 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (120 + \max(3, 83)) / 2$$

$$= 101.04 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 299784 / 413.3153$$

$$= 71.13 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (373244 / 413.3153)$$

$$= 88.56 \text{ N./mm}^2$$



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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	68.25	207.00	119.58	207.00
Radial Flange	1.72	138.00	3.01	138.00
Tangential Flange	47.10	138.00	82.51	138.00
Maximum Average	57.67	138.00	101.04	138.00
Bolting	71.13	172.00	88.56	172.00

Minimum Required Flange Thickness [Rigidity] 152.425 mm

Estimated M.A.W.P. (Operating) 1454 KPa.
 Estimated Finished Weight of Flange at given Thk. 745.4 kg.
 Estimated Unfinished Weight of Forging at given Thk 925.9 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	344.74	374.99	0.919	344.74	374.99	0.919
Tangential Flange	Sto	158.71	250.00	0.635	158.71	250.00	0.635
Combined	Sh/f + St	747.97	749.99	0.997	747.97	749.99	0.997
Combined	Sr + St	357.30	749.99	0.476	357.30	749.99	0.476
Combined	Sh + Sto	658.70	749.99	0.878	658.70	749.99	0.878
Radial Flange	Sr	12.56	499.99	0.025	12.56	499.99	0.025
Maximum Stress Ratio				1.000			1.000



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Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.6321/1.3162)

= 1.2400

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.4991/1.2681) if $h/h0 < 0.35$, else max (0.25; 0.3032/0.7587)

= 0.3936

In this case, the hub ratio $h/h0$, 0.3449 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	718.6 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	40.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Root Area of a Single Bolt	Abs	4.6968 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.639 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (1825.0^2 - 1787.0^2) + (0.5 \cdot 1787.0 \cdot 5.0)$
 $= 1122.7 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 207.0 \cdot 1122.683 / (88 \cdot 4.697)$
 $= 562.3 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(562.271; 507.5)$
 $= 507.5 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(507.5; 290.0)$
 $= 507.5 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(507.5; 718.624)$
 $= 507.5 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$
 $:: 507.5 \geq 40.0(1122.683 / (88 \cdot 4.697))$
 $:: 507.5 \geq 108.7 \text{ N./mm}^2$ Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 507.5 \geq (97.0 \cdot 1122.683 + \pi/4 \cdot 1017.203 \cdot 1787.0^2) / (0.7 \cdot 4.697 \cdot 88)$
 $:: 507.5 \geq 464.6 \text{ N./mm}^2$ Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$
 $:: 507.5 \leq 276.0(1122.683 / (88 \cdot 4.697))$
 $:: 507.5 \leq 749.7 \text{ N./mm}^2$ Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax(\text{thetagmax} / \text{thetafmax})$
 $:: 507.5 \leq 718.624(1.0/0.639)$
 $:: 507.5 \leq 1123.9 \text{ N./mm}^2$ Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 507.5 \cdot 0.2 \cdot 4.697 \cdot 28.575$
 $= 138.9 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 25565.2 / 1.0 \cdot 9806.637 \cdot 0.33 / (4.234 \cdot 201334 \cdot 11.0^2 \cdot 139.18 \cdot 0.3)$
 $= 1.000$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 14592.1 / 1.0 \cdot 9806.637 \cdot 0.33 / (4.234 \cdot 197847 \cdot 11.0^2 \cdot 139.18 \cdot 0.3)$
 $= 0.581$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 25565.2 / 1.0 \cdot 9806.637 \cdot 0.33 / (4.553 \cdot 201334 \cdot 11.0^2 \cdot 139.18 \cdot 0.3)$
 $= 0.930$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 14592.1 / 1.0 \cdot 9806.637 \cdot 0.33 / (4.553 \cdot 197847 \cdot 11.0^2 \cdot 139.18 \cdot 0.3)$
 $= 0.540$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.573, Temperature Reduction per Fig. UCS 66.1 = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	65.29	207.00	111.38	207.00
Radial Flange	2.20	138.00	3.76	138.00
Tangential Flange	40.57	138.00	69.20	138.00
Maximum Average	52.93	138.00	90.29	138.00
Bolting	71.13	172.00	88.56	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 139.421 mm

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Flg Calc [Int P]: RHS Cover

Flng: 12

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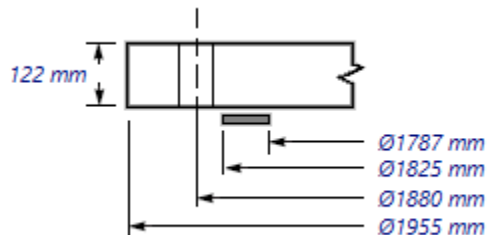
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Flange Input Data Values

Description: RHS Cover :

RHS Blind Flang

Flange Type	TEMA Channel Cover		
Design Pressure	P	1019.33	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1955.000	mm
Flange Thickness	t	122.0000	mm
Flange Material	SA-266 2		
Flange Material UNS number	K03506		
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	5906.194	mm
Youngs Modulus for Blind Flange Matl.	Sy	197846.66	N./mm ²
Allowed Channel Cover Deflection	dfAllow	2.1937	mm
Diameter of Bolt Circle	C	1880.000	mm
Nominal Bolt Diameter	a	28.5750	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	88		
Flange Face Outside Diameter	Fod	1836.000	mm
Flange Face Inside Diameter	Fid	1755.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1825.000	mm
Gasket Inside Diameter	Gi	1787.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	19.000 mm
Basic Gasket Width,	$bo = N/2$	9.500 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.767 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1809.466 mm



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 Flg Calc [Int P]: RHS Cover Flng: 12 9:28am Page 44 of 222
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Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1809.4663^2 \cdot 1019.33$$

$$= 267301.906 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.7669 \cdot \pi \cdot 1809.4663 \cdot 3.0 \cdot 1019.33$$

$$= 27536.676 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(267302 + 27537 + 0, 0)$$

$$= 294838.562 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 7.7669 \cdot \pi \cdot 1809.466 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 310423.531 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg / Sa)$$

$$= \max((294839 + 0 + 4 \cdot 0 / 1809.47) / 172, 310424 / 172)$$

$$= 176.996 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	176.996	413.315	
Radial Distance between Bolts and Edge:	28.575	37.500	
Circ. Spacing between the Bolts:	63.500	67.102	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 413.315 \cdot 172.0 / (68.95 \cdot \pi (1825.0 + 1787.0))$$

$$= 9.086 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 413.3153$$

$$= 724892.12 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 294838.56 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1880.0 - 1809.4663) / 2$$

$$= 35.2668 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 294839 \cdot 35.2668 \cdot 1.0 / (119.0^2 \cdot 1809.4663) + 0.3 \cdot 1.0 \cdot 1019.33 \cdot 1809.4663^2 / 119.0^2$$

$$= 78.27 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:



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Flg Calc [Int P]: RHS Cover

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$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 724892 \cdot 35.2668 \cdot 1.0 / (119.0^2 \cdot 1809.4663)$$

$$= 18.59 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 294839 / 413.3153$$

$$= 69.96 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (310424 / 413.3153)$$

$$= 73.65 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	78.27	138.00	18.59	138.00
Bolting	69.96	172.00	73.65	172.00

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 2.1061 mm

Allowed deflection at center of Cover 2.1937 mm

Reqd. Blind Flange Thickness at Center 100.207 mm

Reqd. Blind Flange Thickness at Gasket 50.374 mm

Min. Required Flange Thickness per TEMA 117.394 mm

Estimated M.A.W.P. (Operating) 1082 KPa.

Estimated Finished Weight of Flange at given Thk. 2838.4 kg.

Estimated Unfinished Weight of Forging at given Thk 2838.4 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.861, Temperature Reduction per Fig. UCS 66.1 = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 4 °C

Min Metal Temp. at Required thickness (UCS 66.1) -3 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	74.47	138.00	17.69	138.00
Bolting	69.96	172.00	73.65	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 116.713 mm

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 Internal Pressure Calculations: Step: 7 9:28am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Blind Flang		1019.3	131	3	1955	117.3	1
LHS Ch Flg		1017.2	165	3	1755	117.3	2
LHS Ch Shell		1017.2	14	3	1755	117.3	3
Main Shell		1567.2	22	3	1755	138	4
Main Shell		1567.2	14	3	1755	138	5
Main Shell		1567.2	22	3	1755	138	6
RHS Ch Shell		1017.2	14	3	1755	117.3	7
RHS Ch Flg		1017.2	165	3	1755	117.3	8
RHS Blind Flang		1019.3	130	3	1955	117.3	9

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Blind Flang		1000	1102.49	1276.14	123	117.394	1
LHS Ch Flg		1000	1437.2	1853.57	157	152.425	2
LHS Ch Shell		1000	1437.17	1853.57	14	10.6763	3
Main Shell		1550	2922.34	3408.3	22	13.069	4
Main Shell		1550	1693.83	2180.67	14	13.069	5
Main Shell		1550	2922.34	3408.3	22	13.069	6
RHS Ch Shell		1000	1437.17	1853.57	14	10.6763	7
RHS Ch Flg		1000	1437.2	1853.57	157	152.425	8
RHS Blind Flang		1000	1062.57	1233.44	122	117.394	9

Warning: Shell Side MAWP governed by a standard Flange!

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:
 For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 1580 KPa.
 Shell Side MAPnc = 1960 KPa.
 Channel Side MAWP = 1063 KPa.
 Channel Side MAPnc = 1233 KPa.

Elements Suitable for Design Internal Pressure.

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Element: LHS Ch Shell

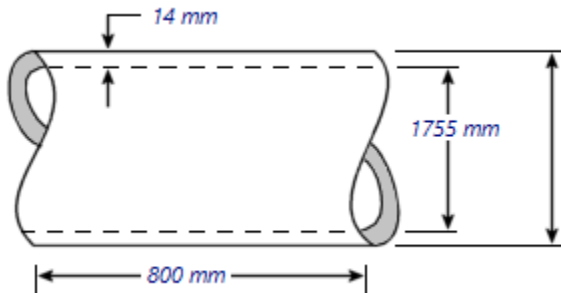
Step: 8

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Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	1017.232 KPa.	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	800.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1755.000 mm	Actual Stress	96.5 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	44811.7 cm ²	SG of Contents	1.000
Element Volume	1935583.5 cm ³	Weight of Contents	1934.4 Kg
Empty Weight	482.4 Kg	Operating Weight	2416.8 Kg
		Comp. All. Ope. Stress	117.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1017.232 \cdot 880.5) / (138.0 \cdot 0.85 - 0.6 \cdot 1017.232) \\
 &= 7.6763 + 3.0000 = 10.6763 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 17.232 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 11.0) / (880.5 + 0.6 \cdot 11.0) \\
 &= 1454 - 17 = 1437 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 14.0) / (877.5 + 0.6 \cdot 14.0) \\
 &= 1854 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1017.232 \cdot (880.5 + 0.6 \cdot 11.0)) / (0.85 \cdot 11.0) \\
 &= 96.518 \text{ N./mm}^2
 \end{aligned}$$



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 Element: LHS Ch Shell Step: 8 9:28am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.791 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14.0$, $t_r = 8.02$, $c = 3.0$ mm, $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.62$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -39 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1062.571 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.620	Temperature Difference	21 °C
UCS-66(c) red.	No	Computed Min. Temp.	-39 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: LHS Ch Shell

Cylindrical Shell From 30 to Tsht Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	1783.00	799.97	162.09	0.4487	0.0015423	97.51

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 97.5051) / (3 \cdot 162.0909) = 802.0145 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.096	1783.00	799.97	435.28	0.4487	0.0003377	33.77

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 33.7664) / (3 \cdot 435.2807) = 103.4259 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	1783.00	8482.19	162.09	4.7573	0.0001258	12.58

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 12.5809) / (3 \cdot 162.0909) = 103.4827 \text{ KPa.}$$



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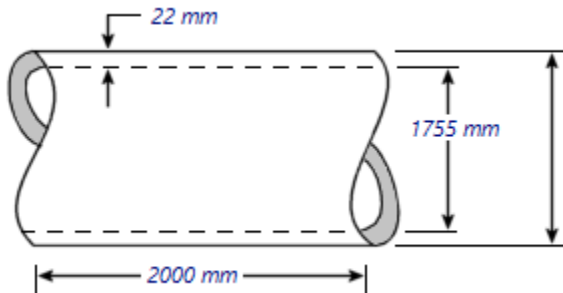
Element: Main Shell

Step: 9 9:28am

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	1567.232 KPa.	Int. Design Temperature	150 °C
Material	SA-516 70	Ext. Design Temperature	150 °C
Length	2000.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1755.000 mm	Actual Stress	73.6 N./mm ²
Spec. Min. Thk t,e, or sa	22.0000 mm	Spec. Nominal Thk.	22.000 mm
Surface Area	113034.5 cm ²	SG of Contents	1.000
Element Volume	3466042.2 cm ³	Weight of Contents	3463.9 Kg
Empty Weight	1903.8 Kg	Operating Weight	5367.7 Kg
		Comp. All. Ope. Stress	131.1 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1567.232 \cdot 880.5) / (138.0 \cdot 1.0 - 0.6 \cdot 1567.232) \\
 &= 10.0690 + 3.0000 = 13.0690 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 17.232 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 19.0) / (880.5 + 0.6 \cdot 19.0) \\
 &= 2940 - 17 = 2922 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 22.0) / (877.5 + 0.6 \cdot 22.0) \\
 &= 3408 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1567.232 \cdot (880.5 + 0.6 \cdot 19.0)) / (1.0 \cdot 19.0) \\
 &= 73.573 \text{ N./mm}^2
 \end{aligned}$$



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Element: Main Shell Step: 9 9:28am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.238 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 22.0$, $t_r = 10.152$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.534$, Temp. Reduction = 28°C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-33 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1580.044 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	22.000 mm	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.534	Temperature Difference	28 °C
UCS-66(c) red.	No	Computed Min. Temp.	-33 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

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FileName : 128-C-AK-With 4 nozzle

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Element: Main Shell

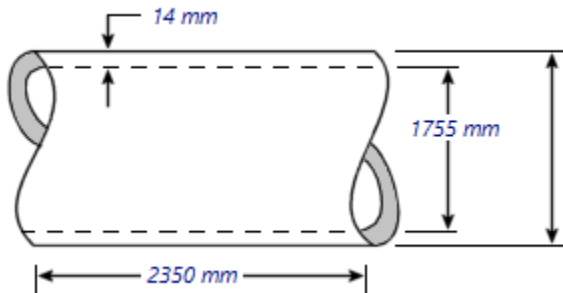
Step: 10

9:28am

Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	1567.232 KPa.	Int. Design Temperature	150 °C
Material	SA-516 70	Ext. Design Temperature	150 °C
Length	2350.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1755.000 mm	Actual Stress	126.4 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	131634.3 cm ²	SG of Contents	1.000
Element Volume	4072599.8 cm ³	Weight of Contents	4070.1 Kg
Empty Weight	1417.1 Kg	Operating Weight	5487.2 Kg
		Comp. All. Ope. Stress	116.7 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1567.232 \cdot 880.5) / (138.0 \cdot 1.0 - 0.6 \cdot 1567.232) \\
 &= 10.0690 + 3.0000 = 13.0690 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 17.232 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 11.0) / (880.5 + 0.6 \cdot 11.0) \\
 &= 1711 - 17 = 1694 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 14.0) / (877.5 + 0.6 \cdot 14.0) \\
 &= 2181 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1567.232 \cdot (880.5 + 0.6 \cdot 11.0)) / (1.0 \cdot 11.0) \\
 &= 126.397 \text{ N./mm}^2
 \end{aligned}$$



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FileName : 128-C-AK-With 4 nozzle

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Element: Main Shell

Step: 10

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.791 %**Minimum Design Metal Temperature Results:**Govrn. thk, $t_g = 14.0$, $t_r = 10.152$, $c = 3.0$ mm, $E^* = 1.0$ Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.923$, Temp. Reduction = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -22 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

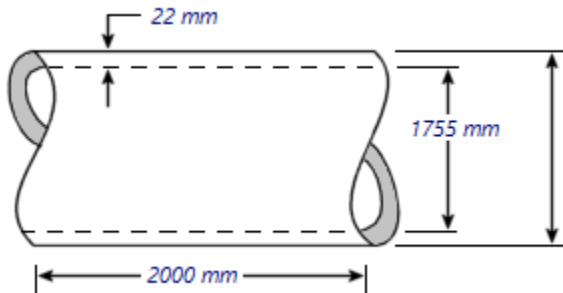
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1580.044 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.923	Temperature Difference	4 °C
UCS-66(c) red.	No	Computed Min. Temp.	-22 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	1567.232 KPa.	Int. Design Temperature	150 °C
Material	SA-516 70	Ext. Design Temperature	150 °C
Length	2000.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1755.000 mm	Actual Stress	73.6 N./mm ²
Spec. Min. Thk t,e, or sa	22.0000 mm	Spec. Nominal Thk.	22.000 mm
Surface Area	113034.5 cm ²	SG of Contents	1.000
Element Volume	3466042.2 cm ³	Weight of Contents	3463.9 Kg
Empty Weight	1903.8 Kg	Operating Weight	5367.7 Kg
		Comp. All. Ope. Stress	131.1 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1567.232 \cdot 880.5) / (138.0 \cdot 1.0 - 0.6 \cdot 1567.232) \\
 &= 10.0690 + 3.0000 = 13.0690 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 17.232 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 19.0) / (880.5 + 0.6 \cdot 19.0) \\
 &= 2940 - 17 = 2922 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 22.0) / (877.5 + 0.6 \cdot 22.0) \\
 &= 3408 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1567.232 \cdot (880.5 + 0.6 \cdot 19.0)) / (1.0 \cdot 19.0) \\
 &= 73.573 \text{ N./mm}^2
 \end{aligned}$$



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FileName : 128-C-AK-With 4 nozzle

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Element: Main Shell

Step: 11

9:28am

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.238 %**Minimum Design Metal Temperature Results:**Govrn. thk, $t_g = 22.0$, $t_r = 10.152$, $c = 3.0$ mm, $E^* = 1.0$ Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.534$, Temp. Reduction = 28 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C

Min Metal Temp. at Required thickness (UCS 66.1) -33 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

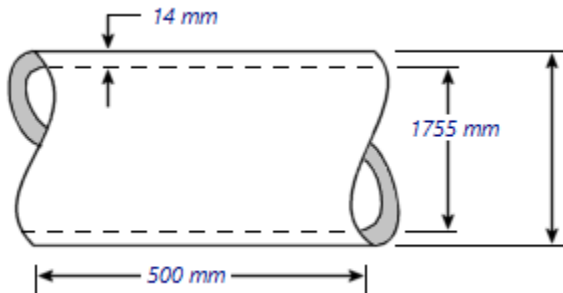
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1580.044 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	22.000 mm	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.534	Temperature Difference	28 °C
UCS-66(c) red.	No	Computed Min. Temp.	-33 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: RHS Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	1017.232 KPa.	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		500.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1755.000 mm	Actual Stress		96.5 N./mm ²
Spec. Min. Thk t,e, or sa		14.0000 mm	Spec. Nominal Thk.		14.000 mm
Surface Area		28007.3 cm ²	SG of Contents		1.000
Element Volume		1209739.8 cm ³	Weight of Contents		1209.0 Kg
Empty Weight		301.5 Kg	Operating Weight		1510.5 Kg
			Comp. All. Ope. Stress		131.1 N./mm ²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		85 %	Long. Joint Eff.	E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1017.232 \cdot 880.5) / (138.0 \cdot 0.85 - 0.6 \cdot 1017.232) \\
 &= 7.6763 + 3.0000 = 10.6763 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 17.232 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 11.0) / (880.5 + 0.6 \cdot 11.0) \\
 &= 1454 - 17 = 1437 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 14.0) / (877.5 + 0.6 \cdot 14.0) \\
 &= 1854 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1017.232 \cdot (880.5 + 0.6 \cdot 11.0)) / (0.85 \cdot 11.0) \\
 &= 96.518 \text{ N./mm}^2
 \end{aligned}$$



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 FileName : 128-C-AK-With 4 nozzle Page 56 of 222
 Element: RHS Ch Shell Step: 12 9:28am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.791 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14.0$, $t_r = 8.02$, $c = 3.0$ mm, $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.62$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -39 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1062.571 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.620	Temperature Difference	21 °C
UCS-66(c) red.	No	Computed Min. Temp.	-39 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: RHS Ch Shell

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	1783.00	500.03	162.09	0.2804	0.0026209	109.41

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 109.4074) / (3 \cdot 162.0909) = 899.915 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.362	1783.00	500.03	530.42	0.2804	0.0004115	41.15

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 41.1453) / (3 \cdot 530.4158) = 103.4231 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	1783.00	8480.55	162.09	4.7563	0.0001259	12.58

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 12.5836) / (3 \cdot 162.0909) = 103.5045 \text{ KPa.}$$



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FileName : 128-C-AK-With 4 nozzle

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Shop Pressure Test Results:

Step: 13

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	2054.057 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2015.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2548.008 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1738.048 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	2216.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	2216.500 KPa.
App 27-4	= M.A.W.P.	1580.044 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1381.342 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1300.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1584.308 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1168.828 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1430.000 KPa.
App 27-4	= M.A.W.P.	1062.571 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	119.2	234.0	0.510	1601.51
Main Shell	103.9	234.0	0.444	2565.21
Main Shell	162.3	234.0	0.694	2565.21
Main Shell	103.9	234.0	0.444	2565.21
RHS Ch Shell	119.2	234.0	0.510	1601.51

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	118.00	118.00	1.000
N1-Tube Inlet	Pad	138.00	138.00	1.000
N2-Tube Outlet	Nozzle	118.00	118.00	1.000
N2-Tube Outlet	Pad	138.00	138.00	1.000
S3-Shell outlet	Nozzle	138.00	138.00	1.000
S1-Shell Inlet	Nozzle	138.00	138.00	1.000
N5-Vent	Nozzle	118.00	118.00	1.000
N6-Drain	Nozzle	118.00	118.00	1.000
S2-Shell Inlet	Nozzle	138.00	138.00	1.000
S4-Shell outlet	Nozzle	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Blind Flang	138.00	138.00	1.000
LHS Ch Flg	138.00	138.00	1.000
LHS Ch Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000



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FileName : 128-C-AK-With 4 nozzle

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Shop Pressure Test Results:

Step: 13

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Main Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
RHS Ch Shell	138.00	138.00	1.000
RHS Ch Flg	138.00	138.00	1.000
RHS Blind Flang	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	13.17	216.00	0.061
N2-Tube Outlet	13.17	216.00	0.061
S3-Shell outlet	39.59	225.00	0.176
S1-Shell Inlet	39.59	225.00	0.176
N5-Vent	5.96	216.00	0.028
N6-Drain	5.96	216.00	0.028
S2-Shell Inlet	39.59	225.00	0.176
S4-Shell outlet	39.59	225.00	0.176

Elements Suitable for Test Pressure.

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 FileName : 128-C-AK-With 4 nozzle Page 59 of 222
 Field Pressure Test Results: Step: 14 9:28am Sep 9,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	2054.057 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2015.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2548.008 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1738.048 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	2216.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	2216.500 KPa.
App 27-4	= M.A.W.P.	1580.044 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1381.342 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1300.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1584.308 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1168.828 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1430.000 KPa.
App 27-4	= M.A.W.P.	1062.571 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 1550.0 • 1.0
 = 2015.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 1000.0 • 1.0
 = 1300.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	125.0	234.0	0.534	1317.20
Main Shell	95.4	234.0	0.408	2032.20
Main Shell	163.9	234.0	0.700	2032.20
Main Shell	95.4	234.0	0.408	2032.20
RHS Ch Shell	125.0	234.0	0.534	1317.20

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	118.00	118.00	1.000
N1-Tube Inlet	Pad	138.00	138.00	1.000
N2-Tube Outlet	Nozzle	118.00	118.00	1.000
N2-Tube Outlet	Pad	138.00	138.00	1.000
S3-Shell outlet	Nozzle	138.00	138.00	1.000
S1-Shell Inlet	Nozzle	138.00	138.00	1.000
N5-Vent	Nozzle	118.00	118.00	1.000
N6-Drain	Nozzle	118.00	118.00	1.000
S2-Shell Inlet	Nozzle	138.00	138.00	1.000
S4-Shell outlet	Nozzle	138.00	138.00	1.000



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Field Pressure Test Results:

Step: 14

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Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Blind Flang	138.00	138.00	1.000
LHS Ch Flg	138.00	138.00	1.000
LHS Ch Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
RHS Ch Shell	138.00	138.00	1.000
RHS Ch Flg	138.00	138.00	1.000
RHS Blind Flang	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	10.83	216.00	0.050
N2-Tube Outlet	10.83	216.00	0.050
S3-Shell outlet	31.36	225.00	0.139
S1-Shell Inlet	31.36	225.00	0.139
N5-Vent	4.72	216.00	0.022
N6-Drain	4.72	216.00	0.022
S2-Shell Inlet	31.36	225.00	0.139
S4-Shell outlet	31.36	225.00	0.139

Elements Suitable for Test Pressure.

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FileName : 128-C-AK-With 4 nozzle

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ASME TS Calc: Pressure Test

Case: 1 9:28a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		2015.00 KPa.
Shell Thickness	ts	22.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1755.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		1300.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1755.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	2408
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	6450.00 mm
Straight Tube Length (bet. inner tubsht faces) L		6350.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	857.961 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	20857.4 cm ²



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Perimeter of Tube Layout Cp 5403.5 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
Thickness of Tubesheet h 50.0000 mm
Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
Tubesheet Outside Diameter A 1783.000 mm

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 573.000 mm
Tube End condition corresponding to Span (l) k 1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
Perform Differential Pressure Design No
Run Multiple Load Cases Yes
Channel Side Min. Design Pressure Ptd,min -103.4190 KPa.

Mean Shell Metal Temp. along Shell len. Tsm 21.11 °C
Mean Tube Metal Temp. along Tube length Ttm 21.11 °C
Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



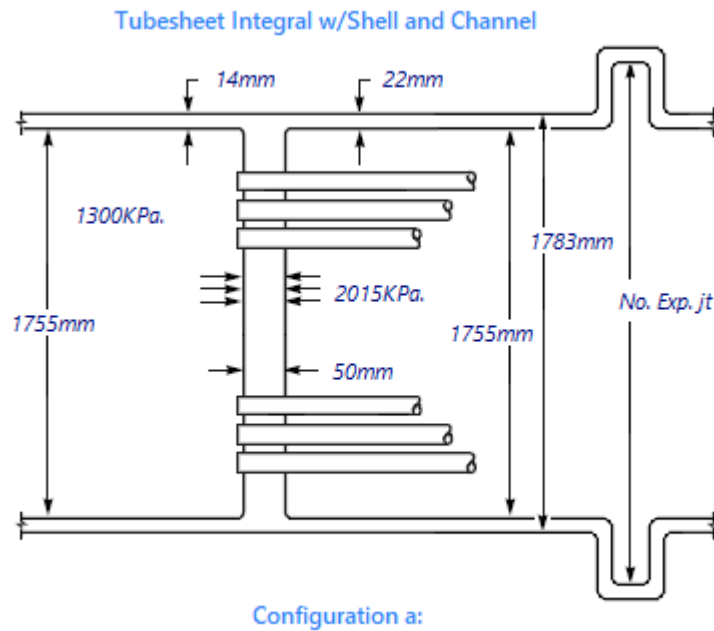
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ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:



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The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1300.0 \cdot (19.05/2 - 0)) / (92.4 \cdot 1.0 + 0.4 \cdot 1300.0)$$

$$= 0.1333 + 0.0000 = 0.1333 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	6350.00	9.03	50.0000	0.0134948	95.84
EMAWP = (2.167/(D/T) - 0.0833) * B = 15019.1992 KPa.						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4598	19.05	6350.00	41.43	50.0000	0.0006407	65.84
EMAWP = (4*B)/(3*(D/T)) = (4 * 65.8397)/(3 * 41.4339) = 2118.5835 KPa.						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4598 mm
Allowable Internal Pressure at Corroded thickness	22456.99 KPa.
Required Internal Design Pressure	1300.00 KPa.
Allowable External Pressure at Corroded thickness	15019.20 KPa.
Required External Design Pressure	2118.42 KPa.
Required Thickness due to Shell Side pressure	0.4598 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/50.0$$

$$= 0.0000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (p); dt - 2 \cdot tt)$$

$$= \max(19.05 - 2 \cdot 2.11 (0.20270E+09/0.20270E+09) \cdot (92/137) \cdot (0); 19.05 - 2 \cdot 2.11)$$

$$= 19.0500 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 857.961 + 19.05$$

$$= 1734.972 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.75 - 19.05) / 31.75$$

$$= 0.400$$



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Equivalent outer tube limit radius [ao]:

$$= D_o / 2$$

$$= 1734.9719 / 2$$

$$= 867.4860 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)}$$

$$= 31.75 / \sqrt{1 - 4 \cdot \min(0 \cdot 100.0 ; 4 \cdot 1734.972 \cdot 31.75) / (\pi \cdot 1734.972^2)}$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.75 - 19.05) / 31.75$$

$$= 0.4000$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 877.5 / 867.486$$

$$= 1.0115$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 877.5 / 867.486$$

$$= 1.0115$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 2408 ((19.05 - 2 \cdot 2.11) / (2 \cdot 867.486))^2$$

$$= 0.8241$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot a_o))^2$$

$$= 1 - 2408 (19.05 / (2 \cdot 867.486))^2$$

$$= 0.7097$$

Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((8.0 - 0); 0)$$

$$= 8.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot t_s(D_s + t_s) E_s / L$$

$$= \pi \cdot 22.0 (1755.0 + 22.0) \cdot 20270E+09 / 6350.0$$

$$= 399796.6875 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot t_t(D_t - t_t) E_t / L$$

$$= \pi \cdot 2.11 (19.05 - 2.11) \cdot 20270E+09 / 6350.0$$

$$= 365.5316 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (Nt \cdot K_t)$$

$$= 399796.688 / (2408 \cdot 365.532)$$

$$= 0.45421$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s / K_j)$$

$$= 1 / (1 + 399796.688 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + t_s) t_s)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1755.0 + 22.0) 22.0)$$



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= 0.0092 1/mm

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.009 \cdot 0.20270E+09 \cdot 22.0^3 / (6(1 - 0.3^2))$$

$$= 3634397184.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1755.0 + 22.0) \cdot 0.36344E+1 / 50.0^3 \cdot (1 + 50.0 \cdot 0.009 + (50.0 \cdot 0.009)^2 / 2)$$

$$= 485259648.0000 \text{ KPa}$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1755.0^2 / (4 \cdot 0.20270E+09 \cdot 22.0) \cdot (1 - 1755.0 \cdot 0.3 / (2(1755.0 + 22.0)))$$

$$= 0.14708E+00 \text{ mm/N} \cdot \text{mm}^2$$

Channel coefficient [βc]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(1755.0 + 14.0)14.0}$$

$$= 0.0116 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.012 \cdot 0.20270E+09 \cdot 14.0^3 / (6(1 - 0.3^2))$$

$$= 1176726656.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1755.0 + 14.0 / \cos(0)) \cdot 0.11767E+1 / 50.0^3 \cdot (1.0 + 50.0 \cdot 0.012 \cdot (1.0 + 1.0) / 2 + (50.0 \cdot 0.012)^2 / 2 \cdot 1.0)$$

$$= 174292240.0 \text{ KPa}$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1755.0^2 / (4 \cdot 0.20270E+09 \cdot 14.0) \cdot (1 - 1755.0 \cdot 0.3 / (2(1755.0 + 14.0)))$$

$$= 0.23095E+00 \text{ mm/N} \cdot \text{mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 1.574803$; $\mu^* = 0.400000$; $E^* = 87293600 \text{ KPa}$.

$$E^*/E = 0.430652$$
 ; $\nu^* = 0.301272$

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^*)Nt \cdot Et \cdot t(d_t - t)_{ao^2}) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.301^2)2408 \cdot 0.20270E+09 \cdot 2.11(19.05 - 2.11)867.486^2) / (87293600 \cdot 6350.0 \cdot 50.0^3))^{1/4}$$

$$= 8.0189$$

Values from Table 4.18.3

$$Z_d = 0.002893$$
 ; $Z_v = 0.015911$; $Z_m = 0.181704$
 $Z_a = 0.107359E+04$; $Z_w = 0.015911$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1783.0 / 1734.9719$$



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= 1.0277

Compute coefficient [F]:

$$= (1 - v) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.3) / 87293600 \cdot (0.48526E+09 + 0.17429E+09 + 0.20270E+09 \cdot \ln(1.03))$$

= 5.3236

Compute parameter [φ]:

$$= (1 + v) F$$

$$= (1 + 0.3013) 5.3236$$

= 6.9274

Compute parameter [Q1]:

$$= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.0115 - 1 - 6.9274 \cdot 0.0159) / (1 + 6.9274 \cdot 0.1817)$$

= -0.043687

Compute Parameter [Qz1]:

$$= (Z_d + Q_1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.00289 + -0.04369 \cdot 0.01591) / 2 \cdot 8.01888^4$$

= 4.5444

Compute parameter [Qz2]:

$$= (Z_v + Q_1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.01591 + -0.04369 \cdot 0.1817) / 2 \cdot 8.01888^4$$

= 16.4829

Compute parameter [U]:

$$= (Z_w + (p_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.0159 + (1.0115 - 1) 0.1817) 8.01888^4 / (1 + 6.9274 \cdot 0.1817)$$

= 32.9658

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

= 0 (config a)

Compute Parameter [γ]:

= 0.000 mm (For Pressure only cases)

Calculate parameter [ws]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.0115 \cdot 0.36344E+1 \cdot 0.0092 \cdot 0.147081 (1 + 50.0 \cdot 0.0092)$$

= 7257.0796 mm²

Calculate parameter [ω*s]:

$$= a o^2 (p_s^2 - 1) \cdot (p_s - 1) / 4 - \omega_s$$

$$= 867.486^2 (1.012^2 - 1) \cdot (1.012 - 1) / 4 - 7257.08$$

= -7206.6494 mm²

Parameter [wc]:

$$= p_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)]$$

$$= 1.012 \cdot 0.11767E+1 [0.230946 (1.0 + 0.012 \cdot 1.0 \cdot 50.0) - 0 (1.0/50.0 + 1.0 \cdot 0.012/2)]$$

= 5009.6675 mm²

Parameter [ω*c]:

$$= a o^2 [(p_c^2 + 1) \cdot (p_c - 1) / 4 - (p_s - 1) / 2] - \omega_c$$

$$= 867.48596^2 [(1.01154^2 + 1) \cdot (1.01154 - 1) / 4 - (1.01154 - 1) / 2] - 5009.66748$$

= -4959.2378 mm²

Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D



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Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= Ps \{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs - [(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \}$$

$$= 2015.0 \{ 0.71 + 2(1 - 0.71) 0.3 + [2/0.454 \cdot (1755.0/1734.972)^2] 0.3 - [(1.012^2 - 1)/(1.0 \cdot 0.454)] - [(1 - 1.0)/(2 \cdot 1.0 \cdot 0.454)] [(0^2 - (1755.0)^2)/1734.972^2] \}$$

$$= 4401.5630 \text{ KPa.}$$

Pressure [P't]:

$$= [xt + 2(1 - xt) vt + 1/(J \cdot Kst)] Pt$$

$$= [0.824 + 2(1 - 0.824) 0.3 + 1/(1.0 \cdot 0.454)] 1300.0$$

$$= 4070.6182 \text{ KPa.}$$

Pressure [Py]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 2408 \cdot 365.532 \cdot 0 / (3.142 \cdot 867.486^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [Pw]:

$$= -\gamma b \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2)$$

$$= -0 \cdot 32.966 \cdot 0 / (2 \cdot \pi \cdot 867.486^2)$$

$$= -0.0000 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega*s \cdot Ps - \omega*c \cdot Pt)$$

$$= -(32.966/867.486^2) \cdot (-7206.649 \cdot 2015.0 - 4959.238 \cdot 1300.0)$$

$$= 353.7108 \text{ KPa.}$$

Calculate the pressure [Pw]:

$$= (U/ao^2) \cdot (\omega*s \cdot P*s - \omega*c \cdot P*c)$$

$$= (32.966/867.486^2) \cdot (7257.0796 \cdot 0 - 5009.6675 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot (P's - P't + Py + Pw + Pw + Prim)$$

$$= 0.1000E+01 \cdot 0.454 / (1 + 1.0 \cdot 0.454 (4.544 + (1.012 - 1)16.483)) \cdot (4401.563 - 4070.618 + 0 + 0 + 0 + 353.711)$$

$$= 98.7062 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega*s \cdot Ps - \omega*c \cdot Pt) - (\omega*s \cdot P*s - \omega*c \cdot P*c)) + W* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-7206.649 \cdot 2015.0 - 4959.238 \cdot 1300.0) - (7257.08 \cdot 0 - 5009.667 \cdot 0)) \cdot 0 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 6.92743 \cdot 0.1817)$$

$$= -364.5350 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.044 + 2 \cdot -364.535 / (98.706 \cdot 867.486^2)$$

$$= -0.13994$$

Fm Value from Table 4.18.3 = 0.069969

Tube sheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu*) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$



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$$= (1.5 \cdot 0.07 / 0.4) \cdot (2 \cdot 867.486 / (50.0 - 8.0))^2 \cdot 98.71$$

$$= 44.1969 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot Sy = 1.5 \cdot 0.95 \cdot 250.0 = 356.24 \text{ N./mm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu \star) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.0271 / 0.4) \cdot (2 \cdot 867.486 / (15.457 - 8.0))^2 \cdot 64.84$$

$$= 356.2436 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc$$

$$= 15.4574 + 0 + 0$$

$$= 15.4574 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta T \cdot Sy)$$

$$= (2 \cdot 0.4 \cdot 50.0 / 867.486) \cdot 189.997$$

$$= 8.7608 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [Pe] <= Taulimit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 3.8100 \text{ mm}$$

$$\text{Tubesheet Shear Stress} = 6.0134 \text{ N./mm}^2$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(HreqB; HreqS)$$

$$= \max(15.4574; 3.81)$$

$$= 15.4574 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -0.1614, F_{tmax} = 2.2116$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot F_{tmin}) / (Xt - Xs)$$

$$= ((2015.0 \cdot 0.7097 - 1300.0 \cdot 0.8241) - (98.706) \cdot -0.161) /$$

$$(0.8241 - 0.7097)$$

$$= 3.2760 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot F_{tmax}) / (Xt - Xs)$$

$$= ((2015.0 \cdot 0.7097 - 1300.0 \cdot 0.8241) - (98.706) \cdot 2.212) /$$

$$(0.8241 - 0.7097)$$

$$= 1.2280 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 3.276 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= \beta T \cdot Sy_{t} = 170.9969 \text{ N./mm}^2$$

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 3.28 \cdot 1.1229$$

$$= 37.51 \text{ Kg}$$

Tube Metal Area [A_t]:

$$= \pi (d_o - t) t$$



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ASME TS Calc: Pressure Test

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$$= \pi (19.05 - 2.11) 2.11$$

$$= 1.123 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (19.05 - 2.11) 170.997$$

$$= 1957.967 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 171.0$$

$$= 1957.9670 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 170.997 / 170.997$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 171.0$$

$$= 1300.8848 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) 1.0 \cdot 1.0} - 0.75 \cdot 19.05 \right)$$

$$= 2.562 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm

Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 1957.97 is $\geq$ tube strength 1957.97 Kg

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= \beta T \cdot S_{y,s} = 0.95 \cdot 260.0 = 370.49 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [$\sigma_{s,m}$]:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 867.486^2 / ((1755.0 + 22.0) 22.0) \cdot [98.71 + (1.012^2 - 1)$$

$$(2015.0 - 1300.0)] + 877.5^2 \cdot 1300.0 / ((1755.0 + 22.0) 22.0)$$

$$= 27.8264 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 22.0 mm, the minimum Shell length adjacent to the tubesheet should be 353.690 mm

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	44.2 $\leq$	356.2 N./mm ²	Pass
Tubesheet Shear Stress	1.9 $\leq$	190.0 N./mm ²	Pass



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ASME TS Calc: Pressure Test

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Maximum Tube Stress	3.3	<=	171.0 N./mm ²	Pass
Maximum Force on any one Tube	37.5	<=	1958.0 Kg	Pass
Axial Membrane Stress in Shell	27.8	<=	370.5 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	15.457	50.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	28.345 ...	112	356	12 190	Ps+Pt-Th D1	Ok
D2uc	33.428 ...	165	356	15 190	Ps+Pt-Th D2	Ok
D3uc	15.457 ...	44	356	2 190	Ps+Pt-Th D3	Ok
Max:	33.4277 ... mm	0.463		0.079 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses Ten Allwd Cmp Allwd	:	---- Shell Band Stress Ten Allwd Cmp Allwd	:	Pass Fail
D1uc	13 370 ...				Ok
D2uc	14 370 ...				Ok
D3uc	28 370 ...				Ok
Max	RATIO 0.075 ...				

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses Ten Allwd Cmp Allwd	Tube Loads Ld Allwd	Shell Stress Stress Allwd	Channel Stress Stress Allwd	Pass Fail
D1uc	13 170 -11 -170	153 1957	... 370	... 370	Ok



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ASME TS Calc: Pressure Test

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D2uc	15	170	-13	-170	172	1957	...	370	...	370	Ok
D3uc	3	170	...	...	38	1957	...	370	...	370	Ok

Max RATIO	0.088	0.073	0.088	...	...
-----------	-------	-------	-------	-----	-----

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	33.428	50.000	Ok
Tube thickness :	0.460	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Min Shell length of thk, (22.000) adj. to tubesheet: 353.690 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm

Due to Pressure Shell Compresses by : -0.076 mm

Due to Pressure + Thermal Shell Compresses by : -0.076 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0136	...	...	...	...	...
2	-0.0758	...	...	...	...	...
3	-0.0611	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

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 External Pressure Calculations: Step: 15 9:28am Sep 9,2026

External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	120	No Calc	No Calc
20	30	No Calc	...	154	No Calc	No Calc
30	Tsht	799.969	1783	11	0.0015423	97.5051
Tsht	40	No Calc	...	11	No Calc	No Calc
40	50	6350	1799	19	0.00038155	38.1208
50	60	6350	1783	11	0.00017168	17.1529
60	Tsht	6350	1799	19	0.00038155	38.1208
Tsht	70	No Calc	...	19	No Calc	No Calc
70	80	500.03	1783	11	0.0026209	109.407
80	90	No Calc	...	154	No Calc	No Calc
90	100	No Calc	...	119	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	123	88.9	103.419	No Calc
20	30	157	152.425	103.419	No Calc
30	Tsht	14	7.09621	103.419	802.015
Tsht	40	14	No Calc	103.419	No Calc
40	50	22	No Calc	...	536.781
50	60	14	No Calc	...	141.089
60	Tsht	22	No Calc	...	536.781
Tsht	70	22	No Calc	...	No Calc
70	80	14	6.36151	103.419	899.915
80	90	157	152.425	103.419	No Calc
90	100	122	88.9	103.419	No Calc

Minimum

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External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	799.969	8482.19	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	6350	No Calc	No Calc	No Calc
50	60	6350	No Calc	No Calc	No Calc
60	Tsht	6350	No Calc	No Calc	No Calc
Tsht	70	No Calc	No Calc	No Calc	No Calc
70	80	500.03	8480.55	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.



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External Pressure Calculations: Step: 15 9:28am Sep 9, 2026

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 Element and Detail Weights: Step: 16 9:28am Sep 9,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	3047.79	...	3047.79	...	304.779
20	30	781.513	518084	754.16	518879	78.1513
30	40	482.422	1935583	379.688	1948841	48.2422
40	50	1903.8	3466042	1646.96	3499186	190.38
50	60	1417.12	4072599	1115.33	4111544	141.712
60	70	1903.8	3466042	1646.96	3499186	190.38
70	80	301.513	1209739	237.305	1218025	30.1513
80	90	781.513	518084	754.16	518879	78.1513
90	100	3024.52	...	3024.52	...	302.452
Total		13643	15186174.00	12606	15314541.00	1364



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Liqd	...	61.5	-8.50006	...	Liquid: 10
20	Liqd	517.767	102.5	...	...	Liquid: 20
30	Liqd	1934.4	400	...	...	Liquid: 30
30	Nozl	53.4174	400	974.337	...	N1-Tube Inlet
30	Nozl	53.4174	400	-974.337	...	N2-Tube Outlet
30	Wght	200	400	...	...	Pass partition
40	Sadl	495.045	1300	1170.75	...	Left Saddle
40	Liqd	3463.93	1000	...	...	Liquid: 40
40	Insl	420.49	1000	...	...	Insulation
40	Nozl	247.476	525	-1116.41	...	S3-Shell outlet
40	Nozl	256.83	525	1116.41	...	S1-Shell Inlet
40	Wght	500	1270	-1270	...	Wght:[1 of 1]
50	Liqd	4070.12	1175	...	...	Liquid: 50
50	Insl	489.913	1175	...	...	Insulation
50	Nozl	4.46483	1259.96	896.588	...	N5-Vent
50	Nozl	4.46483	1343.36	-896.588	...	N6-Drain
60	Sadl	495.045	700	1170.75	...	RH fixed saddle
60	Liqd	3463.93	1000	...	...	Liquid: 40
60	Insl	420.49	1000	...	...	Insulation
60	Nozl	245.32	1575	1116.41	...	S2-Shell Inlet
60	Nozl	247.476	1575	-1116.41	...	S4-Shell outlet
60	Wght	500	730	-1500	...	Wght:[1 of 1]
70	Liqd	1209	250	...	...	Liquid: 50
80	Liqd	517.767	102.5	...	...	Liquid: 60
90	Liqd	...	61	-8.50006	...	Liquid: 70
30	FTsh	771.783	825	...	...	Fixed Tubesheet
30	Tube	14869.1	4025	...	...	Tubes
30	RTsh	771.783	7225	...	...	Tubesheet
40	Bafl	132.284	574	...	...	Baffles
40	Bafl	132.284	1148	...	...	Baffles
40	Bafl	132.284	1722	...	...	Baffles
40	Bafl	132.284	2296	...	...	Baffles
40	Bafl	132.284	2870	...	...	Baffles
40	Bafl	132.284	3444	...	...	Baffles
40	Bafl	132.284	4018	...	...	Baffles



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Element and Detail Weights:

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40 Baf1	132.284	4592	...	...	Baffles
40 Baf1	132.284	5166	...	...	Baffles
40 Baf1	132.284	5740	...	...	Baffles
40 Baf1	132.921	6314	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	990.1
Liquid	15176.9
Insulation	1330.9
Nozzles	1112.9
Weights	1200.0
Exchanger Components	17868.4
Liquid in Tubes	2681.6

Sum of the Detail Weights 40360.8 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	15008.4	15008.4	15008.4	15008.4	15008.4	15008.4
Saddles	990.1	990.1	990.1	990.1	990.1	990.1
Nozzles	1112.9	1112.9	1112.9	1112.9	1112.9	1112.9
Wld Weights	200.0	200.0	200.0	200.0	200.0	200.0
Exchanger	17868.4	17868.4	17868.4	17868.4	17868.4	17868.4
Insulation	...	...	...	...	1330.9	1330.9 *
Test Weights	...	1000.0	...	...	...	...
Ope Weights	...	...	...	...	...	1000.0
Ope. Liquid	...	...	...	...	...	15176.9
Tube Ope Lqd	...	...	...	...	...	2681.6
Test Liquid	...	15176.9	...	...	...	...
Tube Tst Lqd	...	2681.6	...	...	...	...
Totals	35179.8	54038.3	35179.8	35179.8	36510.6	55369.2

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	35179.8 kg.
Shop Test Wt.	- Fabricated weight + water (full)	54038.3 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	35179.8 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	35179.8 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	36510.6 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	55369.2 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	54228.4 kg.
Field Test Wt.	- Empty weight + test medium (full)	55369.2 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	2683281.0 cm ³
Weight of Ope Liq in tubes :	2681.6 kg.
Weight of Water in tubes :	2681.6 kg.

Surface Areas of Elements:

	Outside Surface	Inside Surface
--	-----------------	----------------



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Element and Detail Weights:

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From	To	Area cm ²	Area cm ²
10	20	38063.9	...
20	30	18671.5	11743.7
30	40	44811.7	44108
40	50	113035	110270
50	60	131634	129567
60	70	113035	110270
70	80	28007.3	27567.5
80	90	18671.5	11743.7
90	100	38002.5	...

Total	543931.7	445269.9 cm ²
Total	585.5	479.3 ft ²

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Nozzle Flange MAWP:

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Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Flange Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N1-Tube Inlet	1820.4	1960.0	82	150	GR 1.1	...	...	...	...
N2-Tube Outlet	1820.4	1960.0	82	150	GR 1.1	...	...	...	...
S3-Shell outlet	1580.0	1960.0	150	150	GR 1.1	1657.6	1580.0	1580.0	712.4
S1-Shell Inlet	1580.0	1960.0	150	150	GR 1.1	1657.6	1580.0	1580.0	712.4
N5-Vent	1580.0	1960.0	150	150	GR 1.1	...	...	...	...
N6-Drain	1580.0	1960.0	150	150	GR 1.1	...	...	...	...
S2-Shell Inlet	1580.0	1960.0	150	150	GR 1.1	1657.6	1580.0	1580.0	712.4
S4-Shell outlet	1580.0	1960.0	150	150	GR 1.1	1657.6	1580.0	1580.0	712.4

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 1580.044 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 1960.006 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 1820.434 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 1960.006 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
S3-Shell outlet	551.688	1.20	5046	3824	1580.0	1580.0	Pass
S1-Shell Inlet	551.688	1.20	5046	3824	1580.0	1580.0	Pass
S2-Shell Inlet	551.688	1.20	5046	3824	1580.0	1580.0	Pass
S4-Shell outlet	551.688	1.20	5046	3824	1580.0	1580.0	Pass

Evaluating Flange on Nozzle [S3-Shell outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(5046) + 4 \cdot \text{abs}(3824) \cdot 551.688 \leq \pi \cdot 551.688^3 [(1580.044 - 1580.044) + 1.200 \cdot 1580.044]$$

$$= 0.89171E+05 \leq 0.10200E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle S3-Shell outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1580.044, 1.200 \cdot 1580.044 + 1580.044 - 89170.9 / (\pi \cdot 551.688^3))$$

$$= \min(1580.044, 1818.468)$$

$$= 1580.044 \text{ KPa.}$$

Evaluating Flange on Nozzle [S1-Shell Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(5046) + 4 \cdot \text{abs}(3824) \cdot 551.688 \leq \pi \cdot 551.688^3 [(1580.044 - 1580.044) + 1.200 \cdot 1580.044]$$

$$= 0.89171E+05 \leq 0.10200E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle S1-Shell Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1580.044, 1.200 \cdot 1580.044 + 1580.044 - 89170.9 / (\pi \cdot 551.688^3))$$



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Nozzle Flange MAWP:

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= min(1580.044, 1818.468)
= 1580.044 KPa.

Evaluating Flange on Nozzle [S2-Shell Inlet]:

= $16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$
= $16(5046) + 4 \cdot \text{abs}(3824) 551.688 \leq \pi \cdot 551.688^3 [(1580.044 - 1580.044) + 1.200 \cdot 1580.044]$
= $0.89171E+05 \leq 0.10200E+06$ Check Passes

Maximum Pressure for Flange on Nozzle S2-Shell Inlet [Pmax]:

= min(PR, $FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)$)
= min(1580.044, $1.200 \cdot 1580.044 + 1580.044 - 89170.9 / (\pi \cdot 551.688^3)$)
= min(1580.044, 1818.468)
= 1580.044 KPa.

Evaluating Flange on Nozzle [S4-Shell outlet]:

= $16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$
= $16(5046) + 4 \cdot \text{abs}(3824) 551.688 \leq \pi \cdot 551.688^3 [(1580.044 - 1580.044) + 1.200 \cdot 1580.044]$
= $0.89171E+05 \leq 0.10200E+06$ Check Passes

Maximum Pressure for Flange on Nozzle S4-Shell outlet [Pmax]:

= min(PR, $FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)$)
= min(1580.044, $1.200 \cdot 1580.044 + 1580.044 - 89170.9 / (\pi \cdot 551.688^3)$)
= min(1580.044, 1818.468)
= 1580.044 KPa.

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation:

Step: 18 9:28am Sep 9,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$\begin{aligned}
 &= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar}))) \\
 &= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.228 \cdot 0.978)/(1 + 1.7 \cdot 3.4 \cdot 0.228))) \\
 &= \min(0.85, 0.913) \\
 &= 0.850
 \end{aligned}$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.556
Structure Height to Diameter ratio	4.375

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Zg = 274320.0$ mm

Effective Height [z]:

$$\begin{aligned}
 &= \text{Centroid Height} + \text{Vessel Base Elevation} \\
 &= 1500.0 + 300.0 = 1800.0 \text{ mm} \\
 &= 5.906 \text{ ft. Imperial Units}
 \end{aligned}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\begin{aligned}
 &\text{Because } z (5.906 \text{ ft.}) < 15 \text{ ft.} \\
 &= 2.01(15 / Zg)^{2/\alpha} \\
 &= 2.01(15/900.0)^{2/9.5} \\
 &= 0.849
 \end{aligned}$$

Type of Hill: No Hill



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Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0.0 • 0.0 • 0.0)²

= 1.0

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 0.849 • 1.0 • 1.0 • 1.0 • 60.399²)

= 16.0 psf [78.12] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16.0 • 0.85 • 0.556 • 3.106

= 23.5 lbs. [10.7] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Blind Flang	1800.0	0.000	0.000	0.000	0.849	1.000	78.120
LHS Ch Flg	1800.0	0.000	0.000	0.000	0.849	1.000	78.120
LHS Ch Shell	1800.0	0.000	0.000	0.000	0.849	1.000	78.120
Main Shell	1800.0	0.000	0.000	0.000	0.849	1.000	78.120
Main Shell	1800.0	0.000	0.000	0.000	0.849	1.000	78.120
Main Shell	1800.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS Ch Shell	1800.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS Ch Flg	1800.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS Blind Flang	1800.0	0.000	0.000	0.000	0.849	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Pass partitio	0.00	1800.00	78.12	0.00
Wght:[1 of 1]	0.00	300.00	78.12	0.00
Wght:[1 of 1]	0.00	300.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1800	2346	2885.58	78.12	11.724
20	30	1800	2106	4317.3	78.12	17.541
30	40	1800	2139.6	17116.8	78.12	69.5447
40	50	1800	2398.8	47976	78.12	194.924
50	60	1800	2379.6	55920.6	78.12	227.203
60	70	1800	2398.8	47976	78.12	194.924
70	80	1800	2139.6	10698	78.12	43.4655
80	90	1800	2106	4317.3	78.12	17.541
90	100	1800	2346	2862.12	78.12	11.6287

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.



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Wind Load Calculation:

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(8.12830.75)$
 = 0.096 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.096$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.157 \cdot 54379 \cdot 1.25$
 = 3197.492 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.157 \cdot 54379$
 = 1705.3 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	977.5	4943.56	317.234



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Earthquake Load Calculation:

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20	30	877.5	4943.56	284.781
30	40	877.5	4943.56	284.781
40	50	877.5	4943.56	284.781
40	50	877.5	4943.56	284.781
50	60	877.5	4943.56	284.781
60	0	877.5	4943.56	284.781
60	70	877.5	4943.56	284.781
70	80	877.5	4943.56	284.781
80	90	877.5	4943.56	284.781
90	100	977.5	4943.56	317.234

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation:

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Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	4356.175 mm
Center of Gravity of the Liquid	4160.838 mm
Center of Gravity of the Insulation	4356.175 mm
Center of Gravity of the Nozzles	4026.065 mm
Center of Gravity of the Added Weights (Operating)	3752.008 mm
Center of Gravity of the Added Weights (Empty)	731.175 mm
Center of Gravity of the Tubesheet(s)	4356.175 mm
Center of Gravity of the Tubes	4356.175 mm
Center of Gravity of the Baffles	4626.431 mm
Center of Gravity of Bare Shell New and Cold	4214.566 mm
Center of Gravity of Bare Shell Corroded	4215.403 mm
Vessel CG in the Operating Condition	4252.618 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	4275.894 mm
Vessel CG in the Test Condition	4247.030 mm

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Saddle Calcs: Road Transport:

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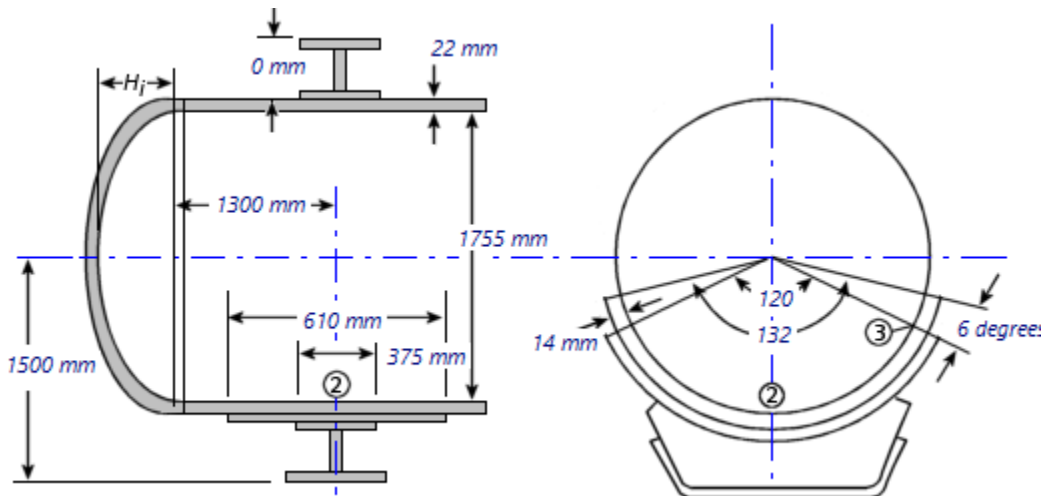
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	888.50	mm
Shell Thickness used in this Case	t	22.000	mm
Stiffened Vessel Length per 4.15.6	L	6350.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1300.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	610.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1500.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		20742.54	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



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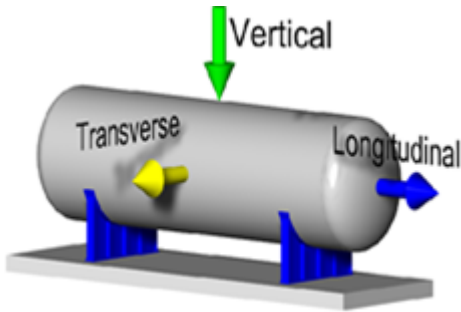
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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-2.05	-208.80
Long. Stress at Top of Midspan	2.05	138.00
Long. Stress at Bottom of Midspan	2.05	138.00
Long. Stress at Top of Saddles	7.13	138.00
Long. Stress at Bottom of Saddles	-3.95	-234.00
Long. Stress at Bottom of Saddles	3.95	138.00
Tangential Shear in Shell	7.19	110.40
Circ. Stress at Horn of Saddle	16.31	207.00
Circ. Compressive Stress in Shell	0.72	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 208.89 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \\
 &= 1774.4 \text{ lbf} \\
 &= 804.871 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 5.902^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \cdot 1.2 \\
 &= 278.9 \text{ lbf} \\
 &= 126.5 \text{ Kg}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$\begin{aligned}
 &= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw} \\
 &= 0.1 \cdot 34189.67 \cdot 1.0 + 804.87 \\
 &= 4223.8 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$\begin{aligned}
 &= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw} \\
 &= 0.1 \cdot 34189.67 \cdot 1.0 + 126.5 \\
 &= 3545.5 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$\begin{aligned}
 &= F_t / \text{Num of Saddles} \cdot B / E \\
 &= 4223.8/2 \cdot 1500.0/1582.2286 \\
 &= 2002.2 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$\begin{aligned}
 &= F_l \cdot B / L_s \\
 &= 3545 \cdot 1500.0/3750.0 \\
 &= 1418.2 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot g_y \\
 &= 17848.0 \cdot 0.05
 \end{aligned}$$



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= 892.399 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= 17848 + 892 + max(1418, 2002)

= 20742.5 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 21237.58 Kgf

Transverse Shear Load 2111.92 Kgf

Longitudinal Shear Load 3545.47 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)

= min(2(1755.0/2 + 22.0 + 14.0)sin(120.0/2), 2•888.5)

= 1582.229 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -20743•1300.0[1-(1-1300.0/6350.0+(888.5²-0.0²)/

(2•1300.0•6350.0))/(1+(4•0.0)/(3•6350.0))]

= -4231.2 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 20743•6350/4((1+2(889²-0²)/(6350²))/(1+(4•0)/

(3•6350))-4•1300/6350)

= 7253.0 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)

= 0.0 • 888.5/(2•14.0) - 7253.0/(π•888.5²•14.0)

= -2.05 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 0.0 • 888.5/(2 • 14.0) + 7253.0/(π • 888.5² • 14.0)

= 2.05 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)

= 0.0•888.5/(2•22.0)--4231.2/(0.1066•π•888.5²•22.0)

= 7.13 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 0.0•888.5/(2•22.0)+-4231.2/(0.1923•π•888.5²•22.0)

= -3.95 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 20743(6350.0 - 2 • 1300.0)/(6350.0 + (4 • 0.0/3))

= 12249.5 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 12249.53 / (888.5 \cdot 22.0)$$

$$= 7.19 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(888.5 \cdot 22.0)}$$

$$= 109.052 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(375.0 + 1.56 \cdot \sqrt{(888.5 \cdot 22.0)}, 2 \cdot 1300.0)$$

$$= 593.10 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 20743 \cdot 0.1 / (593.104(22.0 + 1.0 \cdot 14.0))$$

$$= -0.72 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -20743 / (4(22.0 + 1.0 \cdot 14.0)593.104) -$$

$$12 \cdot 0.053 \cdot 20743 \cdot 888.5 / (6350.0(22.0 + 1.0 \cdot 14.0)^2)$$

$$= -16.31 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1650.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	625.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(877.5 \cdot 22.0)} + 610.0)22.0^3) / 12$$

$$= 73.360 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 73.36 + 181.885(177.164 - 11.0)^2$$

$$= 50292.773 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	826.8	22.0	11.0	181.9	200073.5	0.503E+05
Wearplate	610.0	14.0	29.0	85.4	247660.0	0.188E+05
Web	10.0	566.5	319.2	56.6	1808550.9	0.266E+05
BasePlate	400.0	20.0	612.5	80.0	4899999.0	0.152E+06
Totals	...	...	...	403.9	7156283.5	0.247E+06

Distance to Centroid [C1]:

$$= AY / A$$



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Saddle Calcs: Road Transport: Step: 21 9:28am Sep 9,2026

$$= 7156284/403.935$$

$$= 177.164 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 20742.539$$

$$= 4221.5562 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 1.8644 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1500.0 - 877.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 774.313 \text{ mm}$$

Bending Moment, M = Fh * d = 3268.8718 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 2.2967 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(20743 + 495)400.0) / (4 \cdot 1650.0 \cdot 173.33)}$$

$$= 14.781 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (877.5 + 22.0 + 14.0)$$

$$= 1582.229 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1582.2285 / (4 - 1)$$

$$= 527.409 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 527.4095 \cdot 400.0/2$$

$$= 1054.819 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 21237.584 / (1650.0 \cdot 400.0)$$

$$= 3.218 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1054.8 \cdot 3.22$$

$$= 3394.213 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



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$$= 36.5 + 26.37$$

$$= 62.870 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 3394.2/62.8705$$

$$= 5.294 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 375.0 - 10.0$$

$$= 365.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/62.87$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375.0/2$$

$$= 187.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4394.5/62.87)}$$

$$= 83.605 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1500.0 - \cos(120.0/2) (877.5 + 22.0 + 14.0) - 20.0$$

$$= 1023.250 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1023.25/83.605$$

$$= 12.239$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 3545.5 / (2 \cdot 1650.0) \cdot 527.409 \cdot 1023.25/2$$

$$= 289.913 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($12.2391 < 123.2107$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.24)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (12.24) / (8 \cdot 123.21) - (12.24^3) / (8 \cdot 123.21^3))$$

$$= 151.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 5.29/151.84 + (289.91 \cdot 187.5/43945308) / 173.33$$



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= 0.105

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	527.4	10.0	0.0	52.7	0.0	4.40
Totals	...	...	...	89.2	...	0.440E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/89.241

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 375.0/2

= 187.500 mm

Length of Inner Rib [L]:

= Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk

= 1500.0 - $\sqrt{(913.5 + 14.0)^2 - (527.409/2)^2}$ - 20.0

= 605.390 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$

= $\sqrt{(4398.8/89.241)}$

= 70.208 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 605.39/70.208

= 8.623

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 3545.47/(2 • 1650.0)

= 1.074 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 1.074 • 527.409 • 605.39

= 343.046 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 343.046/234604.938

= 14.340 N./mm²

Compressive Allowable, KL/r < Cc (8.6228 < 123.2107) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (8.62)^2 / (2 \cdot 123.21^2)) 260 /$

$(5/3 + 3 \cdot (8.62) / (8 \cdot 123.21) - (8.62^3) / (8 \cdot 123.21^3))$

= 153.2 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 7.46/153.21 + (343.05 • 187.5/4398.838)/173.33

= 0.131

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:



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Vessel Mean Radius	Rm	888.50	mm
Shell Thickness used in this Case	t	22.000	mm
Stiffened Vessel Length per 4.15.6	L	6350.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1300.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	610.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1500.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		19160.94	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-1.89	-208.80
Long. Stress at Top of Midspan		1.89	138.00
Long. Stress at Bottom of Midspan		1.89	138.00
Long. Stress at Top of Saddles		6.59	138.00
Long. Stress at Bottom of Saddles		-3.65	-234.00
Long. Stress at Bottom of Saddles		3.65	138.00
Tangential Shear in Shell		6.65	110.40
Circ. Stress at Horn of Saddle		15.07	207.00
Circ. Compressive Stress in Shell		0.67	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf$$

$$= 208.89 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0$$

$$= 1774.4 \text{ lbf}$$

$$= 804.871 \text{ Kgf}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult}$$

$$= \pi \cdot 5.902^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \cdot 1.2$$

$$= 278.9 \text{ lbf}$$

$$= 126.5 \text{ Kgf}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$= gz \cdot \text{Shipping Weight} \cdot gw + Ftw$$

$$= 0.1 \cdot 34189.67 \cdot 1.0 + 804.87$$



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= 4223.8 Kgf

Longitudinal Force due to Axial Acceleration [FI]:

= $g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw}$

= $0.1 \cdot 34189.67 \cdot 1.0 + 126.5$

= 3545.5 Kgf

Saddle Reaction Force due to Transverse Load [Fwt]:

= $F_t / \text{Num of Saddles} \cdot B / E$

= $4223.8 / 2 \cdot 1500.0 / 1582.2286$

= 2002.2 Kgf

Saddle Reaction Force due to Longitudinal Load [Fwl]:

= $F_l \cdot B / L_s$

= $3545 \cdot 1500.0 / 3750.0$

= 1418.2 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

= Saddle Load $\cdot g_y$

= $16341.7 \cdot 0.05$

= 817.085 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= $16342 + 817 + \max(1418, 2002)$

= 19160.9 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	19655.98	Kgf
Transverse Shear Load	2111.92	Kgf
Longitudinal Shear Load	3545.47	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= $\min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$

= $\min(2(1755.0/2 + 22.0 + 14.0) \sin(120.0/2), 2 \cdot 888.5)$

= 1582.229 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$

= $-19161 \cdot 1300.0 [1 - (1 - 1300.0/6350.0 + (888.5^2 - 0.0^2) / (2 \cdot 1300.0 \cdot 6350.0)) / (1 + (4 \cdot 0.0) / (3 \cdot 6350.0))]$

= -3908.6 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$

= $19161 \cdot 6350 / 4 ((1 + 2(888.5^2 - 0^2)/(6350^2)) / (1 + (4 \cdot 0) / (3 \cdot 6350)) - 4 \cdot 1300 / 6350)$

= 6700.0 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^3 t)$

= $0.0 \cdot 888.5 / (2 \cdot 14.0) - 6700.0 / (\pi \cdot 888.5^2 \cdot 14.0)$

= -1.89 N./mm²



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Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 888.5 / (2 \cdot 14.0) + 6700.0 / (\pi \cdot 888.5^2 \cdot 14.0)$$

$$= 1.89 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 888.5 / (2 \cdot 22.0) - 3908.6 / (0.1066 \cdot \pi \cdot 888.5^2 \cdot 22.0)$$

$$= 6.59 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 888.5 / (2 \cdot 22.0) + 3908.6 / (0.1923 \cdot \pi \cdot 888.5^2 \cdot 22.0)$$

$$= -3.65 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 19161(6350.0 - 2 \cdot 1300.0) / (6350.0 + (4 \cdot 0.0 / 3))$$

$$= 11315.5 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 11315.52 / (888.5 \cdot 22.0)$$

$$= 6.65 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(888.5 \cdot 22.0)}$$

$$= 109.052 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(375.0 + 1.56 \cdot \sqrt{(888.5 \cdot 22.0)}, 2 \cdot 1300.0)$$

$$= 593.10 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot r))$$

$$= -0.7603 \cdot 19161 \cdot 0.1 / (593.104(22.0 + 1.0 \cdot 14.0))$$

$$= -0.67 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot r)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot r)^2)$$

$$= -19161 / (4(22.0 + 1.0 \cdot 14.0)593.104) -$$

$$12 \cdot 0.053 \cdot 19161 \cdot 888.5 / (6350.0(22.0 + 1.0 \cdot 14.0)^2)$$

$$= -15.07 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1650.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	625.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



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Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(877.5 \cdot 22.0)} + 610.0)^3) / 12$$

$$= 73.360 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 73.36 + 181.885 (177.164 - 11.0)^2$$

$$= 50292.773 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	826.8	22.0	11.0	181.9	200073.5	0.503E+05
Wearplate	610.0	14.0	29.0	85.4	247660.0	0.188E+05
Web	10.0	566.5	319.2	56.6	1808550.9	0.266E+05
BasePlate	400.0	20.0	612.5	80.0	4899999.0	0.152E+06
Totals	...	...	...	403.9	7156283.5	0.247E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7156284 / 403.935$$

$$= 177.164 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 19160.939$$

$$= 3899.6665 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.7223 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1500.0 - 877.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 774.313 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 3019.6233 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.1216 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(19161 + 495) \cdot 400.0) / (4 \cdot 1650.0 \cdot 173.33)}$$

$$= 14.220 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (877.5 + 22.0 + 14.0)$$

$$= 1582.229 \text{ mm}$$

Distance between Ribs [e]:



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$$\begin{aligned}
 &= \text{Web Length} / (\text{Nribs} - 1) \\
 &= 1582.2285 / (4 - 1) \\
 &= 527.409 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 527.4095 \cdot 400.0 / 2 \\
 &= 1054.819 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 19655.982 / (1650.0 \cdot 400.0) \\
 &= 2.978 \text{ Kg/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= \text{Ap} \cdot \text{Bp} \\
 &= 1054.8 \cdot 2.98 \\
 &= 3141.440 \text{ Kg}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 36.5 + 26.37 \\
 &= 62.870 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P / \text{Ar} \\
 &= 3141.4 / 62.8705 \\
 &= 4.900 \text{ N./mm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 375.0 - 10.0 \\
 &= 365.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= \text{AY} / \text{A} \\
 &= 0.0 / 62.87 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 375.0 / 2 \\
 &= 187.500 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ \text{Total Inertia} / \text{Total Area} } \\
 &= \sqrt{ 4394.5 / 62.87 } \\
 &= 83.605 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1500.0 - \cos(120.0 / 2) (877.5 + 22.0 + 14.0) - 20.0 \\
 &= 1023.250 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} } \\
 &= \sqrt{ 2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0 }
 \end{aligned}$$



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$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1023.25/83.605$$

$$= 12.239$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 3545.5 / (2 \cdot 1650.0) \cdot 527.409 \cdot 1023.25/2$$

$$= 289.913 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (12.2391 < 123.2107) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.24)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (12.24) / (8 \cdot 123.21) - (12.24^3) / (8 \cdot 123.21^3))$$

$$= 151.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 4.9/151.84 + (289.91 \cdot 187.5/43945308) / 173.33$$

$$= 0.102$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	527.4	10.0	0.0	52.7	0.0	4.40
Totals	...	...	...	89.2	...	0.440E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/89.241$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375.0/2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 1500.0 - \sqrt{ (913.5 + 14.0)^2 - (527.409/2)^2 } - 20.0$$

$$= 605.390 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 4398.8/89.241 }$$

$$= 70.208 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 605.39/70.208$$

$$= 8.623$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 3545.47 / (2 \cdot 1650.0)$$

$$= 1.074 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.074 \cdot 527.409 \cdot 605.39$$



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= 343.046 Kg-m.

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

= Bending Moment / Section Modulus

= 343.046/234604.938

= 14.340 N./mm²

Compressive Allowable, $KL/r < C_c$ (8.6228 < 123.2107) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)

= (1-(8.62)²/(2 • 123.21²))260/

(5/3+3•(8.62)/(8• 123.21)-(8.62³)/(8•123.21³)

= 153.2 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C1/I)/S_{ba}$

= 6.91/153.21 + (343.05 • 187.5/4398.838)/173.33

= 0.128

Review notes about nozzle loadings and supports in the Warnings report.

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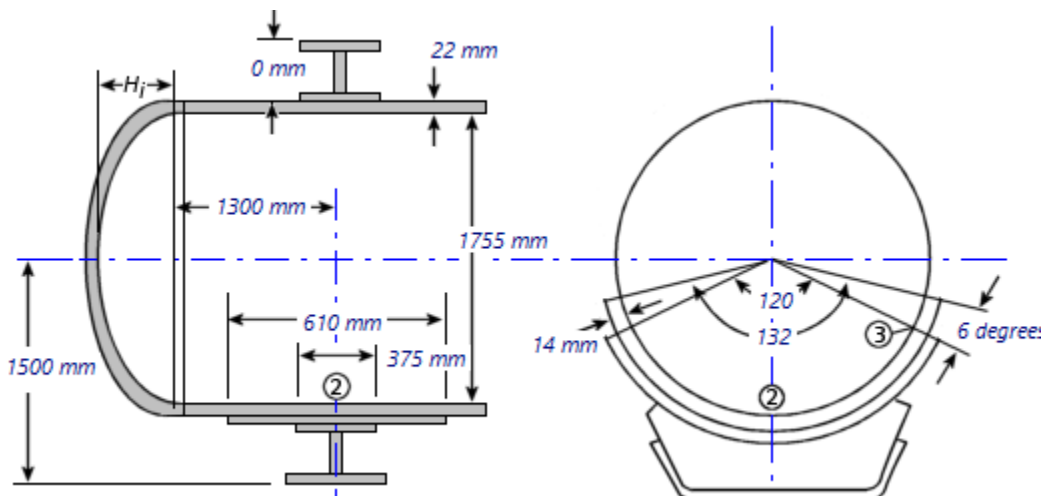
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	890.00	mm
Shell Thickness used in this Case	t	22.000	mm
Stiffened Vessel Length per 4.15.6	L	6350.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1300.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	610.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1500.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	34132.90	Kgf
Pressure used in Analysis	P	1558.601 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	58.78	138.00
Long. Stress at Bottom of Midspan	67.33	138.00
Long. Stress at Top of Saddles	50.04	138.00
Long. Stress at Bottom of Saddles	29.01	138.00
Tangential Shear in Shell	13.69	110.40
Circ. Stress at Horn of Saddle	31.71	172.50
Circ. Compressive Stress in Shell	1.33	138.00



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(788.5/2 + 0) \cdot 1500.0/1582.2286$$

$$= 1121.3 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(314.54, 0) \cdot 1500.0/3750.0$$

$$= 125.8 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(3197.49, 0) \cdot 1500.0/3750.0$$

$$= 1279.0 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(3197/2 + 0) \cdot 1500.0/1582.2286$$

$$= 4547.0 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 29586 + \max(126, 1121, 1279, 4547)$$

$$= 34132.9 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(0.0 + \pi/4(1.799 \cdot 1.2)^2)$$

$$= 3084.499 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	34627.95	Kgf
Transverse Shear Load	1598.75	Kgf
Longitudinal Shear Load	3197.49	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(1755.0/2 + 22.0 + 14.0)\sin(120.0/2), 2 \cdot 890.0)$$

$$= 1582.229 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -34133 \cdot 1300.0 [1 - (1 - 1300.0/6350.0 + (890.0^2 - 0.0^2)/(2 \cdot 1300.0 \cdot 6350.0))/(1 + (4 \cdot 0.0)/(3 \cdot 6350.0))]$$

$$= -6955.5 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 34133 \cdot 6350/4((1 + 2(890^2 - 0^2)/(6350^2))/(1 + (4 \cdot 0)/$$



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$$(3 \cdot 6350) - 4 \cdot 1300 / 6350 \\ = 11942.3 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t) \\ = 1558.601 \cdot 890.0 / (2 \cdot 11.0) - 11942.3 / (\pi \cdot 890.0^2 \cdot 11.0) \\ = 58.78 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 1558.601 \cdot 890.0 / (2 \cdot 11.0) + 11942.3 / (\pi \cdot 890.0^2 \cdot 11.0) \\ = 67.33 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t) \\ = 1558.601 \cdot 890.0 / (2 \cdot 19.0) - 6955.5 / (0.1066 \cdot \pi \cdot 890.0^2 \cdot 19.0) \\ = 50.04 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 1558.601 \cdot 890.0 / (2 \cdot 19.0) + 6955.5 / (0.1923 \cdot \pi \cdot 890.0^2 \cdot 19.0) \\ = 29.01 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 34133(6350.0 - 2 \cdot 1300.0) / (6350.0 + (4 \cdot 0.0 / 3)) \\ = 20157.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t) \\ = 1.1707 \cdot 20157.23 / (890.0 \cdot 19.0) \\ = 13.69 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{890.0 \cdot 19.0} \\ = 101.430 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(375.0 + 1.56 \cdot \sqrt{890.0 \cdot 19.0}, 2 \cdot 1300.0) \\ = 577.86 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1) \\ = \min(137.997 / 137.997, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr)) \\ = -0.7603 \cdot 34133 \cdot 0.1 / (577.86(19.0 + 1.0 \cdot 14.0)) \\ = -1.33 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2) \\ = -34133 / (4(19.0 + 1.0 \cdot 14.0)577.86) - 12 \cdot 0.053 \cdot 34133 \cdot 890.0 / (6350.0(19.0 + 1.0 \cdot 14.0)^2) \\ = -31.71 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 3750.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature}) \\ = 0.000012 \cdot 3750.0(150.0 - 21.1)$$



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= 6.006 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1650.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	232.0	N./mm ²
Height of Web at Center	Hw,c	625.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(880.5 \cdot 19.0)} + 610.0)^3) / 12$$

$$= 46.400 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 46.4 + 154.237 (186.988 - 9.5)^2$$

$$= 48634.152 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	811.8	19.0	9.5	154.2	146525.3	0.486E+05
Wearplate	610.0	14.0	26.0	85.4	222040.0	0.221E+05
Web	10.0	566.5	316.2	56.6	1791555.8	0.246E+05
BasePlate	400.0	20.0	609.5	80.0	4875999.5	0.143E+06
Totals	...	...	...	376.3	7036120.0	0.238E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7036120 / 376.287$$

$$= 186.988 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 34132.902$$

$$= 6946.7861 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 3.0680 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 139.1985 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1500.0 - 880.5 \cdot \sin(120.0 / 2) / 1.0472$$

$$= 771.832 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 5361.8599 \text{ Kg-m.}$$



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Bending Stress, $S_b = (M * C_1 / I) = 4.1271 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 * \text{Yield Str} = 154.6650 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{(3(34133 + 495)400.0 / (4 * 1650.0 * 154.665))}$
 $= 19.980 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120.0/2) (877.5 + 22.0 + 14.0)$
 $= 1582.229 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nribs} - 1)$
 $= 1582.2285 / (4 - 1)$
 $= 527.409 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 527.4095 * 400.0 / 2$
 $= 1054.819 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 34627.945 / (1650.0 * 400.0)$
 $= 5.247 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p * B_p$
 $= 1054.8 * 5.25$
 $= 5534.275 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 36.5 + 26.37$
 $= 62.870 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 5534.3 / 62.8705$
 $= 8.633 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 375.0 - 10.0$
 $= 365.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0 / 62.87$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:



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= Saddle Width / 2
= 375.0/2
= 187.500 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{4394.5/62.87}$
= 83.605 mm

Length of Outer Rib [L]:
= Saddle Height - $\cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
= 1500.0 - $\cos(120.0/2) (877.5 + 22.0 + 14.0) - 20.0$
= 1023.250 mm

Intermediate Term [Cc]:
= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/232.0}$
= 130.434

Slenderness ratio [KL/r]:
= KL/r
= 1 • 1023.25/83.605
= 12.239

Bending Moment [Rm]:
= $FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$
= 3197.5/(2 • 1650.0) • 527.409 • 1023.25/2
= 261.459 Kg-m.

Compressive Allowable, $KL/r < Cc$ (12.2391 < 130.4336) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2/(2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r)/(8 \cdot Cc) - (KL/r)^3/(8 \cdot Cc^3))$
= $(1 - (12.24)^2/(2 \cdot 130.43^2)) 232 / (5/3 + 3 \cdot (12.24)/(8 \cdot 130.43) - (12.24)^3/(8 \cdot 130.43^3))$
= 135.7 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
= $Sc/Sca + (Rm \cdot C1 / I) / Sba$
= 8.63/135.73 + (261.46 • 187.5/43945308)/154.66
= 0.134

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	527.4	10.0	0.0	52.7	0.0	4.40
Totals	...	...	...	89.2	...	0.440E+04

Distance to Centroid from Datum [ytot]:
= AY / A
= 0.0/89.241
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 375.0/2
= 187.500 mm

Length of Inner Rib [L]:
= Saddle Height - $\sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$
= 1500.0 - $\sqrt{(913.5 + 14.0)^2 - (527.409/2)^2} - 20.0$
= 605.390 mm

Radius of Gyration [r]:



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$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{4398.8 / 89.241}$$

$$= 70.208 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 605.39 / 70.208$$

$$= 8.623$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 3197.493 / (2 \cdot 1650.0)$$

$$= 0.969 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.969 \cdot 527.409 \cdot 605.39$$

$$= 309.377 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 309.377 / 234604.938$$

$$= 12.932 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($8.6228 < 130.4336$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.62)^2 / (2 \cdot 130.43^2)) 232 / (5/3 + 3 \cdot (8.62) / (8 \cdot 130.43) - (8.62^3) / (8 \cdot 130.43^3))$$

$$= 136.9 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 12.17 / 136.86 + (309.38 \cdot 187.5 / 4398.838) / 154.66$$

$$= 0.173$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0428 cm ²
Saddle Load Q0 (Weight)	Q0	30081.0 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1279.0 Kg
Maximum Transverse Force	Ft	1598.7 Kg
Maximum Longitudinal Force	F1	3197.5 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3197.5 / (3.043 \cdot 6)$$

$$= 17.2 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1598.7 / (3.043 \cdot 6)$$

$$= 8.6 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (BoltShearAllowable \cdot Nbolts)$
 = $3197.49 / (82.8 \cdot 6.0)$
 = 0.6312 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + \text{Sum of X Moments}$
 = $1500.0 \cdot 1598.75 + 0.0$
 = 2398.17 Kg-m.

Eccentricity (e):
 = $Rmom / QO$
 = $2398.17 / 30080.97$
 = 79.72 mm < $B_{plen} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0.0, 0.6312, 0.0]$
 = 0.6312 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 14.0$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.534$, Temp. Reduction = 28 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	890.00	mm
Shell Thickness used in this Case	t	22.000	mm
Stiffened Vessel Length per 4.15.6	L	6350.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1300.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	610.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1500.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:



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Saddle Force Q, Operating Case 31045.50 Kg
 Pressure used in Analysis P 1558.601 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	59.16	138.00
Long. Stress at Bottom of Midspan	66.95	138.00
Long. Stress at Top of Saddles	48.81	138.00
Long. Stress at Bottom of Saddles	29.68	138.00
Tangential Shear in Shell	12.45	110.40
Circ. Stress at Horn of Saddle	28.84	172.50
Circ. Compressive Stress in Shell	1.21	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (788.5/2 + 0) \cdot 1500.0/1582.2286$
 $= 1121.3 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(314.54, 0) \cdot 1500.0/3750.0$
 $= 125.8 \text{ Kg}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(3197.49, 0) \cdot 1500.0/3750.0$
 $= 1279.0 \text{ Kg}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (3197/2 + 0) \cdot 1500.0/1582.2286$
 $= 4547.0 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 26499 + \max(126, 1121, 1279, 4547)$
 $= 31045.5 \text{ Kg}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766.08 (0.0 + \pi/4 (1.799 \cdot 1.2)^2)$
 $= 3084.499 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	31540.54	Kg
Transverse Shear Load	1598.75	Kg
Longitudinal Shear Load	3197.49	Kg

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2 (1755.0/2 + 22.0 + 14.0) \sin(120.0/2), 2 \cdot 890.0)$
 $= 1582.229 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434



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$$K7p = 0.0434$$

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned} &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)] \\ &= -31045 \cdot 1300.0 [1 - (1 - 1300.0/6350.0 + (890.0^2 - 0.0^2)/(2 \cdot 1300.0 \cdot 6350.0))/(1 + (4 \cdot 0.0)/(3 \cdot 6350.0))] \\ &= -6326.3 \text{ Kg-m.} \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned} &= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L] \\ &= 31045 \cdot 6350/4 [(1 + 2(890^2 - 0^2)/(6350^2))/(1 + (4 \cdot 0)/(3 \cdot 6350)) - 4 \cdot 1300/6350] \\ &= 10862.1 \text{ Kg-m.} \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned} &= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t) \\ &= 1558.601 \cdot 890.0/(2 \cdot 11.0) - 10862.1/(\pi \cdot 890.0^2 \cdot 11.0) \\ &= 59.16 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned} &= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t) \\ &= 1558.601 \cdot 890.0/(2 \cdot 11.0) + 10862.1/(\pi \cdot 890.0^2 \cdot 11.0) \\ &= 66.95 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned} &= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t) \\ &= 1558.601 \cdot 890.0/(2 \cdot 19.0) - 6326.3/(0.1066 \cdot \pi \cdot 890.0^2 \cdot 19.0) \\ &= 48.81 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned} &= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t) \\ &= 1558.601 \cdot 890.0/(2 \cdot 19.0) + 6326.3/(0.1923 \cdot \pi \cdot 890.0^2 \cdot 19.0) \\ &= 29.68 \text{ N./mm}^2 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned} &= Q(L - 2a)/(L + (4 \cdot hm/3)) \\ &= 31045(6350.0 - 2 \cdot 1300.0)/(6350.0 + (4 \cdot 0.0/3)) \\ &= 18334.0 \text{ Kg} \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned} &= K2 \cdot T / (\pi \cdot Rm \cdot t) \\ &= 1.1707 \cdot 18333.96/(\pi \cdot 890.0 \cdot 19.0) \\ &= 12.45 \text{ N./mm}^2 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned} &= 0.78 \cdot \sqrt{(\pi \cdot Rm \cdot t)} \\ &= 0.78 \cdot \sqrt{(\pi \cdot 890.0 \cdot 19.0)} \\ &= 101.430 \text{ mm} \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned} &= \min(b + 1.56 \cdot \sqrt{(\pi \cdot Rm \cdot t)}, 2a) \\ &= \min(375.0 + 1.56 \cdot \sqrt{(\pi \cdot 890.0 \cdot 19.0)}, 2 \cdot 1300.0) \\ &= 577.86 \text{ mm} \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$\begin{aligned} &= \min(Sr/S, 1) \\ &= \min(137.997/137.997, 1) \\ &= 1.0000 \end{aligned}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$



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$$= -0.7603 \cdot 31045 \cdot 0.1 / (577.86(19.0 + 1.0 \cdot 14.0))$$

$$= -1.21 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -31045 / (4(19.0 + 1.0 \cdot 14.0)577.86) -$$

$$12 \cdot 0.053 \cdot 31045 \cdot 890.0 / (6350.0(19.0 + 1.0 \cdot 14.0)^2)$$

$$= -28.84 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1650.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	232.0	N./mm ²
Height of Web at Center	Hw, c	625.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(880.5 \cdot 19.0) + 610.0})^3) / 12$$

$$= 46.400 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 46.4 + 154.237(186.988 - 9.5)^2$$

$$= 48634.152 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	811.8	19.0	9.5	154.2	146525.3	0.486E+05
Wearplate	610.0	14.0	26.0	85.4	222040.0	0.221E+05
Web	10.0	566.5	316.2	56.6	1791555.8	0.246E+05
BasePlate	400.0	20.0	609.5	80.0	4875999.5	0.143E+06
Totals	...	...	...	376.3	7036120.0	0.238E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7036120 / 376.287$$

$$= 186.988 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$

$$= 0.204 \cdot 31045.496$$

$$= 6318.4316 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.7905 \text{ N./mm}^2$$



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Allowed Stress, $S_a = 0.6 \cdot \text{Yield Str} = 139.1985 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1500.0 - 880.5 \cdot \sin(120.0/2) / 1.0472$
 $= 771.832 \text{ mm}$

Bending Moment, $M = F_h \cdot d = 4876.8662 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 3.7538 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 154.6650 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(31045 + 495) \cdot 400.0) / (4 \cdot 1650.0 \cdot 154.665)}$
 $= 19.069 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120.0/2) (877.5 + 22.0 + 14.0)$
 $= 1582.229 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1582.2285 / (4 - 1)$
 $= 527.409 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 527.4095 \cdot 400.0 / 2$
 $= 1054.819 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 31540.541 / (1650.0 \cdot 400.0)$
 $= 4.779 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 1054.8 \cdot 4.78$
 $= 5040.842 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 36.5 + 26.37$
 $= 62.870 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 5040.8 / 62.8705$
 $= 7.863 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 375.0 - 10.0$



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= 365.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/62.87

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 375.0/2

= 187.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{4394.5/62.87}$

= 83.605 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 1500.0 - cos(120.0/2)(877.5 + 22.0 + 14.0) - 20.0

= 1023.250 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/232.0}$

= 130.434

Slenderness ratio [KL/r]:

= KL/r

= 1 • 1023.25/83.605

= 12.239

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 3197.5/(2 • 1650.0) • 527.409 • 1023.25/2

= 261.459 Kg-m.

Compressive Allowable, KL/r < Cc (12.2391 < 130.4336) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (12.24)^2 / (2 \cdot 130.43^2)) 232 /$

$(5/3 + 3 \cdot (12.24) / (8 \cdot 130.43) - (12.24)^3 / (8 \cdot 130.43^3))$

= 135.7 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba

= 7.86/135.73 + (261.46 • 187.5/43945308)/154.66

= 0.129

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	527.4	10.0	0.0	52.7	0.0	4.40
Totals	...	...	...	89.2	...	0.440E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/89.241

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 375.0/2



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= 187.500 mm

Length of Inner Rib [L]:

= Saddle Height - $\sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2}$ - Bpthk

= 1500.0 - $\sqrt{(913.5 + 14.0)^2 - (527.409/2)^2}$ - 20.0

= 605.390 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$

= $\sqrt{(4398.8/89.241)}$

= 70.208 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 605.39/70.208

= 8.623

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 3197.493/(2 • 1650.0)

= 0.969 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.969 • 527.409 • 605.39

= 309.377 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 309.377/234604.938

= 12.932 N./mm²

Compressive Allowable, KL/r < Cc (8.6228 < 130.4336) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r³)/(8•Cc³))

= (1-(8.62)²/(2 • 130.43²))232/

(5/3+3•(8.62)/(8• 130.43)-(8.62³)/(8•130.43³))

= 136.9 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 11.08/136.86 + (309.38 • 187.5/4398.838)/154.66

= 0.165

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0428 cm ²
Saddle Load Q0 (Weight)	Q0	26993.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1279.0 Kg
Maximum Transverse Force	Ft	1598.7 Kg
Maximum Longitudinal Force	F1	3197.5 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

= 9024.3 Kg

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:



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= max(Ff; FI) / (Bolt Area • Number of Bolts)
 = 9024.3 / (3.043 • 6)
 = 48.5 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 = Ft / (Bolt Area • Number of Bolts)
 = 1598.7 / (3.043 • 6)
 = 8.6 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = FI / (BoltShearAllowable • Nbolts)
 = 3197.49 / (82.8 • 6.0)
 = 0.6312 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = B • Ft + Sum of X Moments
 = 1500.0 • 1598.75 + 0.0
 = 2398.17 Kg-m.

Eccentricity (e):
 = Rmom / QO
 = 2398.17 / 26993.56
 = 88.84 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 0.6312, 0.0]
 = 0.6312 cm²

MDMT Calculations for: RH fixed saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 14.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.534, Temp. Reduction = 28 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

Step: 23

9:28am

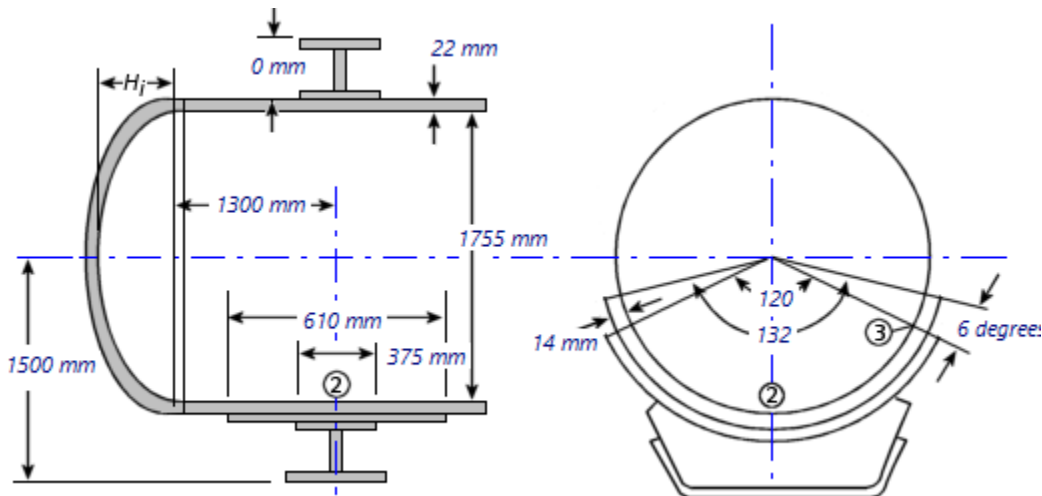
Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	888.50	mm
Shell Thickness used in this Case	t	22.000	mm
Stiffened Vessel Length per 4.15.6	L	6350.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1300.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	610.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1500.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		28628.45	Kgf
Pressure used in Analysis	P	2023.709	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	61.39	234.00
Long. Stress at Bottom of Midspan	67.05	234.00
Long. Stress at Top of Saddles	50.71	234.00
Long. Stress at Bottom of Saddles	35.41	234.00
Tangential Shear in Shell	9.93	138.58
Circ. Stress at Horn of Saddle	22.51	350.99
Circ. Compressive Stress in Shell	1.00	234.00



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(394.2/2 + 0) \cdot 1500.0/1582.2286$$

$$= 560.6 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(157.27, 0) \cdot 1500.0/3750.0$$

$$= 62.9 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 28068 + \max(63, 561, 0, 0)$$

$$= 28628.5 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(0.0 + \pi/4(1.799 \cdot 1.2)^2)$$

$$= 3084.499 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 29123.50 Kgf

Transverse Shear Load 197.12 Kgf

Longitudinal Shear Load 157.27 Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1755.0/2 + 22.0 + 14.0)\sin(120.0/2), 2 \cdot 888.5)$$

$$= 1582.229 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -28628 \cdot 1300.0 [1 - (1 - 1300.0/6350.0 + (888.5^2 - 0.0^2)/(2 \cdot 1300.0 \cdot 6350.0))/(1 + (4 \cdot 0.0)/(3 \cdot 6350.0))]$$

$$= -5839.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 28628 \cdot 6350/4((1 + 2(889^2 - 0^2)/(6350^2))/(1 + (4 \cdot 0)/(3 \cdot 6350)) - 4 \cdot 1300/6350)$$

$$= 10010.4 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 2023.709 \cdot 888.5/(2 \cdot 14.0) - 10010.4/(\pi \cdot 888.5^2 \cdot 14.0)$$

$$= 61.39 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$



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$$= 2023.709 \cdot 888.5 / (2 \cdot 14.0) + 10010.4 / (\pi \cdot 888.5^2 \cdot 14.0)$$

$$= 67.05 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 2023.709 \cdot 888.5 / (2 \cdot 22.0) - 5839.8 / (0.1066 \cdot \pi \cdot 888.5^2 \cdot 22.0)$$

$$= 50.71 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 2023.709 \cdot 888.5 / (2 \cdot 22.0) + 5839.8 / (0.1923 \cdot \pi \cdot 888.5^2 \cdot 22.0)$$

$$= 35.41 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 28628(6350.0 - 2 \cdot 1300.0) / (6350.0 + (4 \cdot 0.0 / 3))$$

$$= 16906.6 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [$\tau2$]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 16906.57 / (888.5 \cdot 22.0)$$

$$= 9.93 \text{ N./mm}^2$$

Decay Length (4.15.20) [$x1, x2$]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(888.5 \cdot 22.0)}$$

$$= 109.052 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(375.0 + 1.56 \cdot \sqrt{(888.5 \cdot 22.0)}, 2 \cdot 1300.0)$$

$$= 593.10 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma6, r$]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 28628 \cdot 0.1 / (593.104(22.0 + 1.0 \cdot 14.0))$$

$$= -1.00 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [$\sigma7, r^*$]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -28628 / (4(22.0 + 1.0 \cdot 14.0)593.104) -$$

$$12 \cdot 0.053 \cdot 28628 \cdot 888.5 / (6350.0(22.0 + 1.0 \cdot 14.0)^2)$$

$$= -22.51 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1650.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	625.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(877.5 \cdot 22.0)} + 610.0)^3) / 12$$

$$= 73.360 \text{ cm}^4$$



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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 73.36 + 181.885(177.164 - 11.0)^2$$

$$= 50292.773 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	826.8	22.0	11.0	181.9	200073.5	0.503E+05
Wearplate	610.0	14.0	29.0	85.4	247660.0	0.188E+05
Web	10.0	566.5	319.2	56.6	1808550.9	0.266E+05
BasePlate	400.0	20.0	612.5	80.0	4899999.0	0.152E+06
Totals	...	...	...	403.9	7156283.5	0.247E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7156284 / 403.935$$

$$= 177.164 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 28628.453$$

$$= 5826.5107 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.5733 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1500.0 - 877.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 774.313 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 4511.6343 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 3.1698 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(28628 + 495)400.0 / (4 \cdot 1650.0 \cdot 173.33))}$$

$$= 17.309 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (877.5 + 22.0 + 14.0)$$

$$= 1582.229 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1582.2285 / (4 - 1)$$

$$= 527.409 \text{ mm}$$



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 527.4095 \cdot 400.0 / 2$$

$$= 1054.819 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 29123.498 / (1650.0 \cdot 400.0)$$

$$= 4.413 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1054.8 \cdot 4.41$$

$$= 4654.548 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 36.5 + 26.37$$

$$= 62.870 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 4654.5 / 62.8705$$

$$= 7.260 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 375.0 - 10.0$$

$$= 365.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 62.87$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 375.0 / 2$$

$$= 187.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (4394.5 / 62.87) }$$

$$= 83.605 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1500.0 - \cos(120.0 / 2) (877.5 + 22.0 + 14.0) - 20.0$$

$$= 1023.250 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 0.19994E+09 / 260.0) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 1023.25 / 83.605$$

$$= 12.239$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 157.3 / (2 \cdot 1650.0) \cdot 527.409 \cdot 1023.25 / 2$$

$$= 12.860 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($12.2391 < 123.2107$) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.24)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (12.24) / (8 \cdot 123.21) - (12.24^3) / (8 \cdot 123.21^3))$$

$$= 151.8 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 7.26 / 151.84 + (12.86 \cdot 187.5 / 43945308) / 173.33$$

$$= 0.051$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	527.4	10.0	0.0	52.7	0.0	4.40
Totals	...	...	...	89.2	...	0.440E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 89.241$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375.0 / 2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(R_o + W_{pdthk})^2 - (\text{Pitch}/2)^2} - B_{pthk}$$

$$= 1500.0 - \sqrt{(913.5 + 14.0)^2 - (527.409/2)^2} - 20.0$$

$$= 605.390 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4398.8 / 89.241)}$$

$$= 70.208 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 605.39 / 70.208$$

$$= 8.623$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 157.268 / (2 \cdot 1650.0)$$

$$= 0.048 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.048 \cdot 527.409 \cdot 605.39$$

$$= 15.217 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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= 15.217/234604.938
 = 0.636 N./mm²

Compressive Allowable, $KL/r < C_c$ (8.6228 < 123.2107) per AISC E2-1 [Sca]:
 = $(1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$
 = $(1 - (8.62)^2 / (2 \cdot 123.21^2)) 260 /$
 $(5/3 + 3 \cdot (8.62) / (8 \cdot 123.21) - (8.62^3) / (8 \cdot 123.21^3))$
 = 153.2 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
 = 10.23/153.21 + (15.22 • 187.5/4398.838)/173.33
 = 0.070

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0428 cm ²
Saddle Load QO (Weight)	QO	28562.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	62.9 Kg
Maximum Transverse Force	Ft	197.1 Kg
Maximum Longitudinal Force	F1	157.3 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 = $F1 / (Bolt\ Area \cdot Number\ of\ Bolts)$
 = 157.3 / (3.043 • 6)
 = 0.8 N./mm². Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = $Ft / (Bolt\ Area \cdot Number\ of\ Bolts)$
 = 197.1 / (3.043 • 6)
 = 1.1 N./mm². Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F1 / (BoltShearAllowable \cdot Nbolts)$
 = 157.27 / (103.2 • 6.0)
 = 0.0249 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + Sum\ of\ X\ Moments$
 = 1500.0 • 197.12 + 0.0
 = 295.69 Kg-m.

Eccentricity (e):
 = $Rmom / QO$
 = 295.69/28562.86
 = 10.35 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)



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Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 0.0249, 0.0]
 = 0.0249 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	888.50	mm
Shell Thickness used in this Case	t	22.000	mm
Stiffened Vessel Length per 4.15.6	L	6350.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1300.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	610.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1500.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		25541.05	Kgf
Pressure used in Analysis	P	2023.709	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	61.70	234.00
Long. Stress at Bottom of Midspan	66.74	234.00
Long. Stress at Top of Saddles	49.65	234.00
Long. Stress at Bottom of Saddles	36.00	234.00
Tangential Shear in Shell	8.86	138.58
Circ. Stress at Horn of Saddle	20.08	350.99
Circ. Compressive Stress in Shell	0.89	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 = Ftr(Ft/Num of Saddles + Z Force Load) • B / E
 = 3.0(394.2/2 + 0) • 1500.0/1582.2286
 = 560.6 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
 = max(Fl,wind; Sum of X Forces) • B / Ls
 = max(157.27, 0) • 1500.0/3750.0
 = 62.9 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 24980 + max(63, 561, 0, 0)



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= 25541.0 Kgf

Longitudinal Wind Force [FI]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))

= 1.1 • 766.08(0.0 + $\pi/4$ (1.799 • 1.2)²)

= 3084.499 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 26036.09 Kgf

Transverse Shear Load 197.12 Kgf

Longitudinal Shear Load 157.27 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)

= min(2(1755.0/2 + 22.0 + 14.0)sin(120.0/2), 2•888.5)

= 1582.229 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -25541•1300.0[1-(1-1300.0/6350.0+(888.5²-0.0²)/

(2•1300.0•6350.0))/(1+(4•0.0)/(3•6350.0))]

= -5210.0 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 25541•6350/4((1+2(889²-0²)/(6350²))/(1+(4•0)/

(3•6350))-4•1300/6350)

= 8930.9 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)

= 2023.709 • 888.5/(2•14.0) - 8930.9/(π •888.5²•14.0)

= 61.70 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 2023.709 • 888.5/(2 • 14.0) + 8930.9/(π • 888.5² • 14.0)

= 66.74 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)

= 2023.709•888.5/(2•22.0)-5210.0/(0.1066• π •888.5²•22.0)

= 49.65 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 2023.709•888.5/(2•22.0)+5210.0/(0.1923• π •888.5²•22.0)

= 36.00 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 25541(6350.0 - 2 • 1300.0)/(6350.0 + (4 • 0.0/3))

= 15083.3 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 15083.3 / (888.5 \cdot 22.0)$$

$$= 8.86 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(888.5 \cdot 22.0)}$$

$$= 109.052 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(375.0 + 1.56 \cdot \sqrt{(888.5 \cdot 22.0)}, 2 \cdot 1300.0)$$

$$= 593.10 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 25541 \cdot 0.1 / (593.104(22.0 + 1.0 \cdot 14.0))$$

$$= -0.89 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -25541 / (4(22.0 + 1.0 \cdot 14.0)593.104) -$$

$$12 \cdot 0.053 \cdot 25541 \cdot 888.5 / (6350.0(22.0 + 1.0 \cdot 14.0)^2)$$

$$= -20.08 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1650.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	625.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(877.5 \cdot 22.0)} + 610.0)22.0^3) / 12$$

$$= 73.360 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 73.36 + 181.885(177.164 - 11.0)^2$$

$$= 50292.773 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	826.8	22.0	11.0	181.9	200073.5	0.503E+05
Wearplate	610.0	14.0	29.0	85.4	247660.0	0.188E+05
Web	10.0	566.5	319.2	56.6	1808550.9	0.266E+05
BasePlate	400.0	20.0	612.5	80.0	4899999.0	0.152E+06
Totals	...	...	...	403.9	7156283.5	0.247E+06

Distance to Centroid [C1]:



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$$\begin{aligned}
 &= AY / A \\
 &= 7156284 / 403.935 \\
 &= 177.164 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Angle } [\beta]: \\
 &= 180 - \text{Saddle Angle} / 2 \\
 &= 180 - 120.0 / 2 \\
 &= 120.0
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Coefficient [K1]:} \\
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\
 &= 0.2035
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Force [Fh]:} \\
 &= K1 \cdot Q \\
 &= 0.204 \cdot 25541.047 \\
 &= 5198.1567 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 2.2958 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Dimension [d]:} \\
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1500.0 - 877.5 \cdot \sin(120.0/2) / 1.0472 \\
 &= 774.313 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 4025.0815 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 2.8280 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{(3(25541 + 495)400.0) / (4 \cdot 1650.0 \cdot 173.33)} \\
 &= 16.366 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned}
 \text{Web Length Dimension [Web Length]:} \\
 &= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120.0/2) (877.5 + 22.0 + 14.0) \\
 &= 1582.229 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance between Ribs [e]:} \\
 &= \text{Web Length} / (N_{\text{ribs}} - 1) \\
 &= 1582.2285 / (4 - 1) \\
 &= 527.409 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Baseplate Pressure Area [Ap]:} \\
 &= e \cdot B_{\text{pwid}} / 2 \\
 &= 527.4095 \cdot 400.0 / 2 \\
 &= 1054.819 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Bearing Pressure [Bp]:} \\
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 26036.092 / (1650.0 \cdot 400.0) \\
 &= 3.945 \text{ Kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Axial Load [P]:} \\
 &= Ap \cdot Bp \\
 &= 1054.8 \cdot 3.94 \\
 &= 4161.116 \text{ Kgf}
 \end{aligned}$$

Area of the Rib and Web [Ar]:



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= Rib Area + Web Area
= 36.5 + 26.37
= 62.870 cm²

Compressive Stress [Sc]:
= P/Ar
= 4161.1/62.8705
= 6.491 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:
= Saddle Width - Web Thickness
= 375.0 - 10.0
= 365.000 mm

Distance to Centroid from Datum [ytot]:
= AY / A
= 0.0/62.87
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 375.0/2
= 187.500 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{4394.5/62.87}$
= 83.605 mm

Length of Outer Rib [L]:
= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 1500.0 - cos(120.0/2)(877.5 + 22.0 + 14.0) - 20.0
= 1023.250 mm

Intermediate Term [Cc]:
= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
= $\sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)}$
= 123.211

Slenderness ratio [KL/r]:
= KL/r
= 1 • 1023.25/83.605
= 12.239

Bending Moment [Rm]:
= FI / (2 • Bplen) • e • L / 2
= 157.3/ (2 • 1650.0) • 527.409 • 1023.25/2
= 12.860 Kg-m.

Compressive Allowable, KL/r < Cc (12.2391 < 123.2107) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (12.24)^2 / (2 \cdot 123.21^2)) 260 / (5/3 + 3 \cdot (12.24) / (8 \cdot 123.21) - (12.24^3) / (8 \cdot 123.21^3))$
= 151.8 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
= Sc/Sca + (Rm • C1 / I) / Sba



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$$= 6.49/151.84 + (12.86 \cdot 187.5/43945308)/173.33$$

$$= 0.046$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	527.4	10.0	0.0	52.7	0.0	4.40
Totals	...	...	...	89.2	...	0.440E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/89.241$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375.0/2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1500.0 - \sqrt{(913.5 + 14.0)^2 - (527.409/2)^2} - 20.0$$

$$= 605.390 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{4398.8/89.241}$$

$$= 70.208 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 605.39/70.208$$

$$= 8.623$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 157.268/(2 \cdot 1650.0)$$

$$= 0.048 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.048 \cdot 527.409 \cdot 605.39$$

$$= 15.217 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 15.217/234604.938$$

$$= 0.0636 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (8.6228 < 123.2107) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2/(2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r)/(8 \cdot C_c) - (KL/r)^3/(8 \cdot C_c^3))$$

$$= (1 - (8.62)^2/(2 \cdot 123.21^2)) 260 / (5/3 + 3 \cdot (8.62)/(8 \cdot 123.21) - (8.62^3)/(8 \cdot 123.21^3))$$

$$= 153.2 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 9.15/153.21 + (15.22 \cdot 187.5/4398.838)/173.33$$

$$= 0.063$$

Input Data for Base Plate Bolting Calculations:



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Saddle Calcs: Test Case:

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Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0428 cm ²
Saddle Load Q0 (Weight)	Q0	25475.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	62.9 Kg
Maximum Transverse Force	Ft	197.1 Kg
Maximum Longitudinal Force	F1	157.3 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

= $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $157.3 / (3.043 \cdot 6)$

= 0.8 N./mm². Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

= $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $197.1 / (3.043 \cdot 6)$

= 1.1 N./mm². Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$

= $157.27 / (103.2 \cdot 6.0)$

= 0.0249 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + \text{Sum of X Moments}$

= $1500.0 \cdot 197.12 + 0.0$

= 295.69 Kg-m.

Eccentricity (e):

= $Rmom / Q0$

= $295.69 / 25475.45$

= 11.61 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$

= $\max[0.0, 0.0249, 0.0]$

= 0.0249 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: N1-Tube Inlet

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Input, Nozzle Desc: N1-Tube Inlet

From: 30

Pressure for Reinforcement Calculations	P	1000.000	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1755.00	mm
Design Length of Section	L	799.9689	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		731.18	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		200.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	379.0750	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		80.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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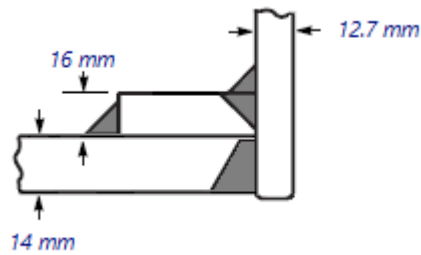
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Nozzle Calcs.: N1-Tube Inlet

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Reinforcement CALCULATION, Description: N1-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	193.675 mm.
Actual Thickness Used in Calculation	12.700 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1000.0 \cdot 880.5 / (138 \cdot 1.0 - 0.6 \cdot 1000.0)$
 $= 6.4088 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1000.0 \cdot 99.84 / (118 \cdot 1.0 - 0.6 \cdot 1000.0)$
 $= 0.8505 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8003 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	399.3500 mm
Parallel to Vessel Wall, opening length	d	199.6750 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



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Nozzle Calcs.: N1-Tube Inlet

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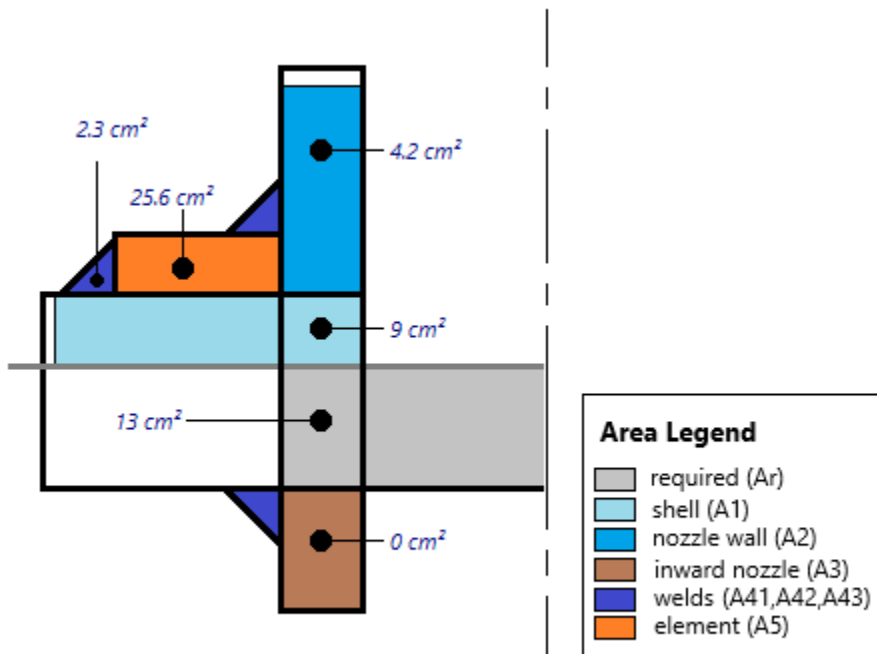


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	12.977	4.147	NA
Area in Shell	A1	9.038	13.591	NA
Area in Nozzle Wall	A2	4.162	4.185	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.260	2.260	NA
Area in Element	A5	25.600	25.600	NA
TOTAL AREA AVAILABLE	Atot	41.061	45.637	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (199.675 \cdot 6.4088 \cdot 1.0 + 2 \cdot 9.7 \cdot 6.4088 \cdot 1.0 \cdot (1 - 0.86)) \\
 &= 12.977 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 199.675(1.0 \cdot 11.0 - 1.0 \cdot 6.409) - 2 \cdot 9.7 \\
 &\quad (1.0 \cdot 11.0 - 1.0 \cdot 6.4088) \cdot (1 - 0.855) \\
 &= 9.038 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 27.5) \cdot (9.7 - 0.85) \cdot 0.8551
 \end{aligned}$$



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Nozzle Calcs.: N1-Tube Inlet

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$$= 4.162 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - A_r \text{ Lost}) \cdot Fr_3 + ((W_i - c_{an}/0.707)^2 - A_r \text{ Lost}) \cdot Fr_2 + \text{Trap} Fr_4$$

$$= (1.) \cdot 0.86 + (0.0) \cdot 0.86 + 140.5311^2 \cdot 1.0$$

$$= 2.260 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot Fr_4$$

$$= (379.075 - 219.075) \cdot 16.0 \cdot 1.0$$

$$= 25.600 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.8505 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 5.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb_1 = 9.4088 \text{ mm}$

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 9.4088 \text{ mm}$

Wall Thickness, shell/head, external pressure $trb_2 = 3.6602 \text{ mm}$

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb_3 = 10.1600 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[10.16, \max(9.4088, 5.5)]$$

$$= 9.4088 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.8505, 9.4088)$$

$$= 9.4088 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 12.700 = 11.113 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 27.3,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 267.7 N./mm ²	Passed
Occasional	: 4.9,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 17.8,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 11.113$, $tr = 0.85$, $c = 3.0 \text{ mm}$, $E* = 1.0$

Thickness Ratio = $tr \cdot E*/(t_g - c) = 0.105$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A -5°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C

Min Metal Temp. w/o impact per UG-20(f) -29°C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 11.113$, $tr = 0.85$, $c = 3.0 \text{ mm}$, $E* = 1.0$

Thickness Ratio = $tr \cdot E*/(t_g - c) = 0.105$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve B -26°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C

Min Metal Temp. w/o impact per UG-20(f) -29°C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Nozzle Calcs.: N1-Tube Inlet

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Govrn. thk, $t_g = 11.113$, $t_r = 0.85$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.105$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 14.0$, $t_r = 6.409$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.583$, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 11.113$, $t_r = 0.85$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.105$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-104 °C
Gov. MDMT of the Reinforcement Pad	:	-42 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-42 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1000.00 / 1960.01 = 0.510$

Weld Size Calculations, Description: N1-Tube Inlet

Intermediate Calc. for nozzle/shell Welds	T_{min}	9.7000 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$5.5000 = 0.5 * T_{minPad}$	$8.4840 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (12.9769 - 9.0384 + 2 \cdot 9.7 \cdot 0.8551 \cdot (1.0 \cdot 11.0 - 6.4088)) \cdot 138)$
 $= 6613.82$ Kg

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_r2) \cdot S_v$
 $= (4.1618 + 25.6 + 2.2604 - 0.0 \cdot 0.86) \cdot 138$
 $= 45060.30$ Kg

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$



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$$= (4.1618 + 0.0 + 0.8551 + (1.8247)) \cdot 138$$

$$= 9627.26 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$

$$= (4.1618 + 0.0 + 2.2604 + 25.6 + (1.8247)) \cdot 138$$

$$= 47627.98 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 219.075 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 20289. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 379.075 \cdot 12.0 \cdot 0.49 \cdot 138$$

$$= 49268. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 104.6875) \cdot (12.7 - 3.0) \cdot 0.7 \cdot 118$$

$$= 26870. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 219.075 \cdot 16.0 \cdot 0.74 \cdot 138$$

$$= 57333. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 219.075 \cdot (14.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 39417. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (49268 + 26870) = 76138 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (20289 + 57333 + 39417 + 0) = 117039 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (49268 + 39417 + 0) = 88685 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 76137 Kgf, must exceed W = 6613 Kgf	or W1 = 45060 Kgf
Path 2-2 = 117038 Kgf, must exceed W = 6613 Kgf	or W2 = 9627 Kgf
Path 3-3 = 88684 Kgf, must exceed W = 6613 Kgf	or W3 = 47627 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1437 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 6.8636 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 320.8636 mm



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 Nozzle Calcs.: N1-Tube Inlet Nozl: 17 9:28am Sep 9,2026

Input Echo, WRC297 Item 1, Description: N1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1755.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	193.675	mm
Nozzle Thickness	12.7000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	379.075	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	1000.00	KPa.
Radial Load	P	-1529.68	Kgf
Circumferential Shear	Vc	1019.78	Kgf
Longitudinal Shear	Vl	1019.78	Kgf
Circumferential Moment	Mc	570.86	Kg-m.
Longitudinal Moment	Ml	570.86	Kg-m.
Torsional Moment	Mt	815.51	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{894.0(14.0 - 3.0)}$$

$$= 546.325 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1788.000	mm
Nozzle Outside Diameter	(d) =	219.075	mm
Vessel Thickness used	(T) =	27.000	mm
Nozzle Thickness used	(t) =	9.700	mm
T / t	=	2.784	
d / t	=	22.585	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.997$$



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Nr/P = 0.095
 Mr/P = 0.111
 MØ/P = 0.095
 NØ/P = 0.207
 MrD/Mc = 0.184
 NrDL/Mc = 0.136
 MØd/Mc = 0.184
 NØDL/Mc = 0.107
 MrD/Ml = 0.148
 NrDL/Ml = 0.116
 MØD/Ml = 0.142
 NØDL/Ml = 0.249

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-29	29	29	-29
Normal Pressure Stress	32	33	32	33
Outplane Stress Summary	9	46	86	4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-31	31	31	-31
Inplane Pressure Stress	16	16	16	16
Inplane Stress Summary	-3	31	66	-22

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	16	48	87	27

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-38	38	38	-38
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16

Outplane Stress Summary	-10	43	74	-26
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-38	38	38	-38
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	32	33	32	33

Inplane Stress Summary	4	55	91	-12
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	15	55	92	28
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-8	-8	8	8
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	12	11	12

Hoop Stress Summary	7	8	24	25
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	2	2	2	2
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Axial Bending (P)	60	-60	60	-60
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-17	-17	17	17
Axial Bending (ML)	-146	146	146	-146
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-95	76	232	-181

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	13	13	7	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	106	79	233	208

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	12	11	12
Hoop Stress Summary	12	13	19	20

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	60	-60	60	-60
Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-188	188	188	-188
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-137	118	274	-223

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	10	10	10	10
Shear Stress Summary	7	7	13	13



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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	150	119	275	245

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{886.0(14.0 - 3.0)}$$

$$= 544.857 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1772.000 mm
Nozzle Outside Diameter	(d) =	379.075 mm
Vessel Thickness used	(T) =	11.000 mm
Nozzle Thickness used	(t) =	89.700 mm
T / t	=	0.123
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.715$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.066
Mr/P	=	0.116
MØ/P	=	0.034
NØ/P	=	0.049
MrD/Mc	=	0.275
NrDL/Mc	=	0.107
MØd/Mc	=	0.083
NØDL/Mc	=	0.068
MrD/Ml	=	0.087
NrDL/Ml	=	0.087
MØD/Ml	=	0.026
NØDL/Ml	=	0.064

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	25	-25	25	-25
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-18	18	18	-18
Normal Pressure Stress	79	80	79	80
Outplane Stress Summary	84	72	137	50

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	86	-86	86	-86
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10



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Inplane Bending (ML)	-63	63	63	-63
Inplane Pressure Stress	39	39	39	39
Inplane Stress Summary	59	15	208	-91

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	85	72	208	141

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	25	-25	25	-25
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-60	60	60	-60
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	39	39	39	39
Outplane Stress Summary	2	72	140	-31

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	86	-86	86	-86
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-201	201	201	-201
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	79	80	79	80
Inplane Stress Summary	-40	190	388	-186

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	1	1	4	4

Vessel Stresses (N./mm²):



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CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	43	190	388	186

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32	33	32	33	32	33	32	33
Circ. Pl (SUS)	-4	-4	12	12	-3	-3	5	5
Circ. Q (SUS)	-18	18	40	-40	-25	25	51	-51
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-3	-3	5	5	1	1	7	7
Long. Q (SUS)	-18	18	44	-44	-27	27	49	-49
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	32.0	33.0	32.0	33.0	32.0	33.0	32.0	33.0
Pm+Pl (SUS)	28.1	29.1	44.0	45.0	29.1	30.1	37.1	38.1
Pm+Pl+Q (Total)	17.0	47.9	84.2	28.3	14.6	55.4	88.9	27.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33.00	138.00	Passed
Pm+Pl (SUS)	45.04	207.00	Passed
Pm+Pl+Q (TOTAL)	88.94	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	79	80	79	80	79	80	79	80
Circ. Pl (SUS)	-1	-1	13	13	-5	-5	21	21
Circ. Q (SUS)	7	-7	43	-43	-115	115	287	-287
Long. Pm (SUS)	39	39	39	39	39	39	39	39
Long. Pl (SUS)	-2	-2	18	18	-2	-2	14	14
Long. Q (SUS)	23	-23	149	-149	-35	35	85	-85
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	79.0	80.0	79.0	80.0	79.0	80.0	79.0	80.0



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Pm+Pl (SUS)	78.0	79.0	92.0	93.0	74.0	75.0	100.0	101.0
Pm+Pl+Q (Total)	85.6	72.3	206.1	142.1	43.2	190.0	387.1	186.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	80.00	138.00	Passed
Pm+Pl (SUS)	101.02	207.00	Passed
Pm+Pl+Q (TOTAL)	387.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	12	11	12	11	12	11	12
Circ. Pl (SUS)	-4	-4	12	12	1	1	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-15	-15	19	19	-15	-15	19	19
Long. Q (SUS)	-86	86	206	-206	-128	128	248	-248
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	11.0	12.0	11.0	12.0	11.0	12.0	11.0	12.0
Pm+Pl (SUS)	17.5	18.4	25.6	26.0	22.4	23.3	24.6	24.7
Pm+Pl+Q (Total)	105.8	78.1	230.3	206.6	150.9	118.6	272.6	244.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.00	118.00	Passed
Pm+Pl (SUS)	26.00	177.00	Passed
Pm+Pl+Q (TOTAL)	272.57	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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Nozzle Calcs.: N2-Tube Outlet

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Input, Nozzle Desc: N2-Tube Outlet

From: 30

Pressure for Reinforcement Calculations	P	1017.203	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1755.00	mm
Design Length of Section	L	799.9689	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		731.18	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		200.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	379.0750	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		80.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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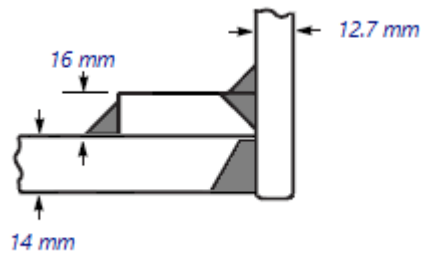
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Nozzle Calcs.: N2-Tube Outlet

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Reinforcement CALCULATION, Description: N2-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	193.675 mm.
Actual Thickness Used in Calculation	12.700 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1017.2 \cdot 880.5 / (138 \cdot 1.0 - 0.6 \cdot 1017.2)$
 $= 6.5195 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1017.2 \cdot 99.84 / (118 \cdot 1.0 - 0.6 \cdot 1017.2)$
 $= 0.8652 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8003 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	399.3500 mm
Parallel to Vessel Wall, opening length	d	199.6750 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



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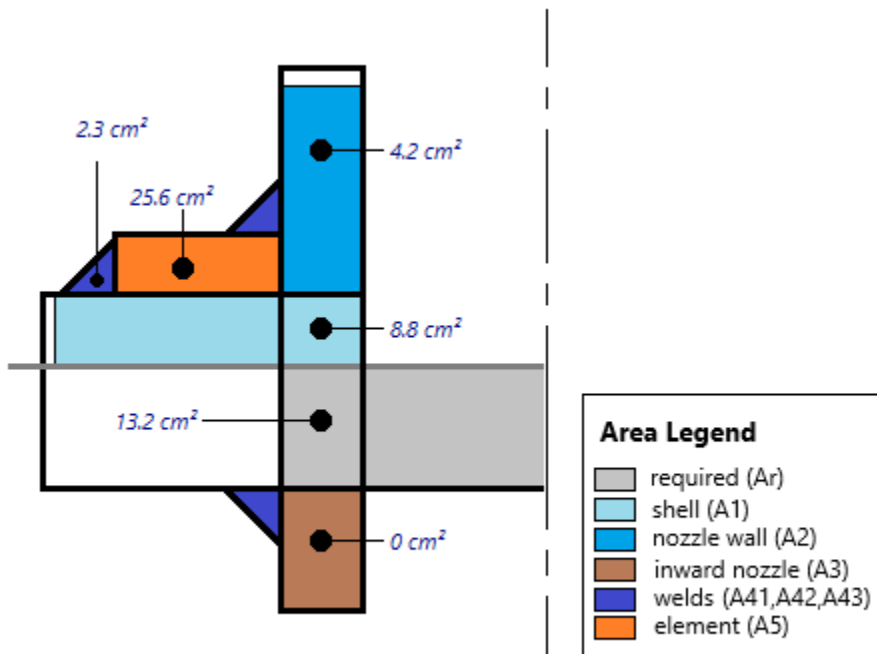


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	13.201	4.147	NA
Area in Shell	A1	8.820	13.591	NA
Area in Nozzle Wall	A2	4.155	4.185	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.260	2.260	NA
Area in Element	A5	25.600	25.600	NA
TOTAL AREA AVAILABLE	Atot	40.836	45.637	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (199.675 \cdot 6.5195 \cdot 1.0 + 2 \cdot 9.7 \cdot 6.5195 \cdot 1.0 \cdot (1 - 0.86)) \\
 &= 13.201 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 199.675(1.0 \cdot 11.0 - 1.0 \cdot 6.52) - 2 \cdot 9.7 \\
 &\quad (1.0 \cdot 11.0 - 1.0 \cdot 6.5195) \cdot (1 - 0.855) \\
 &= 8.820 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 27.5) \cdot (9.7 - 0.87) \cdot 0.8551
 \end{aligned}$$



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$$= 4.155 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - A_r \text{ Lost}) \cdot Fr_3 + ((W_i - \text{can}/0.707)^2 - A_r \text{ Lost}) \cdot fr_2 + \text{Trap} fr_4$$

$$= (1.) \cdot 0.86 + (0.0) \cdot 0.86 + 140.5311^2 \cdot 1.0$$

$$= 2.260 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$$

$$= (379.075 - 219.075) \cdot 16.0 \cdot 1.0$$

$$= 25.600 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.8652 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 5.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb_1 = 9.5195 \text{ mm}$

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 9.5195 \text{ mm}$

Wall Thickness, shell/head, external pressure $trb_2 = 3.6602 \text{ mm}$

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb_3 = 10.1600 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[10.16, \max(9.5195, 5.5)]$$

$$= 9.5195 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.8652, 9.5195)$$

$$= 9.5195 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 12.700 = 11.113 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained : 27.4, Allowable : 118.0 N./mm² Passed

Expansion : 0.0, Allowable : 267.6 N./mm² Passed

Occasional : 5.0, Allowable : 156.9 N./mm² Passed

Shear : 17.8, Allowable : 82.6 N./mm² Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 11.113$, $tr = 0.865$, $c = 3.0 \text{ mm}$, $E* = 1.0$

Thickness Ratio = $tr \cdot E*/(t_g - c) = 0.107$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -5 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 11.113$, $tr = 0.865$, $c = 3.0 \text{ mm}$, $E* = 1.0$

Thickness Ratio = $tr \cdot E*/(t_g - c) = 0.107$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Nozzle Calcs.: N2-Tube Outlet

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Govrn. thk, $t_g = 11.113$, $t_r = 0.865$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.107$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 14.0$, $t_r = 6.52$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.593$, Temp. Reduction = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-41 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 11.113$, $t_r = 0.865$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.107$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -41 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -41 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1017.20 / 1960.01 = 0.519$

Weld Size Calculations, Description: N2-Tube Outlet

Intermediate Calc. for nozzle/shell Welds	T_{min}	9.7000 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$5.5000 = 0.5 * T_{minPad}$	$8.4840 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (13.2012 - 8.8204 + 2 \cdot 9.7 \cdot 0.8551 \cdot (1.0 \cdot 11.0 - 6.5195)) \cdot 138)$
 $= 7210.25$ Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f r2) \cdot S_v$
 $= (4.1549 + 25.6 + 2.2604 - 0.0 \cdot 0.86) \cdot 138$
 $= 45050.57$ Kgf

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r1)) \cdot S_v$



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$$= (4.1549 + 0.0 + 0.8551 + (1.8247)) \cdot 138$$

$$= 9617.53 \text{ Kg}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$

$$= (4.1549 + 0.0 + 2.2604 + 25.6 + (1.8247)) \cdot 138$$

$$= 47618.25 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 219.075 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 20289. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 379.075 \cdot 12.0 \cdot 0.49 \cdot 138$$

$$= 49268. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 104.6875) \cdot (12.7 - 3.0) \cdot 0.7 \cdot 118$$

$$= 26870. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 219.075 \cdot 16.0 \cdot 0.74 \cdot 138$$

$$= 57333. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 219.075 \cdot (14.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 39417. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (49268 + 26870) = 76138 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (20289 + 57333 + 39417 + 0) = 117039 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (49268 + 39417 + 0) = 88685 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 76137	Kgf, must exceed W = 7210 Kg	or W1 = 45050 Kg
Path 2-2 = 117038	Kgf, must exceed W = 7210 Kg	or W2 = 9617 Kg
Path 3-3 = 88684	Kgf, must exceed W = 7210 Kg	or W3 = 47618 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1437 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 6.8636 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 320.8636 mm



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Input Echo, WRC297 Item 1, Description: N2-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1755.000	mm	
Shell Thickness	14.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²	
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	193.675	mm	
Nozzle Thickness	12.7000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-106 B		
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²	
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	379.075	mm	
Thickness of Reinforcing Pad	16.0000	mm	

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1017.20	KPa.
Radial Load	P	-1529.68	Kgf
Circumferential Shear	Vc	1019.78	Kgf
Longitudinal Shear	Vl	1019.78	Kgf
Circumferential Moment	Mc	570.86	Kg-m.
Longitudinal Moment	Ml	570.86	Kg-m.
Torsional Moment	Mt	815.51	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{894.0(14.0 - 3.0)}$$

$$= 546.325 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1788.000	mm
Nozzle Outside Diameter	(d) =	219.075	mm
Vessel Thickness used	(T) =	27.000	mm
Nozzle Thickness used	(t) =	9.700	mm
T / t	=	2.784	
d / t	=	22.585	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.997$$



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Nozzle Calcs.: N2-Tube Outlet

Nozl: 18

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Nr/P = 0.095
 Mr/P = 0.111
 MØ/P = 0.095
 NØ/P = 0.207
 MrD/Mc = 0.184
 NrDL/Mc = 0.136
 MØd/Mc = 0.184
 NØDL/Mc = 0.107
 MrD/Ml = 0.148
 NrDL/Ml = 0.116
 MØD/Ml = 0.142
 NØDL/Ml = 0.249

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-29	29	29	-29
Normal Pressure Stress	32	33	32	33
Outplane Stress Summary	10	47	87	5

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-31	31	31	-31
Inplane Pressure Stress	16	16	16	16
Inplane Stress Summary	-3	31	67	-22

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	16	48	87	28

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Nozzle Calcs.: N2-Tube Outlet

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Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-38	38	38	-38
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16

Outplane Stress Summary	-10	43	74	-26
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-38	38	38	-38
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	32	33	32	33

Inplane Stress Summary	4	55	91	-11
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	16	56	93	27
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-8	-8	8	8
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	13	11	13

Hoop Stress Summary	7	8	24	26
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	2	2	2	2
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Nozzle Calcs.: N2-Tube Outlet

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Axial Bending (P)	60	-60	60	-60
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-17	-17	17	17
Axial Bending (ML)	-146	146	146	-146
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-95	77	233	-181

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	13	13	7	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	106	79	233	208

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	13	11	13
Hoop Stress Summary	12	13	19	21

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	60	-60	60	-60
Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-188	188	188	-188
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-137	118	274	-223

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	10	10	10	10
Shear Stress Summary	7	7	13	13



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Nozzle Calcs.: N2-Tube Outlet

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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	150	119	275	246

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{886.0(14.0 - 3.0)}$$

$$= 544.857 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1772.000 mm
Nozzle Outside Diameter	(d) =	379.075 mm
Vessel Thickness used	(T) =	11.000 mm
Nozzle Thickness used	(t) =	89.700 mm
T / t	=	0.123
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.715$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.066
Mr/P	=	0.116
MØ/P	=	0.034
NØ/P	=	0.049
MrD/Mc	=	0.275
NrDL/Mc	=	0.107
MØd/Mc	=	0.083
NØDL/Mc	=	0.068
MrD/Ml	=	0.087
NrDL/Ml	=	0.087
MØD/Ml	=	0.026
NØDL/Ml	=	0.064

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	25	-25	25	-25
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-18	18	18	-18
Normal Pressure Stress	80	81	80	81
Outplane Stress Summary	85	73	139	51

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	86	-86	86	-86
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10



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Nozzle Calcs.: N2-Tube Outlet

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Inplane Bending (ML)	-63	63	63	-63
Inplane Pressure Stress	40	40	40	40

Inplane Stress Summary	60	15	209	-90
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Two * Max Shear Stress	86	73	209	142
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	25	-25	25	-25
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-60	60	60	-60
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	40	40	40	40

Outplane Stress Summary	3	73	140	-31
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	86	-86	86	-86
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-201	201	201	-201
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	80	81	80	81

Inplane Stress Summary	-39	192	390	-184
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	1	1	4	4
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Vessel Stresses (N./mm²):



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 Nozzle Calcs.: N2-Tube Outlet Nozl: 18 9:28am Sep 9,2026

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	42	192	390	184

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32	33	32	33	32	33	32	33
Circ. Pl (SUS)	-4	-4	12	12	-3	-3	5	5
Circ. Q (SUS)	-18	18	40	-40	-25	25	51	-51
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-3	-3	5	5	1	1	7	7
Long. Q (SUS)	-18	18	44	-44	-27	27	49	-49
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	32.0	33.0	32.0	33.0	32.0	33.0	32.0	33.0
Pm+Pl (SUS)	28.1	29.1	44.0	45.0	29.1	30.1	37.1	38.1
Pm+Pl+Q (Total)	17.0	47.9	84.2	28.3	14.6	55.4	88.9	27.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33.00	138.00	Passed
Pm+Pl (SUS)	45.04	207.00	Passed
Pm+Pl+Q (TOTAL)	88.94	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	80	81	80	81	80	81	80	81
Circ. Pl (SUS)	-1	-1	13	13	-5	-5	21	21
Circ. Q (SUS)	7	-7	43	-43	-115	115	287	-287
Long. Pm (SUS)	40	40	40	40	40	40	40	40
Long. Pl (SUS)	-2	-2	18	18	-2	-2	14	14
Long. Q (SUS)	23	-23	149	-149	-35	35	85	-85
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	80.0	81.0	80.0	81.0	80.0	81.0	80.0	81.0



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 Nozzle Calcs.: N2-Tube Outlet Nozl: 18 9:28am Sep 9,2026

Pm+Pl (SUS)	79.0	80.0	93.0	94.0	75.0	76.0	101.0	102.0
Pm+Pl+Q (Total)	86.6	73.3	207.1	142.1	43.2	191.0	388.1	185.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	81.00	138.00	Passed
Pm+Pl (SUS)	102.02	207.00	Passed
Pm+Pl+Q (TOTAL)	388.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	13	11	13	11	13	11	13
Circ. Pl (SUS)	-4	-4	12	12	1	1	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-15	-15	19	19	-15	-15	19	19
Long. Q (SUS)	-86	86	206	-206	-128	128	248	-248
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	11.0	13.0	11.0	13.0	11.0	13.0	11.0	13.0
Pm+Pl (SUS)	17.5	19.4	25.6	26.6	22.4	24.3	24.6	24.8
Pm+Pl+Q (Total)	105.8	78.1	230.3	207.6	150.9	118.6	272.6	245.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	13.00	118.00	Passed
Pm+Pl (SUS)	26.56	177.00	Passed
Pm+Pl+Q (TOTAL)	272.57	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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Nozzle Calcs.: S3-Shell outlet

Nozl: 19

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Input, Nozzle Desc: S3-Shell outlet

From: 40

Pressure for Reinforcement Calculations	P	1567.203	KPa.
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1755.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1706.18	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		477.8200	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	15.0900	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	32.5890	mm
Height of Beveled Transition	L'	34.5890	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		75.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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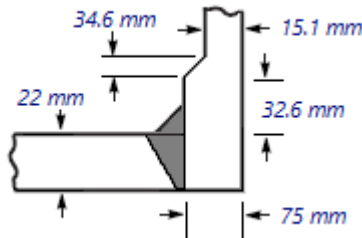
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Nozzle Calcs.: S3-Shell outlet

Nozl: 19

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Reinforcement CALCULATION, Description: S3-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	477.820 mm.
Actual Thickness Used in Calculation	15.090 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1567.2 \cdot 880.5 / (138 \cdot 1.0 - 0.6 \cdot 1567.2)$
 $= 10.0688 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1567.2 \cdot 238.91 / (138 \cdot 1.0 + 0.4 \cdot 1567.2)$
 $= 2.7011 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 59.91 / 0.5773$
 $= 103.7672 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	907.2801 mm
Parallel to Vessel Wall, opening length	d	453.6400 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	47.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



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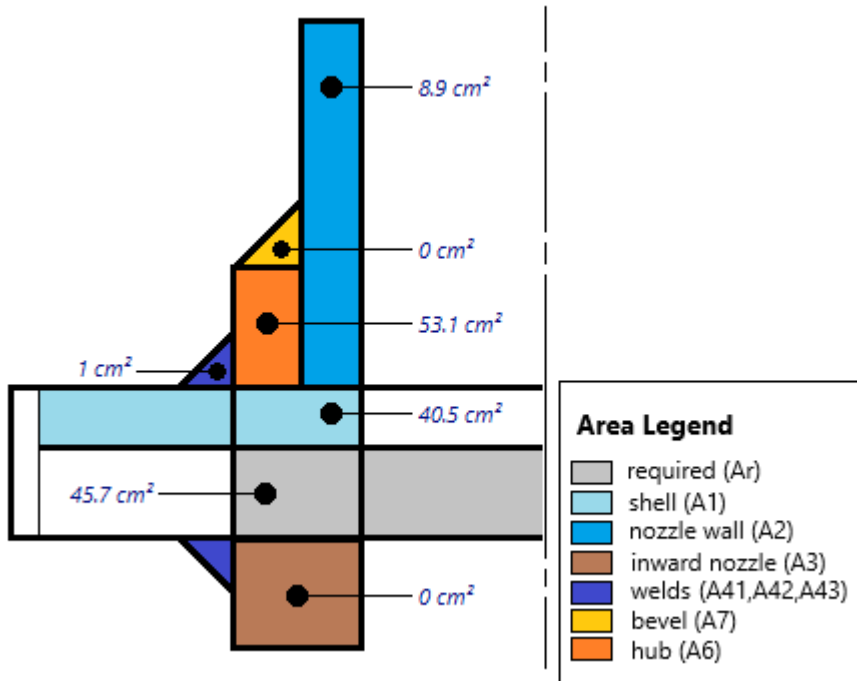
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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	45.676	NA	NA
Area in Shell	A1	40.515	NA	NA
Area in Nozzle Wall	A2	8.919	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	53.063	NA	NA
TOTAL AREA AVAILABLE	Atot	103.498	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (453.64 \cdot 10.0688 \cdot 1.0 + 2 \cdot 72.0 \cdot 10.0688 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 45.676 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 453.64(1.0 \cdot 19.0 - 1.0 \cdot 10.069) - 2 \cdot 72.0 \\
 &\quad (1.0 \cdot 19.0 - 1.0 \cdot 10.0688) \cdot (1 - 1.0) \\
 &= 40.515 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tnp)(tn - trn)fr2 \\
 &= (2 \cdot 47.5)(12.09 - 2.7)1.0 \\
 &= 8.919 \text{ cm}^2
 \end{aligned}$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(47.5, 400.0, 32.6)) \cdot (75.0 - 15.1) \cdot 1.0$$

$$= 53.063 \text{ cm}^2 \text{ Includes Bevel Area of } 14.015 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.7011 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.0688 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.0688 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(13.0688, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.7011, 11.3312)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 15.0900 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 36.8,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 308.2 N./mm ²	Passed
Occasional	: 14.3,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 16.9,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 15.09, tr = 2.701, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.223, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 22.0, tr = 10.069, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.53, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-33 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -33 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -29 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1567.20/1960.01 = 0.800

Weld Size Calculations, Description: S3-Shell outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (45.6761 - 40.5155 + 2 \cdot 72.0 \cdot 1.0 \cdot (1.0 \cdot 19.0 - 10.0688)) \cdot 138)$
 $= 25359.22 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (61.9829 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 88626.86 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (61.9829 + 0.0 + 1. + (27.36)) \cdot 138$
 $= 127126.69 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (61.9829 + 0.0 + 1. + 0.0 + (27.36)) \cdot 138$
 $= 127126.69 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 477.82 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 51751. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 232.865) \cdot (75.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 518833. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 477.82 \cdot (14.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 109417. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (51751 + 518833) = 570584 \text{ Kgf}$



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Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (51751 + 0 + 109417 + 0) = 161169 Kgf

Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (51751 + 109417 + 0) = 161169 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 570584 Kgf, must exceed W = 25359 Kgf or W1 = 88626 Kgf

Path 2-2 = 161168 Kgf, must exceed W = 25359 Kgf or W2 = 127126 Kgf

Path 3-3 = 161168 Kgf, must exceed W = 25359 Kgf or W3 = 127126 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2505 KPa.

The Drop for this Nozzle is : 33.1492 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 455.1493 mm

Input Echo, WRC297 Item 1, Description: S3-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1755.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	597.640	mm
Nozzle Thickness	75.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1567.20	KPa.
Radial Load	P	-3824.19	Kgf
Circumferential Shear	Vc	2549.46	Kgf
Longitudinal Shear	Vl	2549.46	Kgf
Circumferential Moment	Mc	3567.87	Kg-m.
Longitudinal Moment	Ml	3567.87	Kg-m.
Torsional Moment	Mt	5096.96	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (7.969) is < 10
check the Limitations of WRC bulletin 297.



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Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 597.64 + 2 \cdot 1.65 \cdot \sqrt{890.0(22.0 - 3.0)}$
 $= 1026.767 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1780.000 mm
 Nozzle Outside Diameter (d) = 597.640 mm
 Vessel Thickness used (T) = 19.000 mm
 Nozzle Thickness used (t) = 72.000 mm
 T / t = 0.264
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.250$$

Nr/P = 0.056
 Mr/P = 0.098
 M0/P = 0.031
 N0/P = 0.065
 MrD/Mc = 0.262
 NrDL/Mc = 0.093
 M0d/Mc = 0.079
 N0DL/Mc = 0.086
 MrD/Ml = 0.081
 NrDL/Ml = 0.057
 M0D/Ml = 0.023
 N0DL/Ml = 0.075

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-12	-12	12	12
Outplane Bending (ML)	-22	22	22	-22
Normal Pressure Stress	71	73	71	73
Outplane Stress Summary	62	71	132	50

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-78	78	78	-78
Inplane Pressure Stress	35	35	35	35
Inplane Stress Summary	15	50	191	-89

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	6	6	3	3
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress	63	73	191	140
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	-13	-13	13	13
Outplane Bending (Mc)	-76	76	76	-76
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	35	35	35	35

Outplane Stress Summary	-28	86	152	-39
-------------------------	-----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-255	255	255	-255
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	71	73	71	73

Inplane Stress Summary	-131	258	409	-222
------------------------	------	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	6	6
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	131	258	409	222
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the	Au Top	Al Top	Bu Bottom	Bl Bottom
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hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	6	5	6
Hoop Stress Summary	0	1	23	25

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	0	0	11	-1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	3	3	24	27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-13	-13	13	13
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	6	5	6
Hoop Stress Summary	-2	0	25	27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3



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Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-14	14	14	-14
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-11	11	23	-12
----------------------	-----	----	----	-----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	11	12	26	40

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	71	73	71	73	71	73	71	73
Circ. Pl (SUS)	-6	-6	18	18	-10	-10	20	20
Circ. Q (SUS)	-3	3	41	-41	-194	194	316	-316
Long. Pm (SUS)	35	35	35	35	35	35	35	35
Long. Pl (SUS)	-4	-4	14	14	-7	-7	19	19
Long. Q (SUS)	-17	17	139	-139	-57	57	95	-95
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	71.0	73.0	71.0	73.0	71.0	73.0	71.0	73.0
Pm+Pl (SUS)	65.0	67.0	89.0	91.0	61.0	63.0	91.0	93.0
Pm+Pl+Q (Total)	62.5	71.1	188.2	140.1	133.1	257.1	407.1	223.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	73.00	138.00	Passed
Pm+Pl (SUS)	93.03	207.00	Passed
Pm+Pl+Q (TOTAL)	407.10	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):



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Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	6	5	6	5	6	5	6
Circ. Pl (SUS)	-6	-6	18	18	-7	-7	19	19
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	0	0	6	-6	-11	11	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	5.0	6.0	5.0	6.0	5.0	6.0	5.0	6.0
Pm+Pl (SUS)	1.0	0.0	23.0	24.0	2.0	1.0	24.0	25.0
Pm+Pl+Q (Total)	2.2	2.0	23.1	26.1	11.1	12.2	24.3	38.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	138.00	Passed
Pm+Pl (SUS)	25.00	207.00	Passed
Pm+Pl+Q (TOTAL)	38.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: S1-Shell Inlet

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Input, Nozzle Desc: S1-Shell Inlet

From: 40

Pressure for Reinforcement Calculations	P	1550.000	KPa.
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1755.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1706.18	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		477.8200	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	15.0900	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	40.0000	mm
Height of Beveled Transition	L'	42.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		75.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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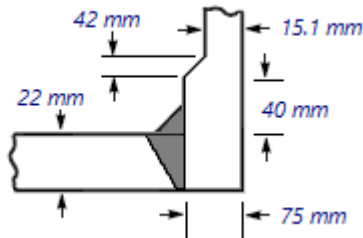
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Nozzle Calcs.: S1-Shell Inlet

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Reinforcement CALCULATION, Description: S1-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	477.820 mm.
Actual Thickness Used in Calculation	15.090 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $1550.0 \cdot 880.5 / (138 \cdot 1.0 - 0.6 \cdot 1550.0)$

= 9.9575 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $1550.0 \cdot 238.91 / (138 \cdot 1.0 + 0.4 \cdot 1550.0)$

= 2.6716 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

= (Hub Thickness - Neck Thickness) / $\tan(30)$

= $59.91 / 0.5773$

= 103.7672 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	907.2801 mm
--	----	-------------

Parallel to Vessel Wall, opening length	d	453.6400 mm
---	---	-------------

Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	47.5000 mm
---	------	------------

Weld Strength Reduction Factor [fr1]:

= $\min(1, S_n / S_v)$

= $\min(1, 138.0 / 138.0)$

= 1.000

Weld Strength Reduction Factor [fr2]:

= $\min(1, S_n / S_v)$

= $\min(1, 138.0 / 138.0)$

= 1.000

Weld Strength Reduction Factor [fr3]:

= $\min(fr2, fr4)$

= $\min(1.0, 1.0)$

= 1.000



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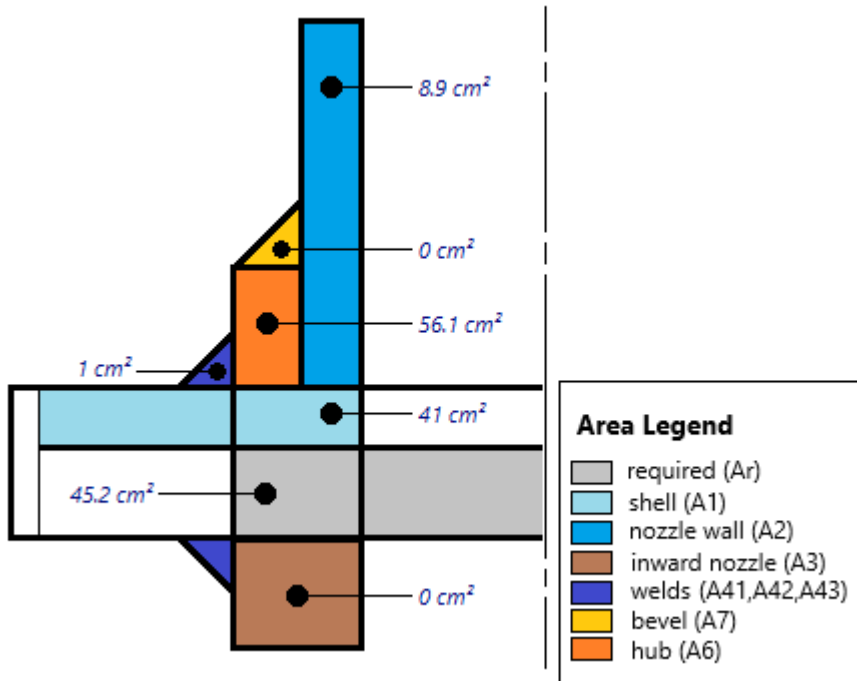
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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	45.171	NA	NA
Area in Shell	A1	41.020	NA	NA
Area in Nozzle Wall	A2	8.947	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	56.112	NA	NA
TOTAL AREA AVAILABLE	Atot	107.080	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \cdot UG-37(c)$$

$$= (453.64 \cdot 9.9575 \cdot 1.0 + 2 \cdot 72.0 \cdot 9.9575 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 45.171 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 453.64(1.0 \cdot 19.0 - 1.0 \cdot 9.958) - 2 \cdot 72.0$$

$$(1.0 \cdot 19.0 - 1.0 \cdot 9.9575) \cdot (1 - 1.0)$$

$$= 41.020 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 47.5)(12.09 - 2.67)1.0$$

$$= 8.947 \text{ cm}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(47.5, 400.0, 40.0)) \cdot (75.0 - 15.1) \cdot 1.0$$

$$= 56.112 \text{ cm}^2 \text{ Includes Bevel Area of } 8.184 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.6716 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 12.9575 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 12.9575 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(12.9575, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.6716, 11.3312)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 15.0900 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 36.7,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 308.3 N./mm ²	Passed
Occasional	: 14.2,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 16.9,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 15.09, tr = 2.672, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.221, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 22.0, tr = 9.958, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.524, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-34 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -34 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -29 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1550.00/1960.01 = 0.791

Weld Size Calculations, Description: S1-Shell Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (45.1714 - 41.0202 + 2 \cdot 72.0 \cdot 1.0 \cdot (1.0 \cdot 19.0 - 9.9575)) \cdot 138)$
 $= 24164.12 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (65.0596 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 92956.27 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (65.0596 + 0.0 + 1. + (27.36)) \cdot 138$
 $= 131456.08 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (65.0596 + 0.0 + 1. + 0.0 + (27.36)) \cdot 138$
 $= 131456.08 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 477.82 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 51751. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 232.865) \cdot (75.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 518833. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 477.82 \cdot (14.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 109417. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (51751 + 518833) = 570584 \text{ Kgf}$



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Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (51751 + 0 + 109417 + 0) = 161169 Kgf

Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (51751 + 109417 + 0) = 161169 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 570584 Kgf, must exceed W = 24164 Kgf or W1 = 92956 Kgf

Path 2-2 = 161168 Kgf, must exceed W = 24164 Kgf or W2 = 131456 Kgf

Path 3-3 = 161168 Kgf, must exceed W = 24164 Kgf or W3 = 131456 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2572 KPa.

The Drop for this Nozzle is : 33.1492 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 455.1493 mm

Input Echo, WRC297 Item 1, Description: S1-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1755.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	597.640	mm
Nozzle Thickness	75.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1550.00	KPa.
Radial Load	P	-3824.19	Kgf
Circumferential Shear	Vc	2549.46	Kgf
Longitudinal Shear	Vl	2549.46	Kgf
Circumferential Moment	Mc	3567.87	Kg-m.
Longitudinal Moment	Ml	3567.87	Kg-m.
Torsional Moment	Mt	5096.96	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (7.969) is < 10
check the Limitations of WRC bulletin 297.



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Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 597.64 + 2 \cdot 1.65 \cdot \sqrt{890.0(22.0 - 3.0)}$$

$$= 1026.767 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1780.000 mm

Nozzle Outside Diameter (d) = 597.640 mm

Vessel Thickness used (T) = 19.000 mm

Nozzle Thickness used (t) = 72.000 mm

T / t = 0.264

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.250$$

Nr/P = 0.056

Mr/P = 0.098

M0/P = 0.031

N0/P = 0.065

MrD/Mc = 0.262

NrDL/Mc = 0.093

M0d/Mc = 0.079

N0DL/Mc = 0.086

MrD/Ml = 0.081

NrDL/Ml = 0.057

M0D/Ml = 0.023

N0DL/Ml = 0.075

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-12	-12	12	12
Outplane Bending (ML)	-22	22	22	-22
Normal Pressure Stress	71	72	71	72
Outplane Stress Summary	62	70	131	49

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-78	78	78	-78
Inplane Pressure Stress	35	35	35	35
Inplane Stress Summary	14	49	190	-89

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	6	6	3	3
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	62	72	190	139
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	-13	-13	13	13
Outplane Bending (Mc)	-76	76	76	-76
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	35	35	35	35

Outplane Stress Summary	-29	85	151	-39
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-255	255	255	-255
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	71	72	71	72

Inplane Stress Summary	-132	257	408	-223
------------------------	------	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	6	6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	132	257	408	223
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses in the	Top	Top	Bottom	Bottom



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hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	6	5	6
Hoop Stress Summary	0	1	23	25

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	0	0	11	-1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	3	3	23	27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-13	-13	13	13
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	6	5	6
Hoop Stress Summary	-2	0	25	27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3



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Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-14	14	14	-14
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-11	11	23	-12
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	11	12	26	40

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	71	72	71	72	71	72	71	72
Circ. Pl (SUS)	-6	-6	18	18	-10	-10	20	20
Circ. Q (SUS)	-3	3	41	-41	-194	194	316	-316
Long. Pm (SUS)	35	35	35	35	35	35	35	35
Long. Pl (SUS)	-4	-4	14	14	-7	-7	19	19
Long. Q (SUS)	-17	17	139	-139	-57	57	95	-95
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	71.0	72.0	71.0	72.0	71.0	72.0	71.0	72.0
Pm+Pl (SUS)	65.0	66.0	89.0	90.0	61.0	62.0	91.0	92.0
Pm+Pl+Q (Total)	62.5	70.1	188.2	139.1	133.1	256.1	407.1	224.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	72.00	138.00	Passed
Pm+Pl (SUS)	92.03	207.00	Passed
Pm+Pl+Q (TOTAL)	407.10	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):



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FileName : 128-C-AK-With 4 nozzle

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Nozzle Calcs.: S1-Shell Inlet

Nozl: 20

9:28am

Sep 9, 2026

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	6	5	6	5	6	5	6
Circ. Pl (SUS)	-6	-6	18	18	-7	-7	19	19
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	0	0	6	-6	-11	11	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	5.0	6.0	5.0	6.0	5.0	6.0	5.0	6.0
Pm+Pl (SUS)	1.0	0.0	23.0	24.0	2.0	1.0	24.0	25.0
Pm+Pl+Q (Total)	2.2	2.0	23.1	26.1	11.1	12.2	24.3	38.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	138.00	Passed
Pm+Pl (SUS)	25.00	207.00	Passed
Pm+Pl+Q (TOTAL)	38.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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FileName : 128-C-AK-With 4 nozzle

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Nozzle Calcs.: N5-Vent

Nozl: 21

9:28am

Sep 9, 2026

Input, Nozzle Desc: N5-Vent

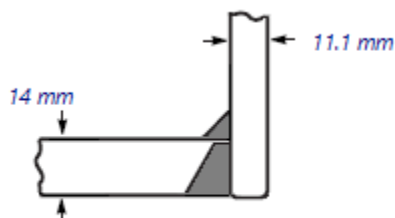
From: 50

Pressure for Reinforcement Calculations	P	1550.000	KPa.
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1755.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4441.13	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





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FileName : 128-C-AK-With 4 nozzle

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Nozzle Calcs.: N5-Vent

Nozl: 21

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ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 38.176 mm.
Actual Thickness Used in Calculation 11.074 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1550.0 \cdot 880.5 / (138 \cdot 1.0 - 0.6 \cdot 1550.0)$
= 9.9575 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1550.0 \cdot 22.09 / (118 \cdot 1.0 - 0.6 \cdot 1550.0)$
= 0.2925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	88.3524	mm
Parallel to Vessel Wall, opening length	d	44.1762	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	20.1860	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N5-Vent.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.2925	mm
Wall Thickness per UG16(b),	tr16b = 5.5000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 12.9575	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 12.9575	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000	mm
Wall Thickness per table UG-45	tb3 = 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.42, max(12.9575, 5.5)]
= 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.2925, 6.42)
= 6.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 11.074 = 9.690$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.69, tr = 0.292, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.044$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C



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 FileName : 128-C-AK-With 4 nozzle Page 181 of 222
 Nozzle Calcs.: N5-Vent Nozl: 21 9:28am Sep 9,2026

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 9.69, tr = 0.292, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.044$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -29 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1550.00 / 1960.01 = 0.791$

Weld Size Calculations, Description: N5-Vent

Intermediate Calc. for nozzle/shell Welds Tmin 8.0744 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6521 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5186 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 164.5186 mm

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FileName : 128-C-AK-With 4 nozzle

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Nozzle Calcs.: N6-Drain

Nozl: 22

9:28am

Sep 9, 2026

Input, Nozzle Desc: N6-Drain

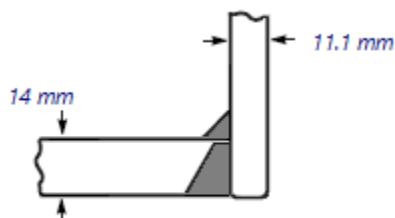
From: 50

Pressure for Reinforcement Calculations	P	1567.203	KPa.
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1755.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4524.53	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





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 FileName : 128-C-AK-With 4 nozzle
 Nozzle Calcs.: N6-Drain Nozl: 22 9:28am Page 183 of 222
 Sep 9, 2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 38.176 mm.
 Actual Thickness Used in Calculation 11.074 mm

Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1567.2 \cdot 880.5 / (138 \cdot 1.0 - 0.6 \cdot 1567.2)$
 $= 10.0688 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1567.2 \cdot 22.09 / (118 \cdot 1.0 - 0.6 \cdot 1567.2)$
 $= 0.2957 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	88.3524	mm
Parallel to Vessel Wall, opening length	d	44.1762	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.1860	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N6-Drain.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2957	mm
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 13.0688	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.0688	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(13.0688, 5.5)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.2957, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 11.074 = 9.690 \text{ mm} \rightarrow \text{OK}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.69, tr = 0.296, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.044$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C



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 FileName : 128-C-AK-With 4 nozzle Page 184 of 222
 Nozzle Calcs.: N6-Drain Nozl: 22 9:28am Sep 9,2026

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 9.69, tr = 0.296, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.044$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -29 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1567.20 / 1960.01 = 0.800$

Weld Size Calculations, Description: N6-Drain

Intermediate Calc. for nozzle/shell Welds Tmin 8.0744 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6521 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5186 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 164.5186 mm

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FileName : 128-C-AK-With 4 nozzle

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Nozzle Calcs.: S2-Shell Inlet

Nozl: 23

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Sep 9, 2026

Input, Nozzle Desc: S2-Shell Inlet

From: 60

Pressure for Reinforcement Calculations	P	1550.000	KPa.
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1755.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7106.18	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		477.8200	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	15.0900	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	30.0904	mm
Height of Beveled Transition	L'	34.5890	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		75.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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FileName : 128-C-AK-With 4 nozzle

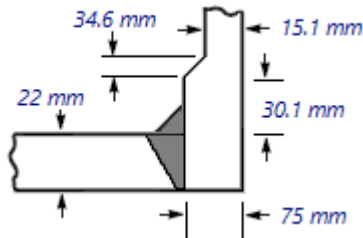
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Nozzle Calcs.: S2-Shell Inlet

Nozl: 23

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Reinforcement CALCULATION, Description: S2-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	477.820 mm.
Actual Thickness Used in Calculation	15.090 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $1550.0 \cdot 880.5 / (138 \cdot 1.0 - 0.6 \cdot 1550.0)$

= 9.9575 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $1550.0 \cdot 238.91 / (138 \cdot 1.0 + 0.4 \cdot 1550.0)$

= 2.6716 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

= (Hub Thickness - Neck Thickness) / tan(30)

= 59.91 / 0.5773

= 103.7672 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	907.2801 mm
--	----	-------------

Parallel to Vessel Wall, opening length	d	453.6400 mm
---	---	-------------

Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	47.5000 mm
---	------	------------

Weld Strength Reduction Factor [fr1]:

= min(1, S_n / S_v)

= min(1, 138.0 / 138.0)

= 1.000

Weld Strength Reduction Factor [fr2]:

= min(1, S_n / S_v)

= min(1, 138.0 / 138.0)

= 1.000

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(1.0, 1.0)

= 1.000



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FileName : 128-C-AK-With 4 nozzle

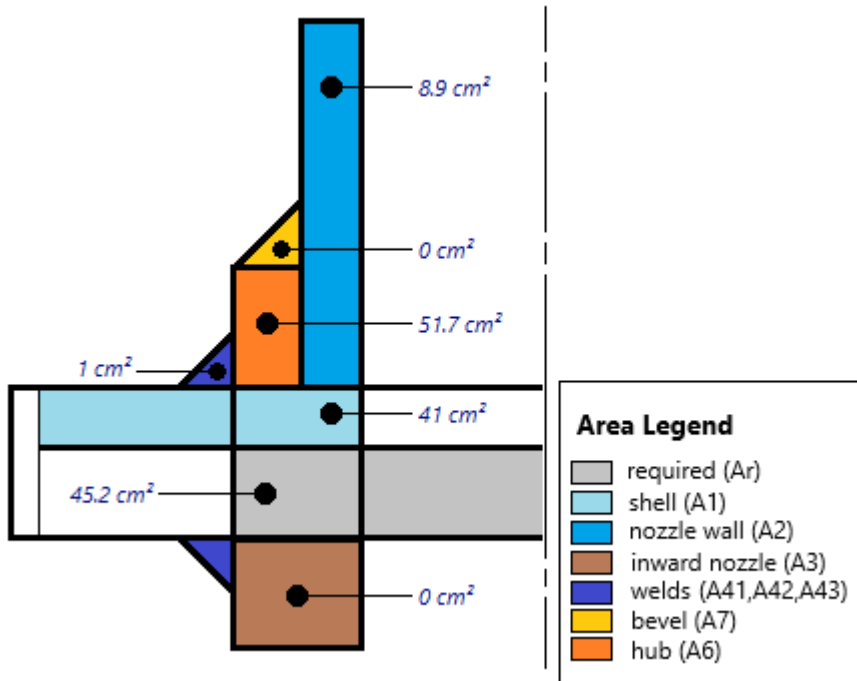
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Nozzle Calcs.: S2-Shell Inlet

Noz1: 23

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	45.171	NA	NA
Area in Shell	A1	41.020	NA	NA
Area in Nozzle Wall	A2	8.947	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	51.665	NA	NA
TOTAL AREA AVAILABLE	Atot	102.632	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \cdot UG-37(c)$$

$$= (453.64 \cdot 9.9575 \cdot 1.0 + 2 \cdot 72.0 \cdot 9.9575 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 45.171 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 453.64(1.0 \cdot 19.0 - 1.0 \cdot 9.958) - 2 \cdot 72.0$$

$$(1.0 \cdot 19.0 - 1.0 \cdot 9.9575) \cdot (1 - 1.0)$$

$$= 41.020 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 47.5)(12.09 - 2.67)1.0$$

$$= 8.947 \text{ cm}^2$$



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FileName : 128-C-AK-With 4 nozzle

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Nozzle Calcs.: S2-Shell Inlet

Nozl: 23

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(47.5, 400.0, 30.1)) \cdot (75.0 - 15.1) \cdot 1.0$$

$$= 51.665 \text{ cm}^2 \text{ Includes Bevel Area of } 15.610 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 5.6716 \text{ mm}$

Wall Thickness per UG16(b), $tr16b = 5.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb1 = 12.9575 \text{ mm}$

Wall Thickness $tb1 = \max(trb1, tr16b) = 12.9575 \text{ mm}$

Wall Thickness $tb2 = \max(trb2, tr16b) = 5.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(12.9575, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(5.6716, 11.3312)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 15.0900 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	36.7,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	308.3 N./mm ²	Passed
Occasional	:	14.2,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	16.9,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 15.09$, $t_r = 2.672$, $c = 3.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.221$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 4 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 22.0$, $t_r = 9.958$, $c = 3.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.524$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C

Min Metal Temp. at Required thickness (UCS 66.1) -34 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -34 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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Nozzle Calcs.: S2-Shell Inlet

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Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -29 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 1550.00/1960.01 = 0.791

Weld Size Calculations, Description: S2-Shell Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (45.1714 - 41.0202 + 2 • 72.0 • 1.0 •
(1.0 • 19.0 - 9.9575))138)
= 24164.12 Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (60.6122 + 1. - 0.0 • 1.0) • 138
= 86698.08 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (60.6122 + 0.0 + 1. + (27.36)) • 138
= 125197.88 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (60.6122 + 0.0 + 1. + 0.0 + (27.36)) • 138
= 125197.88 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 477.82 • 10.0 • 0.49 • 138
= 51751. Kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 232.865) • (75.0 - 3.0) • 0.7 • 138
= 518833. Kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 477.82 • (14.0 - 0.0) • 0.74 • 138
= 109417. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (51751 + 518833) = 570584 Kgf



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Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (51751 + 0 + 109417 + 0) = 161169 Kgf

Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (51751 + 109417 + 0) = 161169 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 570584 Kgf, must exceed W = 24164 Kgf or W1 = 86698 Kgf

Path 2-2 = 161168 Kgf, must exceed W = 24164 Kgf or W2 = 125197 Kgf

Path 3-3 = 161168 Kgf, must exceed W = 24164 Kgf or W3 = 125197 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2499 KPa.

The Drop for this Nozzle is : 33.1492 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 455.1493 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 453.64 mm

Input Echo, WRC297 Item 1, Description: S2-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1755.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	597.640	mm
Nozzle Thickness	75.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	1550.00	KPa.
Radial Load	P	-3824.19	Kgf
Circumferential Shear	Vc	2549.46	Kgf
Longitudinal Shear	Vl	2549.46	Kgf
Circumferential Moment	Mc	3567.87	Kg-m.
Longitudinal Moment	Ml	3567.87	Kg-m.
Torsional Moment	Mt	5096.96	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No



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 Nozzle Calcs.: S2-Shell Inlet Nozl: 23 9:28am Sep 9,2026

*Warning - The ratio, Dn/Tn (7.969) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 597.64 + 2 \cdot 1.65 \cdot \sqrt{890.0(22.0 - 3.0)}$
 $= 1026.767 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1780.000 mm
 Nozzle Outside Diameter (d) = 597.640 mm
 Vessel Thickness used (T) = 19.000 mm
 Nozzle Thickness used (t) = 72.000 mm
 $T / t = 0.264$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.250$$

Nr/P = 0.056
 Mr/P = 0.098
 M0/P = 0.031
 N0/P = 0.065
 MrD/Mc = 0.262
 NrDL/Mc = 0.093
 M0d/Mc = 0.079
 N0DL/Mc = 0.086
 MrD/Ml = 0.081
 NrDL/Ml = 0.057
 M0D/Ml = 0.023
 N0DL/Ml = 0.075

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-12	-12	12	12
Outplane Bending (ML)	-22	22	22	-22
Normal Pressure Stress	71	72	71	72
Outplane Stress Summary	62	70	131	49

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-78	78	78	-78
Inplane Pressure Stress	35	35	35	35
Inplane Stress Summary	14	49	190	-89



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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			4	4	4	4
Shear Stress Summary			6	6	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			62	72	190	139

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			6	6	6	6
Outplane Bending (P)			19	-19	19	-19
Outplane Membrane (Mc)			-13	-13	13	13
Outplane Bending (Mc)			-76	76	76	-76
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			35	35	35	35
Outplane Stress Summary			-29	85	151	-39

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			5	5	5	5
Inplane Bending (P)			61	-61	61	-61
Inplane Membrane (Mc)			-15	-15	15	15
Inplane Bending (Mc)			-255	255	255	-255
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			71	72	71	72
Inplane Stress Summary			-132	257	408	-223

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			4	4	4	4
Shear Stress Summary			3	3	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress			132	257	408	223



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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	6	5	6
Hoop Stress Summary	0	1	23	25

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	0	0	11	-1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	3	3	23	27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-13	-13	13	13
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	6	5	6
Hoop Stress Summary	-2	0	25	27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu	Cl	Du	Dl
	Left	Left	Right	Right



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Nozzle Calcs.: S2-Shell Inlet

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pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-14	14	14	-14
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-11	11	23	-12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)				
	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)				
	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	11	12	26	40

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	71	72	71	72	71	72	71	72
Circ. Pl (SUS)	-6	-6	18	18	-10	-10	20	20
Circ. Q (SUS)	-3	3	41	-41	-194	194	316	-316
Long. Pm (SUS)	35	35	35	35	35	35	35	35
Long. Pl (SUS)	-4	-4	14	14	-7	-7	19	19
Long. Q (SUS)	-17	17	139	-139	-57	57	95	-95
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	71.0	72.0	71.0	72.0	71.0	72.0	71.0	72.0
Pm+Pl (SUS)	65.0	66.0	89.0	90.0	61.0	62.0	91.0	92.0
Pm+Pl+Q (Total)	62.5	70.1	188.2	139.1	133.1	256.1	407.1	224.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	72.00	138.00	Passed
Pm+Pl (SUS)	92.03	207.00	Passed
Pm+Pl+Q (TOTAL)	407.10	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	6	5	6	5	6	5	6
Circ. Pl (SUS)	-6	-6	18	18	-7	-7	19	19
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	0	0	6	-6	-11	11	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	5.0	6.0	5.0	6.0	5.0	6.0	5.0	6.0
Pm+Pl (SUS)	1.0	0.0	23.0	24.0	2.0	1.0	24.0	25.0
Pm+Pl+Q (Total)	2.2	2.0	23.1	26.1	11.1	12.2	24.3	38.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	138.00	Passed
Pm+Pl (SUS)	25.00	207.00	Passed
Pm+Pl+Q (TOTAL)	38.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: S4-Shell outlet

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Input, Nozzle Desc: S4-Shell outlet

From: 60

Pressure for Reinforcement Calculations	P	1567.203	KPa.
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1755.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7106.18	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		477.8200	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	15.0900	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	32.5890	mm
Height of Beveled Transition	L'	34.5890	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		75.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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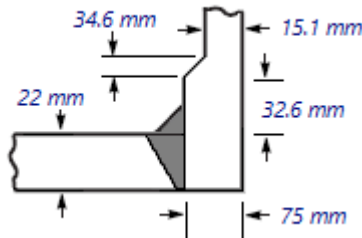
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Nozzle Calcs.: S4-Shell outlet

Nozl: 24

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Reinforcement CALCULATION, Description: S4-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	477.820 mm.
Actual Thickness Used in Calculation	15.090 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $1567.2 \cdot 880.5 / (138 \cdot 1.0 - 0.6 \cdot 1567.2)$

= 10.0688 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $1567.2 \cdot 238.91 / (138 \cdot 1.0 + 0.4 \cdot 1567.2)$

= 2.7011 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

= (Hub Thickness - Neck Thickness) / $\tan(30)$

= $59.91 / 0.5773$

= 103.7672 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	907.2801 mm
--	----	-------------

Parallel to Vessel Wall, opening length	d	453.6400 mm
---	---	-------------

Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	47.5000 mm
---	------	------------

Weld Strength Reduction Factor [fr1]:

= $\min(1, S_n / S_v)$

= $\min(1, 138.0 / 138.0)$

= 1.000

Weld Strength Reduction Factor [fr2]:

= $\min(1, S_n / S_v)$

= $\min(1, 138.0 / 138.0)$

= 1.000

Weld Strength Reduction Factor [fr3]:

= $\min(fr2, fr4)$

= $\min(1.0, 1.0)$

= 1.000



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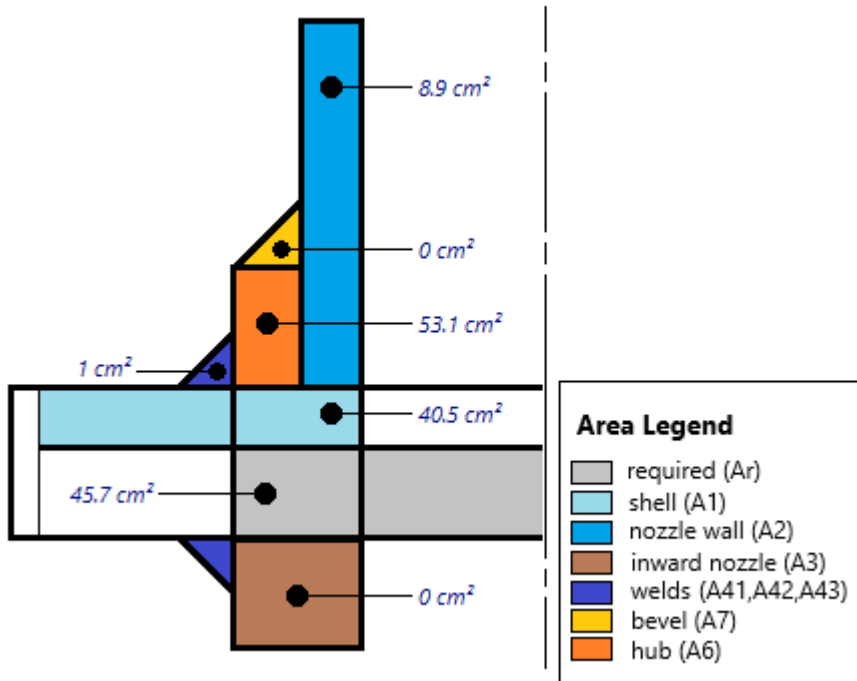
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Nozzle Calcs.: S4-Shell outlet

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	45.676	NA	NA
Area in Shell	A1	40.515	NA	NA
Area in Nozzle Wall	A2	8.919	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	53.063	NA	NA
TOTAL AREA AVAILABLE	Atot	103.498	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \cdot UG-37(c)$$

$$= (453.64 \cdot 10.0688 \cdot 1.0 + 2 \cdot 72.0 \cdot 10.0688 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 45.676 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 453.64(1.0 \cdot 19.0 - 1.0 \cdot 10.069) - 2 \cdot 72.0$$

$$(1.0 \cdot 19.0 - 1.0 \cdot 10.0688) \cdot (1 - 1.0)$$

$$= 40.515 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 47.5)(12.09 - 2.7)1.0$$

$$= 8.919 \text{ cm}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - c_{an}/0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(47.5, 400.0, 32.6)) \cdot (75.0 - 15.1) \cdot 1.0$$

$$= 53.063 \text{ cm}^2 \text{ Includes Bevel Area of } 14.015 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 5.7011 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 5.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb_1 = 13.0688 \text{ mm}$

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 13.0688 \text{ mm}$

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.331, \max(13.0688, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(5.7011, 11.3312)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 15.0900 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	36.8,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	308.2 N./mm ²	Passed
Occasional	:	14.3,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	16.9,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 15.09$, $t_r = 2.701$, $c = 3.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.223$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 4 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 22.0$, $t_r = 10.069$, $c = 3.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.53$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C

Min Metal Temp. at Required thickness (UCS 66.1) -33 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -33 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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Nozzle Calcs.: S4-Shell outlet Nozl: 24 9:28am Sep 9, 2026

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -29 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1567.20/1960.01 = 0.800

Weld Size Calculations, Description: S4-Shell outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (45.6761 - 40.5155 + 2 \cdot 72.0 \cdot 1.0 \cdot (1.0 \cdot 19.0 - 10.0688)) \cdot 138)$
 $= 25359.22 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (61.9829 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 88626.86 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (61.9829 + 0.0 + 1. + (27.36)) \cdot 138$
 $= 127126.69 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (61.9829 + 0.0 + 1. + 0.0 + (27.36)) \cdot 138$
 $= 127126.69 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 477.82 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 51751. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 232.865) \cdot (75.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 518833. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 477.82 \cdot (14.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 109417. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (51751 + 518833) = 570584 \text{ Kgf}$



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Failure Path [Path 2-2]:
 = (Sonw + Tpgw + Tngw + Sinw)
 = (51751 + 0 + 109417 + 0) = 161169 Kgf

Failure Path [Path 3-3]:
 = (Sonw + Tngw + Sinw)
 = (51751 + 109417 + 0) = 161169 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 570584 Kgf, must exceed W = 25359 Kgf or W1 = 88626 Kgf
 Path 2-2 = 161168 Kgf, must exceed W = 25359 Kgf or W2 = 127126 Kgf
 Path 3-3 = 161168 Kgf, must exceed W = 25359 Kgf or W3 = 127126 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2505 KPa.

The Drop for this Nozzle is : 33.1492 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 455.1493 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 307.35 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 453.64 mm

Input Echo, WRC297 Item 1, Description: S4-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1755.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	597.640	mm
Nozzle Thickness	75.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	1567.20	KPa.
Radial Load	P	-3824.19	Kgf
Circumferential Shear	Vc	2549.46	Kgf
Longitudinal Shear	Vl	2549.46	Kgf
Circumferential Moment	Mc	3567.87	Kg-m.
Longitudinal Moment	Ml	3567.87	Kg-m.
Torsional Moment	Mt	5096.96	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No



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*Warning - The ratio, Dn/Tn (7.969) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 597.64 + 2 \cdot 1.65 \cdot \sqrt{890.0(22.0 - 3.0)}$$

$$= 1026.767 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1780.000 mm

Nozzle Outside Diameter (d) = 597.640 mm

Vessel Thickness used (T) = 19.000 mm

Nozzle Thickness used (t) = 72.000 mm

T / t = 0.264

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.250$$

Nr/P = 0.056

Mr/P = 0.098

M0/P = 0.031

N0/P = 0.065

MrD/Mc = 0.262

NrDL/Mc = 0.093

M0d/Mc = 0.079

N0DL/Mc = 0.086

MrD/Ml = 0.081

NrDL/Ml = 0.057

M0D/Ml = 0.023

N0DL/Ml = 0.075

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-12	-12	12	12
Outplane Bending (ML)	-22	22	22	-22
Normal Pressure Stress	71	73	71	73
Outplane Stress Summary	62	71	132	50

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-78	78	78	-78
Inplane Pressure Stress	35	35	35	35
Inplane Stress Summary	15	50	191	-89



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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			4	4	4	4
Shear Stress Summary			6	6	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			63	73	191	140

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			6	6	6	6
Outplane Bending (P)			19	-19	19	-19
Outplane Membrane (Mc)			-13	-13	13	13
Outplane Bending (Mc)			-76	76	76	-76
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			35	35	35	35
Outplane Stress Summary			-28	86	152	-39

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			5	5	5	5
Inplane Bending (P)			61	-61	61	-61
Inplane Membrane (Mc)			-15	-15	15	15
Inplane Bending (Mc)			-255	255	255	-255
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			71	73	71	73
Inplane Stress Summary			-131	258	409	-222

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			4	4	4	4
Shear Stress Summary			3	3	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress			131	258	409	222



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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	6	5	6
Hoop Stress Summary	0	1	23	25

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	0	0	11	-1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	B1
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	B1
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	3	3	24	27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	D1
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-13	-13	13	13
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	6	5	6
Hoop Stress Summary	-2	0	25	27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu	Cl	Du	D1
	Left	Left	Right	Right



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pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-14	14	14	-14
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-11	11	23	-12
----------------------	-----	----	----	-----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	0	0	1	1
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	11	12	26	40
------------------------	----	----	----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	71	73	71	73	71	73	71	73
Circ. Pl (SUS)	-6	-6	18	18	-10	-10	20	20
Circ. Q (SUS)	-3	3	41	-41	-194	194	316	-316
Long. Pm (SUS)	35	35	35	35	35	35	35	35
Long. Pl (SUS)	-4	-4	14	14	-7	-7	19	19
Long. Q (SUS)	-17	17	139	-139	-57	57	95	-95
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	71.0	73.0	71.0	73.0	71.0	73.0	71.0	73.0
Pm+Pl (SUS)	65.0	67.0	89.0	91.0	61.0	63.0	91.0	93.0
Pm+Pl+Q (Total)	62.5	71.1	188.2	140.1	133.1	257.1	407.1	223.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	73.00	138.00	Passed
Pm+Pl (SUS)	93.03	207.00	Passed
Pm+Pl+Q (TOTAL)	407.10	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	6	5	6	5	6	5	6
Circ. Pl (SUS)	-6	-6	18	18	-7	-7	19	19
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	0	0	6	-6	-11	11	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	5.0	6.0	5.0	6.0	5.0	6.0	5.0	6.0
Pm+Pl (SUS)	1.0	0.0	23.0	24.0	2.0	1.0	24.0	25.0
Pm+Pl+Q (Total)	2.2	2.0	23.1	26.1	11.1	12.2	24.3	38.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	138.00	Passed
Pm+Pl (SUS)	25.00	207.00	Passed
Pm+Pl+Q (TOTAL)	38.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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ASME TS Calc:

Case: 1 9:28a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	1550.00 KPa.
Shell Maximum Operating Pressure	Psox,max	1550.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	22.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1755.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	150.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	1000.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	103.42 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1755.000 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	2408
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	6450.00 mm
Straight Tube Length (bet. inner tubsht faces) L		6350.00 mm
Design Temperature of the Tubes		150.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	158.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	857.961 mm



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Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	20857.4 cm ²
Perimeter of Tube Layout	Cp	5403.5 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	150.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	50.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1783.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

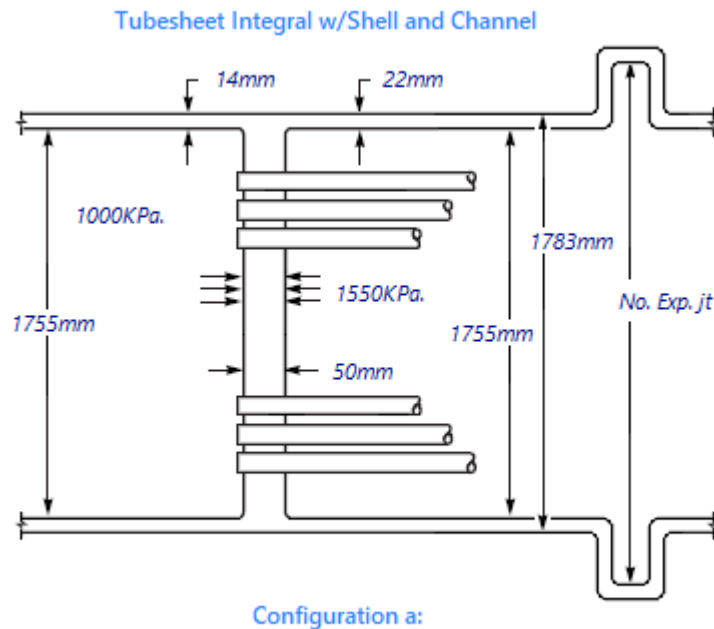
Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	573.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Channel Side Min. Design Pressure	Ptd,min	-103.4190 KPa.
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



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ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	150.0 °C	0.19506E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	150.0 °C	0.19506E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	150.0 °C	0.19506E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	150.0 °C	0.19506E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:



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The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1000.0 \cdot (19.05/2 - 0)) / (92.4 \cdot 1.0 + 0.4 \cdot 1000.0)$$

$$= 0.1026 + 0.0000 = 0.1026 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 150.00 °C
Elastic Modulus for Material 199807168.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	6350.00	9.03	50.0000	0.0134948	95.79
EMAWP = (2.167/(D/T) - 0.0833) * B = 15010.6582 KPa.						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4273	19.05	6350.00	44.58	50.0000	0.0005534	55.29
EMAWP = (4*B)/(3*(D/T)) = (4 * 55.2909)/(3 * 44.5834) = 1653.4625 KPa.						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4273 mm
Allowable Internal Pressure at Corroded thickness	22456.99 KPa.
Required Internal Design Pressure	1000.00 KPa.
Allowable External Pressure at Corroded thickness	15010.66 KPa.
Required External Design Pressure	1653.42 KPa.
Required Thickness due to Shell Side pressure	0.4273 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/50.0$$

$$= 0.0000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (p); dt - 2 \cdot tt)$$

$$= \max(19.05 - 2 \cdot 2.11 (0.19506E+09/1.9506E+09) \cdot (92/137) \cdot (0); 19.05 - 2 \cdot 2.11)$$

$$= 19.0500 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 857.961 + 19.05$$

$$= 1734.972 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.75 - 19.05) / 31.75$$

$$= 0.400$$



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Equivalent outer tube limit radius [ao]:

$$= D_o / 2$$

$$= 1734.9719 / 2$$

$$= 867.4860 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)}$$

$$= 31.75 / \sqrt{1 - 4 \cdot \min(0 \cdot 100.0 ; 4 \cdot 1734.972 \cdot 31.75) / (\pi \cdot 1734.972^2)}$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.75 - 19.05) / 31.75$$

$$= 0.4000$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 877.5 / 867.486$$

$$= 1.0115$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 877.5 / 867.486$$

$$= 1.0115$$

Compute parameter [xt]:

$$= 1 - Nt / ((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 2408 / ((19.05 - 2 \cdot 2.11) / (2 \cdot 867.486))^2$$

$$= 0.8241$$

Determine parameter [xs]:

$$= 1 - Nt / (dt / (2 \cdot a_o))^2$$

$$= 1 - 2408 / (19.05 / (2 \cdot 867.486))^2$$

$$= 0.7097$$

Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((8.0 - 0); 0)$$

$$= 8.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot t_s (D_s + t_s) E_s / L$$

$$= \pi \cdot 22.0 (1755.0 + 22.0) \cdot 19506 \text{E}+09 / 6350.0$$

$$= 384730.8438 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot t_t (D_t - t_t) E_t / L$$

$$= \pi \cdot 2.11 (19.05 - 2.11) \cdot 19506 \text{E}+09 / 6350.0$$

$$= 351.7570 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (Nt \cdot K_t)$$

$$= 384730.844 / (2408 \cdot 351.757)$$

$$= 0.45421$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s / K_j)$$

$$= 1 / (1 + 384730.844 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + t_s) t_s)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1755.0 + 22.0) 22.0)$$



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= 0.0092 1/mm

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.009 \cdot 0.19506E+09 \cdot 22.0^3 / (6(1 - 0.3^2))$$

$$= 3497439488.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1755.0 + 22.0) \cdot 0.34974E+1 / 50.0^3 \cdot (1 + 50.0 \cdot 0.009 + (50.0 \cdot 0.009)^2 / 2)$$

$$= 466973248.0000 \text{ KPa}$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1755.0^2 / (4 \cdot 0.19506E+09 \cdot 22.0) \cdot (1 - 1755.0 \cdot 0.3 / (2(1755.0 + 22.0)))$$

$$= 0.15284E+00 \text{ mm/N./mm}^2$$

Channel coefficient [βc]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(1755.0 + 14.0)14.0}$$

$$= 0.0116 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.012 \cdot 0.19921E+09 \cdot 14.0^3 / (6(1 - 0.3^2))$$

$$= 1156483072.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1755.0 + 14.0 / \cos(0)) \cdot 0.11565E+1 / 50.0^3 \cdot (1.0 + 50.0 \cdot 0.012 \cdot (1.0 + 1.0) / 2 + (50.0 \cdot 0.012)^2 / 2 \cdot 1.0)$$

$$= 171293840.0 \text{ KPa}$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1755.0^2 / (4 \cdot 0.19921E+09 \cdot 14.0) \cdot (1 - 1755.0 \cdot 0.3 / (2(1755.0 + 14.0)))$$

$$= 0.23499E+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

h/p = 1.574803 ; μ* = 0.400000 ; E* = 84004048 KPa.

$$E^*/E = 0.430652 ; \nu^* = 0.301272$$

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^*)Nt \cdot Et \cdot t(d_t - t_t) a_o^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.301^2)2408 \cdot 0.19506E+09 \cdot 2.11(19.05 - 2.11)867.486^2) / (84004048 \cdot 6350.0 \cdot 50.0^3))^{1/4}$$

$$= 8.0189$$

Values from Table 4.18.3

$$Z_d = 0.002893 ; Z_v = 0.015911 ; Z_m = 0.181704$$

$$Z_a = 0.107359E+04 ; Z_w = 0.015911$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1783.0 / 1734.9719$$



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ASME TS Calc:

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= 1.0277

Compute coefficient [F]:

$$= (1 - \nu) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.3) / 84004048 \cdot (0.46697E+09 + 0.17129E+09 + 0.19506E+09 \cdot \ln(1.03))$$

= 5.3533

Compute parameter [ϕ]:

$$= (1 + \nu) F$$

$$= (1 + 0.3013) 5.3533$$

= 6.9661

Compute parameter [Q1]:

$$= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.0115 - 1 - 6.9661 \cdot 0.0159) / (1 + 6.9661 \cdot 0.1817)$$

= -0.043823

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.00289 + -0.04382 \cdot 0.0159) / 2 \cdot 8.01888^4$$

= 4.5399

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.0159 + -0.04382 \cdot 0.1817) / 2 \cdot 8.01888^4$$

= 16.4318

Compute parameter [U]:

$$= (Z_w + (\rho_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.0159 + (1.0115 - 1) 0.1817) 8.01888^4 / (1 + 6.9661 \cdot 0.1817)$$

= 32.8637

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

= 0 (config a)

Compute Parameter [γ]:

= 0.000 mm (For Pressure only cases)

Calculate parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.0115 \cdot 0.34974E+1 \cdot 0.0092 \cdot 0.152841 (1 + 50.0 \cdot 0.0092)$$

= 7257.0801 mm²

Calculate parameter [ω^*s]:

$$= a o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s$$

$$= 867.486^2 (1.012^2 - 1) \cdot (1.012 - 1) / 4 - 7257.08$$

= -7206.6499 mm²

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)]$$

$$= 1.012 \cdot 0.11565E+1 [0.234988 (1.0 + 0.012 \cdot 1.0 \cdot 50.0) - 0 (1.0/50.0 + 1.0 \cdot 0.012/2)]$$

= 5009.6670 mm²

Parameter [ω^*c]:

$$= a o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c$$

$$= 867.48596^2 [(1.01154^2 + 1) \cdot (1.01154 - 1) / 4 - (1.01154 - 1) / 2] - 5009.66699$$

= -4959.2373 mm²

Pressure [P^*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D



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Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= Ps \{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs - [(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \}$$

$$= 1550.0 \{ 0.71 + 2(1 - 0.71) 0.3 + [2/0.454 \cdot (1755.0/1734.972)^2] 0.3 - [(1.012^2 - 1)/(1.0 \cdot 0.454)] - [(1 - 1.0)/(2 \cdot 1.0 \cdot 0.454)] [(0^2 - (1755.0)^2)/1734.972^2] \}$$

$$= 3385.8174 \text{ KPa.}$$

Pressure [P't]:

$$= [xt + 2(1 - xt) vt + 1/(J \cdot Kst)] Pt$$

$$= [0.824 + 2(1 - 0.824) 0.3 + 1/(1.0 \cdot 0.454)] 1000.0$$

$$= 3131.2449 \text{ KPa.}$$

Pressure [Pγ]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 2408 \cdot 351.757 \cdot 0 / (3.142 \cdot 867.486^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [Pw]:

$$= -\gamma b \cdot U \cdot W \cdot / (2 \cdot \pi \cdot ao^2)$$

$$= -0 \cdot 32.864 \cdot 0 / (2 \cdot \pi \cdot 867.486^2)$$

$$= -0.0000 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt)$$

$$= -(32.864/867.486^2) \cdot (-7206.65 \cdot 1550.0 - -4959.237 \cdot 1000.0)$$

$$= 271.2422 \text{ KPa.}$$

Calculate the pressure [Pω]:

$$= (U/ao^2) \cdot (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)$$

$$= (32.864/867.486^2) \cdot (7257.0801 \cdot 0 - 5009.667 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot (Ps - P't + P\gamma + Pw + Pw + Prim)$$

$$= 0.1000E+01 \cdot 0.454 / (1 + 1.0 \cdot 0.454 (4.54 + (1.012 - 1)16.432)) \cdot (3385.817 - 3131.245 + 0 + 0 + 0 + 271.242)$$

$$= 75.8616 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt) - (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)) + W \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-7206.65 \cdot 1550.0 - -4959.237 \cdot 1000.0) - (7257.08 \cdot 0 - 5009.667 \cdot 0)) \cdot 0 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 6.96607 \cdot 0.1817)$$

$$= -279.5427 \text{ Kg}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.044 + 2 \cdot -279.543 / (75.862 \cdot 867.486^2)$$

$$= -0.13986$$

Fm Value from Table 4.18.3 = 0.069930

Tube sheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$



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ASME TS Calc:

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$$= (1.5 \cdot 0.0699 / 0.4) \cdot (2 \cdot 867.486 / (50.0 - 8.0))^2 \cdot 75.86$$

$$= 33.9489 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 138.0 = 207.0 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.0298 / 0.4) \cdot (2 \cdot 867.486 / (17.299 - 8.0))^2 \cdot 53.13$$

$$= 206.9954 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 17.2994 + 0 + 0$$

$$= 17.2994 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y)$$

$$= (2 \cdot 0.4 \cdot 50.0 / 867.486) \cdot 110.398$$

$$= 5.0905 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ limit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 3.8100 \text{ mm}$$

$$\text{Tubesheet Shear Stress} = 4.6175 \text{ N./mm}^2$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(17.2994; 3.81)$$

$$= 17.2994 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -0.1614, F_{tmax} = 2.2121$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((1550.0 \cdot 0.7097 - 1000.0 \cdot 0.8241) - (75.862) \cdot -0.1614) /$$

$$(0.8241 - 0.7097)$$

$$= 2.5199 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((1550.0 \cdot 0.7097 - 1000.0 \cdot 0.8241) - (75.862) \cdot 2.2121) /$$

$$(0.8241 - 0.7097)$$

$$= 0.9456 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 2.52 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 92.3983 \text{ N./mm}^2$$

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 2.52 \cdot 1.1229$$

$$= 28.85 \text{ Kg}$$

Tube Metal Area [At]:

$$= \pi(d_o - t)t$$



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ASME TS Calc:

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$$= \pi (19.05 - 2.11) 2.11$$

$$= 1.123 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (19.05 - 2.11) 92.398$$

$$= 1057.989 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 92.4$$

$$= 1057.9891 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 92.398 / 92.398$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 92.4$$

$$= 702.9343 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) 1.0 \cdot 1.0} - 0.75 \cdot 19.05 \right)$$

$$= 2.562 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm

Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 1057.99 is $\geq$ tube strength 1057.99 Kg

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 138.0 \cdot 1.0 = 138.0 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s ,m]:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 867.486^2 / ((1755.0 + 22.0) 22.0) \cdot [75.86 + (1.012^2 - 1)(1550.0 - 1000.0)] + 877.5^2 \cdot 1000.0 / ((1755.0 + 22.0) 22.0)$$

$$= 21.4036 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 22.0 mm, the minimum Shell length adjacent to the tubesheet should be 353.690 mm

Shell Membrane Stress due to Joint Interaction [σ_s ,m]:

$$= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 867.486^2 / ((1755.0 + 22.0) 22.0) [75.86 + (1.012^2 - 1)(1550.0 - 1000.0)] + 877.5^2 \cdot 1000.0 / ((1755.0 + 22.0) 22.0)$$

$$= 21.4036 \text{ N./mm}^2$$



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ASME TS Calc:

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Shell Bending Stress due to Joint Interaction [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E_s \cdot t_s)] + 6(1 - \nu^2) / (E \cdot h) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 0.34974E+1 / 22.0^2 \{ 0.009 [0.152841 \cdot 1550.0 + 877.5^2 \cdot 0 / (.42914E+1)] + 6(1 - 0.3^2) / (84004048) (867.49 / 50.0)^3 (1 + 50.0 \cdot 0.01 / 2) [75.9 (0.016 + 0.182 \cdot -0.044) + 2 / 867.49^2 \cdot 0.182 \cdot -279.543] \}$$

$$= 81.4077 \text{ N./mm}^2$$

Shell Stress Summation vs. Allowable

$$\therefore | \sigma_{s,m} | + | \sigma_{s,b} | \leq 1.5 \cdot S_s$$

$$\therefore | 21.4 | + | 81.41 | \leq 207.0 \text{ N./mm}^2$$

$$\therefore 102.81, \text{ must be } \leq 207.0 \text{ N./mm}^2$$

Note:

For a given Channel thickness of 14.0 mm, the minimum Channel length adjacent to the tubesheet should be 282.147 mm

Channel Membrane Stress due to Joint Interaction [σ_c, m]:

$$= a_c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta))$$

$$= 877.5^2 \cdot 1000.0 / ((1755.0 + 14.0) 14.0 \cdot \cos(0))$$

$$= 31.093 \text{ N./mm}^2$$

Channel Bending Stress due to Joint Interaction [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot \{ (\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^2) / E \cdot h) \cdot (a_o / h)^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 0.11565E+1 / 14.0^2 \cdot \{ (0.234988 \cdot 1.0 \cdot 0.012 - 0 \cdot 1.0) \cdot 1000.0 - 6((1 - 0.301^2) / 84004048) \cdot (867.486 / 50.0)^3 \cdot (1.0 + 50.0 \cdot 0.012 / 2 \cdot 1.0) \cdot [75.9 (0.01591 + 0.1817 \cdot -0.044) + 2 / 867.486^2 \cdot 0.1817 \cdot -279.543] \}$$

$$= 107.2632 \text{ N./mm}^2$$

Channel Stress Summation vs. Allowable

$$\therefore | \sigma_{c,m} | + | \sigma_{c,b} | \leq 1.5 \cdot S_c$$

$$\therefore | 31.09 | + | 107.26 | \leq 207.0 \text{ N./mm}^2$$

$$\therefore 138.36 \text{ must be } \leq 207.0 \text{ N./mm}^2$$

Computations completed for ASME tubesheet configuration: a

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	33.9 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	1.5 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	2.5 <=	92.4 N./mm ²	Pass
Maximum Force on any one Tube	28.9 <=	1058.0 Kgf	Pass
Axial Membrane Stress in Shell	21.4 <=	138.0 N./mm ²	Pass
Shell Stress (jt. inter.)	102.8 <=	207.0 N./mm ²	Pass
Channel Stress (jt. inter.)	138.4 <=	207.0 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	17.299	50.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 16 Load Cases:



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ASME TS Calc:

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Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tube-sheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	32.476 ...	86	207	9 110	Ps+Pt-Th D1	Ok
D2uc	39.021 ...	129	207	12 110	Ps+Pt-Th D2	Ok
D3uc	17.299 ...	34	207	1 110	Ps+Pt-Th D3	Ok
D4uc	13.316 ...	9	207	1 110	Ps+Pt-Th D4	Ok
O1uc	9.028 ...	61	438	9 110	Ps+Pt+Th O1	Ok
O2uc	11.414 ...	85	438	11 110	Ps+Pt+Th O2	Ok
O3uc	3.810 ...	24	438	1 110	Ps+Pt+Th O3	Ok
O4uc	3.810 ...	...	438	... 110	Ps+Pt+Th O4	Ok
D1c	32.490 ...	77	207	10 110	Ps+Pt-Th-c D1	Ok
D2c	38.627 ...	120	207	13 110	Ps+Pt-Th-c D2	Ok
D3c	18.994 ...	35	207	2 110	Ps+Pt-Th-c D3	Ok
D4c	15.387 ...	8	207	1 110	Ps+Pt-Th-c D4	Ok
O1c	15.366 ...	60	438	10 110	Ps+Pt+Th-c O1	Ok
O2c	17.800 ...	88	438	12 110	Ps+Pt+Th-c O2	Ok
O3c	9.810 ...	28	438	2 110	Ps+Pt+Th-c O3	Ok
O4c	9.810 ...	...	438	... 110	Ps+Pt+Th-c O4	Ok

Max: 39.0213 ... mm 0.624 0.114 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses				----	:	---- Shell Band Stress				----	:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	10	137	...	...		...	...	...	...		Ok	
D2uc	10	137	...	...		...	...	...	...		Ok	
D3uc	21	137	...	...		...	...	...	...		Ok	
D4uc	1	137	-1	-111		...	...	...	...		Ok	
O1uc	10	464	...	...		...	...	...	...		Ok	
O2uc	11	464	...	...		...	...	...	...		Ok	
O3uc	21	464	...	...		...	...	...	...		Ok	
O4uc	...	464	...	...		...	...	...	...		Ok	
D1c	12	137	...	...		...	...	...	...		Ok	
D2c	12	137	...	...		...	...	...	...		Ok	
D3c	25	137	...	...		...	...	...	...		Ok	
D4c	1	137	-1	-109		...	...	...	...		Ok	
O1c	12	464	...	...		...	...	...	...		Ok	



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ASME TS Calc:

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02c	13	464	...	...	...	...	...	...	Ok
03c	25	464	...	...	...	...	...	...	Ok
04c	...	464	...	...	...	...	...	...	Ok

Max RATIO 0.180 0.011 ...

Tube, Shell and Channel Stress Summary:

		---- Tube Stresses				Tube Loads		Shell Stress		Channel Stress	Stress Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd		Stress	Allwd	Stress	Allwd Fail
D1uc	10	92	-8	-87	118	1057	60	206	170	206	Ok
D2uc	12	92	-10	-81	134	1057	147	206	50	206	Ok
D3uc	3	92	...	...	29	1057	103	206	138	206	Ok
D4uc	1	92	-1	-87	12	1057	6	206	18	206	Ok
O1uc	10	184	-8	-87	118	2115	60	464	170	485	Ok
O2uc	11	184	-9	-80	124	2115	143	464	32	485	Ok
O3uc	3	184	...	...	29	2115	103	464	138	485	Ok
O4uc	...	184	...	...	...	2115	...	464	...	485	Ok
D1c	12	92	-9	-87	136	1057	76	206	215	485	Ok
D2c	12	92	-11	-87	135	1057	176	206	62	206	Ok
D3c	3	92	...	...	29	1057	119	206	176	206	Ok
D4c	1	92	-1	-87	14	1057	8	206	22	206	Ok
O1c	12	184	-9	-87	136	2115	76	464	215	485	Ok
O2c	11	184	-10	-87	125	2115	171	464	39	485	Ok
O3c	3	184	...	...	29	2115	119	464	176	485	Ok
O4c	...	184	...	...	...	2115	...	464	...	485	Ok

Max RATIO 0.129 0.129 0.129 0.850 0.850

Summary of Thickness Comparisons for 16 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	39.021	50.000	Ok
Tube thickness :	0.427	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Min Shell length of thk, (22.000) adj. to tubesheet: 353.690 mm

Min Channel length of thk, (14.000) adj. to tubesheet: 282.147 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.061 mm
 Due to Pressure + Thermal Shell Compresses by : -0.061 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	2430.85	0.997	2667.26	1.000
Tubesheet Shear Stress	10955.64	1.000	14776.10	1.000
Tube Tensile Stress	7759.30	1.000	13130.23	1.000
Tube Compressive Stress	7189.46	0.689	13487.00	1.000
Tube-Tubesheet Joint load	7759.30	1.000	13130.23	1.000
Shell Stress (Axial, Junction)	2430.85	0.997	2667.26	0.919
Tubesheet-Channel Junction	2252.69	1.000	2667.26	0.919
Tube Pressure Stress	22456.98	1.000	15010.65	1.000
Minimum MAWP	2252.69		2667.26	



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ASME TS Calc:

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Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	2430.85	0.999	2670.35	1.000
Tubesheet Shear Stress	11848.43	1.000	15869.11	1.000
Tube Tensile Stress	8972.10	1.000	13188.76	1.000
Tube Compressive Stress	10952.23	1.000	15833.80	1.000
Tube-Tubesheet Joint load	8972.10	1.000	13188.76	1.000
Shell Stress (Axial, Junction)	2430.85	0.999	2670.35	0.475
Tubesheet-Channel Junction	2430.85	0.999	2670.35	0.261
Tube Pressure Stress	22456.98	1.000	15019.19	1.000
Minimum MAPnc	2430.85		2670.35	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0109	0.0109	...	...	...	...
2	-0.0609	-0.0609	...	...	...	...
3	-0.0489	-0.0489	...	...	...	...
4	-0.0011	-0.0011	...	...	...	...
5	0.0109	0.0109	...	...	...	...
6	-0.0597	-0.0598	...	...	...	...
7	-0.0489	-0.0489	...	...	...	...
8	...	...	...	...	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness}))$$

$$= \max(50.0/4, \min(50.0, 22.0))$$

$$= 22.000 \text{ mm}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{ps}/\text{Shellside MAPnc}, 1)$$

$$= \min(1550.0/2670.35, 1)$$

$$= 0.580$$

Thickness Ratio = 0.58, Temperature Reduction per Fig. UCS 66.1 = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-45 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness}))$$

$$= \max(50.0/4, \min(50.0, 14.0))$$

$$= 14.000 \text{ mm}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{pt}/\text{Tubeside MAPnc}, 1)$$

$$= \min(1000.0/2430.85, 1)$$



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ASME TS Calc:

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= 0.411

Thickness Ratio = 0.411, Temperature Reduction per Fig. UCS 66.1 = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

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Problems/Failures Summary:

Step: 37

9:28am

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: A standard flange seems to be governing the Shellside MAWP!**

Please review all reports carefully!

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: